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BARRICK GOLD CORPORATION

TECHNICAL REPORT ON THE HEMLO MINE, MARATHON, ONTARIO, CANADA

NI 43-101 Report

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1 SUMMARY

EXECUTIVE SUMMARY

Roscoe Postle Associates Inc. (RPA) was retained by Barrick Gold Corporation (Barrick) to prepare an independent Technical Report on the Hemlo open pit and underground mines (the Hemlo Mine), in Bomby Township, Northwestern Ontario, Canada. The purpose of this report is to support public disclosure of Mineral Resource and Mineral Reserve estimates at the Mine. This Technical Report conforms to National Instrument 43-101 Standards of Disclosure for Mineral Projects (NI 43-101). The effective date of the Mineral Resource and Mineral Reserve estimates in this Technical Report is December 31, 2016, and information in this Technical Report is current as of that date unless otherwise specified. RPA visited the property on December 14 and 15, 2016.

Barrick is a Canadian, publicly-traded mining company with a portfolio of operating mines and projects across five continents. The Hemlo Mine property is located just north of Lake Superior on the Trans-Canada highway approximately 35 km east of the town of Marathon, Ontario, at approximately 85°54' W longitude and 48°41' N latitude.

In production since 1985, operations at the Hemlo Mine have produced over 21 million ounces of gold to December 2016. Past underground production came from Williams, Golden Giant, and David Bell underground mines stretching over a length of two kilometres and a vertical distance of approximately 1,500 m below surface. Golden Giant and David Bell mines are now closed. Underground Mineral Reserves at the Williams Mine are projected to sustain the underground mine operations until 2021 at an average production rate of approximately 3,600 tpd.

The open pit is a traditional truck and shovel/loader operation that has supplied approximately 6,000 tpd of ore to the plant (up to 20,000 tpd moved, including waste and low-grade to stockpile). Open pit Mineral Reserves at the Williams Mine have the potential to support operations until 2027, at rates ranging from 5,000 tpd to 6,000 tpd.

Mineral processing of up to 10,000 tpd at the Hemlo Mine consists of grinding, cyanide leaching, carbon-in-pulp (CIP), carbon stripping and reactivation, electrowinning, and refining.

Table 1-1 summarizes Mineral Resources exclusive of Mineral Reserves.

TABLE 1-1 SUMMARY OF MINERAL RESOURCES – DECEMBER 31, 2016
Barrick Gold Corporation – Hemlo Mine

Category	Tonnage (000 t)	Grade (g/t Au)	Contained Gold (000 oz Au)
Measured	126	2.72	11
Indicated	58,771	0.90	1,709
Total Measured + Indicated	58,897	0.91	1,720
Inferred	7,765	1.94	484

Notes:

1. CIM definitions were followed for Mineral Resources.
2. Mineral Resources are estimated using a gold price of US\$1,500 per ounce, and a US\$/C\$ exchange rate of 1:1.30.
3. Mineral Resources are estimated at gold cut-off grades that vary by material type and mining method, averaging 0.36 g/t Au for open pit and 1.95 g/t Au for underground.
4. Mineral Resources are exclusive of Mineral Reserves.
5. Mineral Resources that are not Mineral Reserves do not have demonstrated economic viability.
6. Totals may not add due to rounding.

The Mineral Reserves are presented in Table 1-2.

TABLE 1-2 MINERAL RESERVES – DECEMBER 31, 2016
Barrick Gold Corporation - Hemlo Mine

Category	Tonnes (000 t)	Grade (g/t Au)	Contained Gold (000 oz Au)
Williams Open Pit			
Proven	6	1.07	0.2
Probable	19,119	1.17	718.6
Total Pit	19,125	1.17	718.8
Williams Underground			
Proven	972	3.76	117.4
Probable	5,645	4.13	750.2
Total Underground	6,617	4.08	867.6
Stockpiles			
Proven	40	1.25	1.6
Total Proven & Probable	25,782	1.92	1,587.9

Notes:

1. CIM definitions were followed for Mineral Reserves.
2. Mineral Reserves are estimated using an average long-term gold price of US\$1,000 per ounce and a US\$/C\$ exchange rate of 1:1.30.
3. Underground Mineral Reserves are estimated at an average break-even cut-off grade of 2.92 g/t Au and an average incremental cut-off grade of 2.43 g/t Au.
4. Open Pit Mineral Reserves are estimated at an average cut-off grade of 0.43 g/t Au.
5. Totals may not add due to rounding.

RPA is not aware of any environmental, permitting, legal, title, taxation, socio-economic, marketing, political, or other relevant factors that could materially affect the Mineral Resource or Mineral Reserve estimate.

CONCLUSIONS

Based on RPA's site visit, interviews with Hemlo personnel, and subsequent review of available information, RPA offers the following conclusions:

GEOLOGY AND MINERAL RESOURCE ESTIMATION

- The Hemlo deposits are best described as amphibolite-facies, disseminated replacement-stockwork Archean load gold deposits.
- Mineral Resource estimates have been prepared utilizing acceptable estimation methodologies. The classification of Measured, Indicated, and Inferred Mineral Resources conforms to Canadian Institute of Mining, Metallurgy and Petroleum (CIM) Definition Standards for Mineral Resources and Mineral Reserves dated May 10, 2014 (CIM definitions).
- The sampling, sample preparation, and analyses are appropriate for the style of mineralization and Mineral Resource estimation. The current drill hole database is reasonable for supporting the Mineral Resource and Mineral Reserve estimates.
- The end of year (EOY) 2016 open pit and underground resource block models are reasonable for supporting the Mineral Resource and Mineral Reserve estimates.
- The resource modelling procedures are well documented.
- Overall, RPA is of the opinion that Barrick has conducted high quality resource modelling work that exceeds industry standard practice.
- Mineral Resources are reported exclusive of Mineral Reserves and are estimated effective December 31, 2016.
- The open pit Measured plus Indicated Mineral Resources total 56.5 million tonnes grading 0.80 g/t Au, containing 1.45 million oz Au. Inferred Mineral Resources are estimated to be 5.1 million tonnes grading 0.62 g/t Au containing 0.1 million oz Au.
- The underground Measured plus Indicated Mineral Resources total 2.4 million tonnes grading 3.49 g/t Au, containing 0.27 million oz Au. Inferred Mineral Resources are estimated to be 2.6 million tonnes grading 4.5 g/t Au containing 0.4 million oz Au.

MINING AND MINERAL RESERVES

- The Mineral Reserve estimates have been prepared utilizing acceptable estimation methodologies and the classification of Proven and Probable Mineral Reserves conforms to CIM definitions.

- Mine planning for the Hemlo open pit and underground mines follows industry standards.
- RPA considers the selection of mining methods and the design practices to be appropriate for the deposits.
- Open pit Proven and Probable Mineral Reserves, consisting of Phases 5 and 6, total 19.1 million tonnes grading 1.17 g/t Au, containing 0.72 million oz Au.
- Underground Proven and Probable Mineral Reserves total 6.6 million tonnes grading 4.08 g/t Au, containing 0.87 million oz Au.
- The Life of Mine (LOM) plan is based upon current operating experience. Recovery and cost estimates are based upon actual operating data and engineering to support a Mineral Reserve statement. Economic analysis using these estimates generates a positive cash flow, which supports a statement of Mineral Reserves.
- Mineral Reserves contained in open pit Phase 6 are supported by a Pre-Feasibility Study. Work on a Feasibility Study and integration with plans for the larger operation are in progress.
- Although Phase 6 has the potential to extend the open pit mine life beyond the underground mine life, there is also good potential for further conversion of underground Mineral Resources to Mineral Reserves, and exploration opportunities, which are likely to result in continued blending of open pit and underground feed.
- Equipment purchases scheduled for the current LOM plan are reasonable.
- The workforce is well trained and capable of achieving the necessary production targets established by the engineering department in a safe manner.
- The reconciliation to 2016 production indicates that the open pit resource model is performing well within the normal range of variability (+/-10%), and that the underground resource model is conservative. Overall, Mineral Reserve estimates are within 10% of production results for tonnes and grade.

PROCESSING

- Ore blending is a well-coordinated and standard operation that ensures that the ore going to the processing operation provides optimum results. The milling operations are well run, safe, environmentally sound, and meet industry standards.
- The adjustments made to the process production data and mill head grades based on actual gold production conform to industry standards.
- The recovery estimates appear to be accurate based on recent data, however, large differences between budgeted head grade and actual head grade of ore from the underground mine has the potential to reduce the accuracy.

ENVIRONMENTAL AND COMMUNITY CONSIDERATIONS

- The Hemlo Mine is within environmental compliance. Permits will need to be updated for the open pit expansion, which may take up to two years to obtain.
- The Hemlo Mine has been a major contributor to the local economy, and Barrick will continue to communicate and work with the First Nations and other communities of interest.
- Reclamation estimates reviewed are realistic, in RPA's opinion.
- The David Bell Mine closed in May 2014, and progressive reclamation has been initiated at both that site and the former Golden Giant Mine sites.

RISKS

The mine has been in production for over 30 years and is a mature operation. RPA has not identified any significant risks and uncertainties that could reasonably be expected to affect the reliability or confidence in the exploration information, the Mineral Resource and Mineral Reserve estimates, or associated projected economic outcomes.

RECOMMENDATIONS

Based on the site visit, discussions with Hemlo personnel, and subsequent review of available information, RPA offers the following recommendations:

OPEN PIT

- Lithological and alteration mapping should be conducted at every opportunity, in order to refine the geological model.
- Continued mapping of discrete structures is recommended, for the purposes of ground control and the mitigation of water infiltration to the underground.
- Blast movement monitoring using electronic markers appears to be providing improvements in open pit dilution control. This should be supported and continued.

UNDERGROUND

- Resources and reserves are sensitive to cut-off grade. There appears to be a "natural cut-off" in the underground resource, which is a result of the lower grade material found in the vicinity of the higher grade ore lenses. Efforts should continue to be made to reduce the capital and operating costs.
- Significant mineral potential has been recognized in newly-acquired claims west of the current underground workings. This will require a concerted effort in a multi-year exploration program to find and define this potential. RPA recommends continued funding of this exploration effort.

- Continued focus should be given to the Lower B Zone, as this area shows significant potential to extend the mine life, yet will require geotechnical modelling and careful stope sequencing to be successful.
- Further conversion of underground resources and development of exploration potential is required to maintain the balance of open pit and underground feed past 2021.
- The 2016 production versus underground resource model results should be reviewed within the same volume, to assess model performance independent of other estimating factors. Consider adjusting some model input parameters to bring the conservative estimate closer to production results.

PROCESSING

- Ore characterization programs should be continued and supported in order to quantify the milling characteristics and achieve good estimates for plant throughput, gold recovery, and plant operating costs in future LOM plans and financial analyses.

ECONOMIC ANALYSES

Under NI 43-101, producing issuers may exclude the information required for Section 22 Economic Analysis on properties currently in production. RPA notes that Barrick is a producing issuer and the Hemlo Mine is currently in production. RPA has performed an economic analysis of the Hemlo Mine using the estimates presented in this report and confirms that the outcome is a positive cash flow that supports the statement of Mineral Reserves.

TECHNICAL SUMMARY

PROPERTY DESCRIPTION AND LOCATION

The Hemlo Mine is located in Bomby Township, in Northwestern Ontario, Canada, just north of Lake Superior on the Trans-Canada highway approximately 35 km east of the town of Marathon, Ontario. The property is located at approximately 85°54' W longitude and 48°41' N latitude.

The Hemlo Mine consists of the Williams Mine and processing facility located at the western end of the property, the Golden Giant Mine in the central position, and the David Bell Mine located at the eastern end of the property. The centre of the property is located at approximately 580,681 mE and 5,394,349 mN (UTM Zone 16, NAD83).

LAND TENURE

Land tenure interests at Hemlo are comprised of freehold patents, leasehold patents, and unpatented mining claims. All claims are contiguous.

In 2009, Barrick acquired Teck Cominco Ltd.'s 50% interest in the Hemlo Mine. In September 2010, Barrick completed the acquisition of Newmont Mining Corporation's (Newmont) interest in the Golden Giant Mine.

In March 2015, Barrick acquired certain lands to the west and north of the Williams Mine, as well as claims underlying the Molson Lake tailings management facility (TMF). Barrick acquired the following properties: CLM 271, CLM 272, CLM 284, CLM 277, CLM 278, and the southern portion of CLM 285. Barrick also acquired the mineral rights on portions of CLM 273, CLM 274, the Sceptre claim, and Horizon claims that it did not already own.

EXISTING INFRASTRUCTURE

The Williams Mine has been active since the start of production in 1985. The David Bell and Golden Giant mines have been closed and are currently in progressive reclamation. Underground workings for these two mines remain accessible through the Williams Mine and are currently used for ventilation for the Williams Mine. The Williams mine site consists of the following major infrastructure:

- A wash bay consisting of two bays for the 777 open pit haul trucks.
- A three bay garage near the open pit to service the pit equipment.
- A 10,000 tpd CIP gold mill with electrowinning and refining capacity.
- A 431 ha tailings management facility with plans to change the deposition method so it has capacity to operate for the LOM.
- A hoistroom and headframe (63.4 m high) with a 1,300 m deep shaft.
- A 4.6 m diameter production hoist with 22 t capacity skips. A 3.7 m service hoist with cage and counterweight.
- A 100 m³/hr paste backfill plant.
- A service building housing the maintenance facilities, offices, and mine dry.
- A four metre by five metre decline and a series of ramp connected levels.

- A 115 kV power line from the Ontario power grid feeding two 33 MVA main transformers.

HISTORY

Production from the Williams Mine began in mid-1985 from the A Zone open pit located at the east end of the property. Soon after, ore was also mined from the underground from the same area to sustain an initial 3,000 tpd mining rate. The completion of the main shaft, the B Zone infrastructure, and a mill expansion program in 1988 facilitated an increase to 6,300 tpd. The closing of the David Bell mill in 1999 and increased production from the C Zone pit brought the mill to a throughput of 10,000 tpd by the end of 2006. For the last five years, production has ranged from 8,000 tpd to 9,500 tpd.

The Williams Mine has been active from 1985 to the present. The David Bell/Golden Giant complex ceased production in 2014 and is currently in surface remediation. The three mines at Hemlo have extracted over 21 million ounces of gold as of the end of 2016.

GEOLOGY AND MINERALIZATION

The Hemlo gold deposit has been interpreted to be an atypical, mesozonal-orogenic, disseminated-replacement-stockwork deposit, broadly synchronous with D₂ and “middle” stage granitoid plutonism, prior to or synchronous with peak regional metamorphism, and involving magmatic ± metamorphic fluids. Much of the mineralization is confined to high strain zones and spatially associated with the contact between felsic volcanic rocks and sedimentary rocks.

The mineralization at the Williams Mine is currently grouped into two main zones: the B and C zones. The A Zone has been mined out. The A and B zones are geologically the same continuous zone and are also known as the Hemlo Main Zone. The C Zone represents multiple sub-parallel lenses of irregular, generally narrow, gold mineralization, divided into two mineralization types (100-series and 300-series). The Footwall Zone is located 30 m to 80 m into the footwall of the B Zone and consists of a series of structurally controlled, discontinuous mineralized lenses. There are three principal rock types at the Williams Mine: (1) feldspathic rock, (2) muscovite schist, and (3) biotite fragmental rock.

The Hemlo deposit is approximately three kilometres long by two kilometres deep and averages 20 m wide with a very small surface expression.

EXPLORATION STATUS

Despite nearly 30 years of mining in the Hemlo camp, significant exploration potential still exists within reach of the current mine workings. Ongoing projects include a data compilation in the lower B Zone area as well as diamond drill programs to explore mineralization located between the C Zone and B Zone. Data acquired with the acquisition of additional land to the west of the current workings suggests significant exploration potential could exist in this area. A number of exploration projects are proposed to test this potential.

Exploration and definition diamond drilling in 2016 totalled over 58 km in 294 holes, targeting several areas of the Williams Mine.

MINERAL RESOURCES

Table 1-1 above summarizes Mineral Resources exclusive of Mineral Reserves as of December 31, 2016, based on a US\$1,500/oz gold price. The EOY 2016 Measured and Indicated Mineral Resources total 58.9 million tonnes averaging 0.91 g/t Au and contain 1.7 million ounces of gold. In addition, the EOY 2016 Inferred Mineral Resources total 7.8 million tonnes averaging 1.94 g/t Au and contain 0.5 million ounces of gold.

The current resources are based on resource models updated in 2016 for the Williams C Zone, Interlake, B Zone West, B Zone Footwall, Lower B Zone, and Beth Adit areas. Resources are based on grade models updated in 2015 for the C Zone open pit area.

Ongoing model reconciliation of the model and production show that the approach and procedures used to create the Hemlo open pit and underground resource models are adequate for planning purposes and are in line with industry standards. The models are created and documented by a qualified person and follow standard industry practice. The model procedure is well documented through the master resource and reserve spreadsheets which includes records of visual checks and economics. RPA is of the opinion that the Mineral Resources as stated by Barrick are estimated in a manner that is appropriate for the style of mineralization and that the resource model is reasonable and acceptable to support the December 2016 Mineral Resource and Mineral Reserve estimates.

RPA is not aware of any environmental, permitting, legal, title, taxation, socio-economic, marketing, political, or other issues that could materially affect the Mineral Resource and Mineral Reserve estimates.

MINERAL RESERVES

The Mineral Reserves as of December 31, 2016, are presented in Table 1-2 above. Overall, RPA finds the Mineral Reserve estimates to be reasonable, acceptable, and compliant with CIM definitions.

RPA verified the Mineral Resource to Mineral Reserve conversions carried out by Hemlo engineering staff. This included review of stope dimensions, dilution factor, dilution grade, average stope widths, diluted tonnes, calculated percent dilution per stope, and mine recovery factors per stope.

MINING METHOD

Mining at Hemlo consists of a combination of open pit and underground operations for a current production rate of approximately 9,500 tpd. The underground mining operation at Williams is accessed via vertical shafts equipped to move personnel and material, and also ore and waste from the underground to the surface.

OPEN PIT

The open pit operation at the Hemlo Mine is a typical truck and loader/hydraulic shovel operation. The mine has an existing fleet of CAT 777 (90 t) trucks and 992 loaders.

During 2012, a Feasibility Study was completed on the open pit expansion at the Williams Mine. The open pit expansion would have added approximately 42.4 million tonnes grading approximately 1.10 g/t Au to Mineral Reserves, and would have extended the mine life from 2017 until 2029. During 2013, low gold prices forced the pit expansion to be suspended. During 2014, plans were made to mine the most readily accessible material within this resource and resulted in a small lay-back on the south wall known as Phase 5. Shortly thereafter, studies began for additional lay-backs on the north wall, which would allow access to deeper ore. A Pre-Feasibility Study on Phase 6 was completed in 2016, and Barrick will proceed with a Feasibility Study this year.

Production from open pit mining currently makes up approximately two-thirds of the mill feed. Over the next few years of Phase 5 mining, production will average approximately 5,700 tpd or 2.1 million tonnes per year ore, with 12,000 tpd of waste for a yearly average of five million tonnes of ore and waste.

Phase 6 is expected to utilize the open pit production capacity as Phase 5 winds down, peaking at a rate of approximately 25,000 tpd moved (ore and waste). Phase 6 has the potential to add five years of additional open pit life, extending the operation to 2026. Further study on Phase 6 is required, and is underway.

UNDERGROUND

Underground stoping at the Williams Mine comprises two methods, the longhole and Alimak methods. Longhole “Longitudinal” stopes are typically 20 m to 25 m high and 15 m to 20 m along strike, with an average ore width of three to eight metres. Longhole “Transverse” stopes are from 15 m to 20 m along the strike, typically 25 m to 30 m high, and three to thirty metres wide. Production drilling is performed with ITH (In-the Hole) drills using four and one half inch diameter holes or Stopemaster drills using three inch diameter holes.

Underground production is forecast to increase from 3,300 tpd to 3,800 tpd between 2017 and 2021 in the LOM.

MINERAL PROCESSING

Ore characterization is a continuous program at Hemlo. Grind-recovery and grade-recovery correlations are updated periodically. Barrick has also developed a relationship between throughput, head grade, and grind size. By combining these relationships, models were developed relating gold recovery to plant throughput. These equations are reported to be valid for milling rates of 270 tonnes per hour (tph) of typical open pit grade ore. Gold recovery from 100-series ore is approximately 92.3% at a milling rate of 270 tph, and the 300-series ore recovery is approximately 85.8% at a milling rate of 270 tph. There is an even split between 100- and 300-series ores in the Mineral Reserves.

The Hemlo processing plant consists of grinding, cyanide leaching, CIP, carbon stripping and reactivation, electrowinning, and refining. The mill has two parallel grinding circuits each consisting of a 6,000 t coarse ore bin, a semi-autogenous grinding (SAG) mill, and a ball mill that is operated in closed circuit with hydrocyclones.

Approximately 30% of the mill tailings are sent to the Williams underground workings as mine paste backfill. Surplus mill tailings are sent to the TMF. Water is reclaimed from the TMF and pumped to the mill; excess water is treated through a seasonally operated effluent treatment plant prior to discharge to the environment.

For the open pit expansion project, a crushing plant will be added to reduce the size of the feed to the SAG mills and, therefore, increase the throughput of the mill, as determined by the secondary crusher trial. The proposed crushing plant is designed to process 1,000 tph at 80% availability.

Completion of the mill upgrades will provide sufficient milling capacity to meet the permitted limit for the processing facilities, which is 11,500 tpd.

ENVIRONMENTAL, PERMITTING, AND SOCIAL CONSIDERATIONS

The Hemlo operations consist of an operating open pit and underground mine, and a CIP process plant facility. Barrick has environmental management groups and management systems to ensure that the necessary permits and licences are obtained and maintained. These groups also carry out the required environmental monitoring and reporting.

TAILINGS STORAGE FACILITY

Tailings from the CIP processing plant are deposited in the TMF located immediately to the south and west of the process plant facility and the Williams and David Bell mines. The TMF will operate as a series of dams, which will be raised periodically to accommodate for the additional tailings from the open pit expansion.

PROJECT PERMITTING

Hemlo is an active mine site and is in compliance with all of its permits and/or approvals. The process facilities are permitted for a production rate of 11,500 tpd. Permits are updated or amended, as necessary, to accommodate changes in the mine life. The Environmental Department at the Hemlo Mine has systems in place to ensure that permits do not come overdue and that they are complied with by the operation.

CAPITAL AND OPERATING COST ESTIMATES

CAPITAL COSTS

LOM capital costs for the operation are estimated to be C\$301 million, as summarized in Table 1-3. Large capital items for 2017 include tailings dam construction for C\$18.8 million, SAG mill shell replacement for C\$5.5 million, crown pillar sequencing for C\$9.6 million, and a Pre-Feasibility Study for underground expansion for C\$11.0 million.

TABLE 1-3 LOM CAPITAL COST ESTIMATE
Barrick Gold Corporation - Hemlo Mine

Description	2017 (C\$ 000)	LOM Total (C\$ 000)
Sustaining Capital		
Open Pit	13,439	59,823
Underground	16,150	52,462
Processing	7,970	16,796
Tailings	18,813	84,236
UG Development & Open Pit Stripping	16,336	75,710
Exploration Drilling	9,588	12,038
Total	82,296	301,066

OPERATING COSTS

Table 1-4 summarizes the reported unit operating costs for 2016, and the projected LOM costs.

TABLE 1-4 OPERATING COST SUMMARY
Barrick Gold Corporation - Hemlo Mine

Item	Units	2016	LOM
Open Pit Mining	C\$/t moved in pit	5.46	4.86
Underground Mining	C\$/t UG ore	86.56	71.23
Mining Total	C\$/t milled	41.10	34.90
Processing	C\$/t milled	12.04	12.53
G & A	C\$/t milled	7.15	7.78
Total	C\$/t milled	60.30	55.21

Open pit costs will decrease on a unit basis as Phase 6 begins, due to slightly higher production rates. Underground unit costs are projected to decrease due to increasing underground production rates over the next five years. Processing and general and administrative (G&A) unit costs are lower in the short-term, then skewed higher by lower volumes once the underground reserves are completed.

2 INTRODUCTION

Roscoe Postle Associates Inc. (RPA) was retained by Barrick Gold Corporation (Barrick) to prepare an independent Technical Report on the Hemlo open pit and underground mines (the Hemlo Mine), in Bomby Township, Northwestern Ontario, Canada. The purpose of this report is to support public disclosure of Mineral Resource and Mineral Reserve estimates at the Mine. This Technical Report conforms to National Instrument 43-101 Standards of Disclosure for Mineral Projects (NI 43-101). The effective date of the Mineral Resource and Mineral Reserve estimates in this Technical Report is December 31, 2016, and information in this Technical Report is current as of that date unless otherwise specified. RPA visited the property on December 14 and 15, 2016.

Barrick is a Canadian, publicly-traded mining company with a portfolio of operating mines and projects across five continents. The Hemlo Mine property is located just north of Lake Superior on the Trans-Canada highway approximately 35 km east of the town of Marathon, Ontario, at approximately 85°54' W longitude and 48°41' N latitude.

In production since 1985, operations at the Hemlo Mine have produced over 21 million ounces of gold to December 2016. Past underground production came from Williams, Golden Giant, and David Bell underground mines stretching over a length of two kilometres and a vertical distance of approximately 1,500 m below surface. Underground Mineral Reserves at the Williams Mine are projected to sustain the underground mine operations until 2021 at an average production rate of approximately 3,600 tpd.

The open pit is a traditional truck and shovel/loader operation that has supplied approximately 6,000 tpd of ore to the plant (up to 20,000 tpd moved, including waste and low-grade to stockpile). Open pit Mineral Reserves at the Williams Mine have the potential to support operations until 2027, at rates ranging from 5,000 tpd to 6,000 tpd.

Mineral processing of up to 10,000 tpd at Hemlo consists of grinding, cyanide leaching, carbon-in-pulp (CIP), carbon stripping and reactivation, electrowinning, and refining.

SOURCES OF INFORMATION

A site visit was carried out by Jason Cox, P. Eng., RPA Principal Mining Engineer, Wayne Valliant, P. Geo., RPA Principal Geologist, and Kathleen Altman, Ph.D., P.E., RPA Principal Metallurgist, on December 14 and 15, 2016.

Discussions were held with personnel from Barrick:

- Andrew Baumen, General Manager
- Rodney Barber, P. Geo., Hemlo Geology Superintendent
- Ben Kowalczyk, Resource Specialist
- Pam Turco, Core Shack Coordinator
- Hal Wells, Senior Production Geologist
- Doug Bruce, Mine Planning Coordinator
- Allison Henstridge, P. Eng., Hemlo Engineering Superintendent
- Angela Chitaroni, Open Pit Long-Range Planner
- Darren Tinney, Mill Superintendent
- Friedrich Po, Senior Metallurgist
- Yonika Wiputri, Metallurgist
- Mohammed Javaid, Tailings Coordinator
- Jeremy Dart, Environmental Superintendent
- Debi Bouchie Employee Relations Superintendent
- Victor Hugo, Site CFO
- Patrick Langlais, P. Eng. Underground Chief Engineer

Mr. Cox is responsible for the overall preparation of this report, and reviewed the estimates of Mineral Reserves, mine planning, and production and is responsible for Sections 15, 16, 18, 19, 21, and 22. Mr. Valliant reviewed the geology, sampling, assaying, and resource estimates and is responsible for Sections 7 to 12, 14, and 23. Dr. Altman reviewed the metallurgical and environmental aspects of the operation and is responsible for Sections 13, 17, 20, and 21. The authors share responsibility for Sections 1 to 6, and 24 to 27 of this Technical Report.

The documentation reviewed, and other sources of information, are listed at the end of this report in Section 27 References.

LIST OF ABBREVIATIONS

Units of measurement used in this report conform to the metric system. All currency in this report is in Canadian dollars (C\$) unless otherwise noted.

a	annum	kWh	kilowatt-hour
A	ampere	L	litre
bbl	barrels	lb	pound
btu	British thermal units	L/s	litres per second
°C	degree Celsius	m	metre
C\$	Canadian dollars	M	mega (million); molar
cal	calorie	m ²	square metre
cfm	cubic feet per minute	m ³	cubic metre
cm	centimetre	μ	micron
cm ²	square centimetre	MASL	metres above sea level
d	day	μg	microgram
dia	diameter	m ³ /h	cubic metres per hour
dmt	dry metric tonne	mi	mile
dwt	dead-weight ton	min	minute
°F	degree Fahrenheit	μm	micrometre
ft	foot	mm	millimetre
ft ²	square foot	mph	miles per hour
ft ³	cubic foot	MVA	megavolt-amperes
ft/s	foot per second	MW	megawatt
g	gram	MWh	megawatt-hour
G	giga (billion)	oz	Troy ounce (31.1035g)
Gal	Imperial gallon	oz/st, opt	ounce per short ton
g/L	gram per litre	ppb	part per billion
Gpm	Imperial gallons per minute	ppm	part per million
g/t	gram per tonne	psia	pound per square inch absolute
gr/ft ³	grain per cubic foot	psig	pound per square inch gauge
gr/m ³	grain per cubic metre	RL	relative elevation
ha	hectare	s	second
hp	horsepower	st	short ton
hr	hour	stpa	short ton per year
Hz	hertz	stpd	short ton per day
in.	inch	t	metric tonne
in ²	square inch	tpa	metric tonne per year
J	joule	tpd	metric tonne per day
k	kilo (thousand)	US\$	United States dollar
kcal	kilocalorie	USg	United States gallon
kg	kilogram	USgpm	US gallon per minute
km	kilometre	V	volt
km ²	square kilometre	W	watt
km/h	kilometre per hour	wmt	wet metric tonne
kPa	kilopascal	wt%	weight percent
kVA	kilovolt-amperes	yd ³	cubic yard
kW	kilowatt	yr	year

3 RELIANCE ON OTHER EXPERTS

This report has been prepared by RPA for Barrick. The information, conclusions, opinions, and estimates contained herein are based on:

- Information available to RPA at the time of preparation of this report,
- Assumptions, conditions, and qualifications as set forth in this report, and
- Data, reports, and other information supplied by Barrick and other third party sources.

For the purpose of this report, RPA has relied on ownership information provided by Barrick in Sections 1 and 4 of this Report. RPA has not researched property title or mineral rights for the Hemlo Mine and expresses no opinion as to the ownership status of the property.

RPA has relied on Barrick for guidance on applicable taxes, royalties, and other government levies or interests, applicable to revenue or income from the Hemlo Mine.

Except for the purposes legislated under provincial securities laws, any use of this report by any third party are at that party's sole risk.

4 PROPERTY DESCRIPTION AND LOCATION

LOCATION

The Hemlo Property is located in Bomby township, in Northwestern Ontario, Canada, just north of Lake Superior on the Trans-Canada highway approximately 35 km east of the town of Marathon, Ontario, at approximately 85°54' W longitude and 48°41' N latitude (Figure 4-1).

The Hemlo Property consists of the Williams Mine located at the western end of the property, the formerly-producing Golden Giant Mine in the centre, and the formerly-producing David Bell Mine located at the eastern end of the property. The centre of the property is located at approximately 580,681 mE and 5,394,349 mN (UTM Zone 16, NAD83).

LAND TENURE

The Hemlo operation consists of the Williams underground and open pit mines, and is comprised of freehold patents, leasehold patents, and unpatented mining claims that are listed and described in Tables 4-1 and 4-2. All claims are contiguous. The Williams Mine is managed by Williams Operating Corporation (WOC).

In 2009, Barrick acquired Teck Cominco Ltd.'s 50% interest in the Hemlo Mine. Barrick now owns 100% of the Hemlo Mine. Ore from Williams underground and open pit operations is processed at the Williams mill and tailings are piped to the Molson Lake tailings impoundment.

In September 2010, Barrick acquired Newmont Mining Corporation's (Newmont) interest in the Golden Giant Mine.

In March 2015, Barrick acquired certain lands to the west and north of the Williams Mine, as well as claims underlying the Molson Lake tailings management facility (TMF). Barrick acquired the following properties: CLM 271, CLM 272, CLM 284, CLM 277, CLM 278, and the southern portion of CLM 285. Barrick also acquired the mineral rights on portions of CLM 273, CLM 274, the Sceptre claim, and Horizon claims that it did not already own. Subsequent to this transaction, no royalty is owed on the Sceptre claims and a 2.5% Gross Revenue Royalty is payable on any future mineral production from the remainder of these claims.

TABLE 4-1 LAND TENURE SUMMARY
Barrick Gold Corporation - Hemlo Mine

	Claim or Mining Lease No.	Area (ha)	Expiry Date
Freehold Patents (Williams)	TB32051	12.78	-
	TB32052	13.69	-
	TB32053	11.82	-
	TB32054	26.38	-
	TB32055	25.69	-
	TB32154	7.78	-
	TB32155	19.10	-
	TB32156	14.82	-
	TB32157	12.54	-
	TB32158	6.48	-
	TB32159	8.00	-
Sub Total		159.08	
Leasehold Patents (Other)	TB646504	0.37	30-Apr-28
	TB609035	0.74	30-Apr-28
	CLM273	438	30-Jun-26
	CLM274	396	30-Jun-26
	CLM302	334	31-May-28
Sub Total		1,169.11	
Unpatented Claims (Staked by Barrick)	1227332	144	5-Oct-19
	1227333	80	5-Oct-20
	1242630	16	18-Sep-20
	4247825	128	29-May-20
	4247826	16	29-May-20
	4266201	256	12-Dec-19
	4266202	144	12-Dec-19
	4266203	192	12-Dec-19
Sub Total		976	
Unpatented Claims (Metalcorp Purchase)	TB4214151	48	8-May-20
	TB4214170	64	29-Jun-20
	TB4222578	80	7-Aug-20
Sub Total		192	
Unpatented Claims (NAE Property)	TB4258148	192	7-Mar-18
	TB4258149	32	7-Mar-18
	TB4258150	48	7-Mar-18
	TB4261118	256	7-Mar-18
	TB4261119	192	7-Mar-18
	TB4261120	240	7-Mar-18

	Claim or Mining Lease No.	Area (ha)	Expiry Date
	TB4261122	80	7-Mar-18
	TB4261123	224	7-Mar-18
	TB4263499	64	11-Jul-17
Sub Total		1,328	
Leasehold Patents (Interlake)			
	TB701683	26	28-Feb-35
	TB701681	23	28-Feb-35
	TB687195	37	28-Feb-35
	TB701682	28	28-Feb-35
	TB701684	27	28-Feb-35
Sub Total		141	
Leasehold Patents (Newmont 2015 Claim Purchase)			
	CLM 277	217	31-Aug-25
	CLM 278	145	30-Jun-26
	CLM 284	191	30-Jun-26
	CLM 271	376	30-Jun-26
	CLM 272	362	30-Jun-26
	Portions of CLM 285, consisting of:	86 ^{Note 1}	31-Oct-34
	TB701701		
	TB701704		
	TB701705		
	TB687196		
	TB701691		
	TB701692 (part)		
	TB701689 (part)		
Sub Total		1,377	

Note:

1. The area for the portions of CLM 285 is an estimate, pending final survey.

The David Bell Mine Expansion active mining claims are listed in Table 4-2.

TABLE 4-2 DAVID BELL LAND TENURE
Barrick Gold Corporation – Hemlo Mine

Description	Claims	Area (ha)
David Bell Mine Active Mining Claims		
Freehold Patents 103959 and 103963	TB549608	275
	TB549609	
	TB549610	
	TB549611	
	TB549612	
	TB554005	
	TB554006	
	TB555061	
	TB555062	
	TB555063	
	TB555064	
	TB555065	
	TB555066	
	TB555067	
	TB577521	
	TB577526	
	TB577527	
David Bell CLM275 Golden Giant Property Active Mining Claims		
Freehold Patent (within CLM275)	TB654945	168.29
	TB656687	
	TB656689	
	TB656690	
	TB673890	
	TB673892	
	TB673893	
	TB673898	
	TB673899	
Freehold Patent F2285	TB673888	13.03
Freehold Patent F2286	TB549612	3.91
Freehold Patent F 2287	TB673887	13.93
Leasehold Patent 107765	TB673886	24.08
	TB673889	

Notes:

1. Freehold Patents do not expire.
2. The Leasehold Patent for David Bell has an expiry date of June 30, 2026.

ROYALTIES

The royalties for the Hemlo Mine are listed in Table 4-3.

TABLE 4-3 SUMMARY OF HEMLO PROPERTY ROYALTIES
Barrick Gold Corporation – Hemlo Mine

Site	Royalty	Type	%	Currency	Comments	Net Royalty %
David Bell	AuRico Metals Canadian Royalty Partnership Gold/10213 Yukon	NSR	3.00	C\$	Covers all David Bell Production	
Golden Giant	Newmont Canada Corporation	GRR	3.00	US\$	Golden Giant SP/QC and Golden Giant Zone Production	
Williams	Lola Williams	NSR	1.50	C\$	Covers all Williams UG & C Zone OP production Sceptre and Horizon pit production not included	1.500
	Barrick buy back of L. Williams		1.00%	C\$	Barrick has bought back 2/3's of the L Williams Royalty making it in effect a 0.5% NSR	1.000
Williams	River Oaks Gold Corporation	Production	0.75	C\$	Covers all Williams UG & C Zone OP production Sceptre and Horizon pit production not included (10213 Yukon, AuRico Metals Canadian Royalty Partnership, International Royalty Corp.)	0.750
Williams	Initial Unit Holders	Production	1.00	C\$	Covers all Williams UG & C Zone OP production Sceptre and Horizon pit production not included	1.000
	Mining tax deduction on IUH		0.085	C\$	Barrick has bought back 8% of the IUH Royalty and the agreement allows for a 8.5% deduction for mining taxes making it in effect a	0.085
	Barrick buy back of IUH		0.073		0.782% NPI	0.073
Total (Williams Only)						2.092
Horizon Block	Newmont Canada Corporation	NRR	3.00	C\$	Covers Horizon Block OP production only, above 10150 elev	3.00
WOC Block (Claims 273-274)	Newmont Canada Corporation	NRR	3.00	C\$	Covers WOC Block OP production only	3.00
Interlake Property	Franco-Nevada Corporation	NPI	50	C\$	50% NPI on the UG Interlake Property becomes effective once WOC has recovered all costs attributable to mining on the property	
Interlake Property	Franco-Nevada Corporation	NSR	3.00	C\$	WOC pays a 3% underlying royalty on all UG Interlake production, these royalty costs are recoverable under the terms of the above 50% NPI	3.00
Claims CLM 273 and CLM 274	Newmont Canada Corporation	GRR	2.50	C\$	March 2015 agreement with Newmont. CLM 273 and CLM 274 (except on Horizon claims only, royalty applies above 10150 elev.)	2.50
Claims CLM 271, 272, 277, 278, 284, and 288 (part)	Newmont Canada Corporation	GRR	2.50	C\$	March 2015 agreement with Newmont	2.50

PERMITS

The active permit list managed by the Environmental department at Hemlo is shown in Table 4-4.

Barrick continues to focus on the site water balance to minimize freshwater usage and water stored in the tailings management facility, while maximizing tailings deposition. In 2015, the Hemlo camp set an objective and target of 80% recycled water, and achieved 92% recycled water. The reclaim system, Actiflo water plant and storm- water system reduces our reliance on water from Cedar Creek, which during drought conditions, cannot supply all of Williams process water needs.

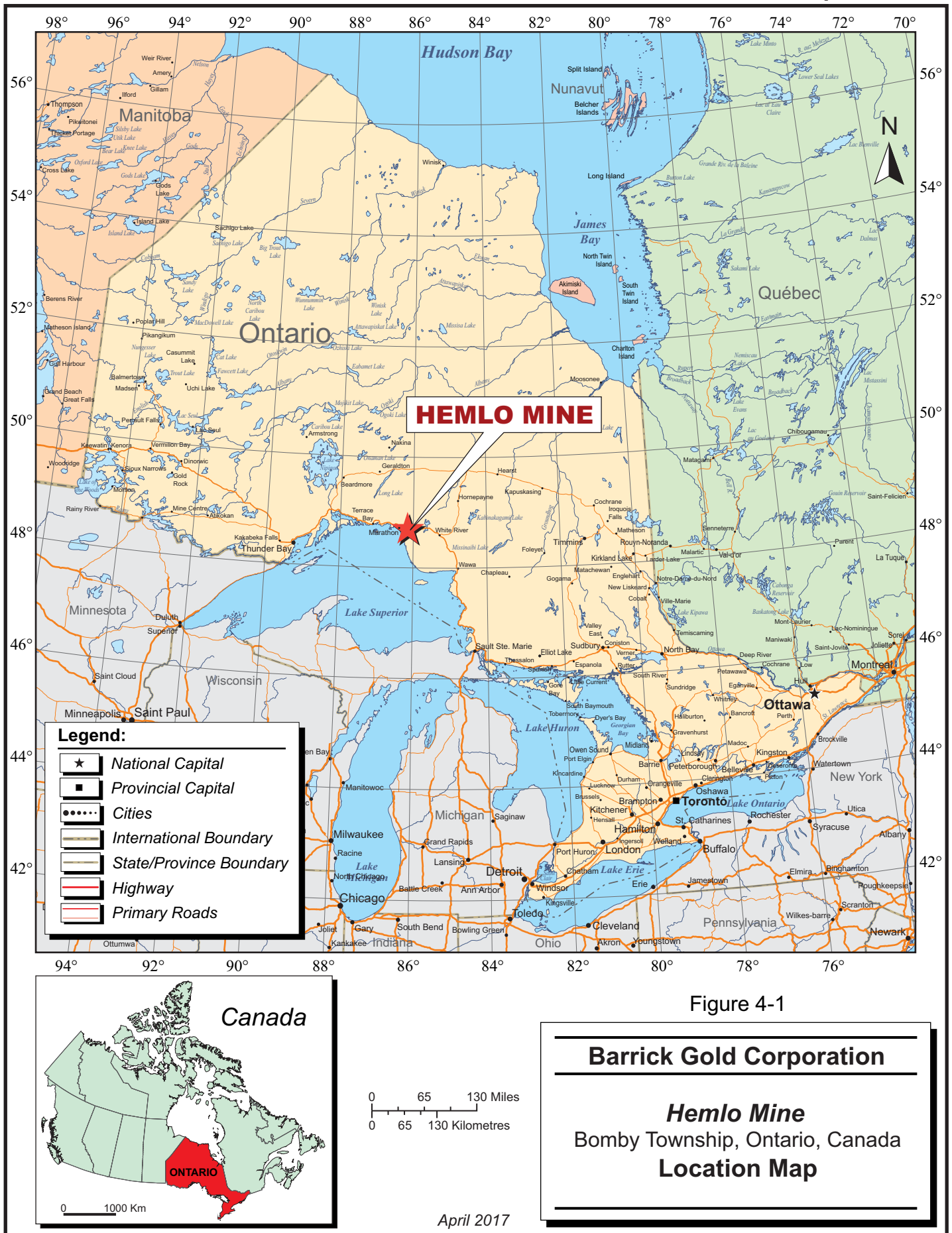
The David Bell and Golden Giant mine sites were the focus of active progressive rehabilitation. A cooperative research initiative is underway to evaluate the use of various engineered soils as covers for rock piles and tailings.

Barrick will also continue to improve its water management strategies at the Hemlo Mine. Ongoing evaluation of the overall water balance, tailings deposition management and ways to reduce water needs will be priority in order to manage the TMF. In order to store tailings, in 2017 and 2018 the TMF will be raised to its maximum elevation of 357.5 MASL as per the design and permit.

RPA is not aware of any environmental liabilities on the property. The Hemlo Mine has all required permits to conduct the proposed work on the property. RPA is not aware of any other significant factors and risks that may affect access, title, or the right or ability to perform the proposed work program on the property.

TABLE 4-4 ACTIVE PERMITS
Barrick Gold Corporation - Hemlo Mine

Permit Title	Permit No.	Issuing Agency
Aggregate Permit - Black River Pit	107926	Ministry of Natural Resources
Aggregate Permit - Cedar Creek Pit	126576	Ministry of Natural Resources
Aggregate Permit - Herrick Lake Sand Pit	80483	Ministry of Natural Resources
Aggregate Permit - Herrick Pit	20203	Ministry of Natural Resources
Aggregate Permit - Philips Creek	20202	Ministry of Natural Resources
Aggregate Permit - Struthers Quarry	20221	Ministry of Natural Resources
Aggregate Permit - Struthers Sand	20819	Ministry of Natural Resources
Aggregate Permit - Wabikoba Lake Area (Pine Grove)	108126	Ministry of Natural Resources
Aggregate Permit - Wabikoba Till Pit	20616	Ministry of Natural Resources
AIR - WOC and DBOC Comprehensive Certificate of Approval	0521-8HZJUT	Ministry of Environment
ECA - Consolidated Tailings	8452-9U3HRU	Ministry of Environment
Encroachment Permit - Emergency Spill Control Pond	EC-2015-61T-1	Ministry of Transportation
Encroachment Permit - Pole Placement on MTO Right-of-Way	EC-2009-61T-2	Ministry of Transportation
Pipe Crossing Agreement With Canadian Pacific	OD 50298	Canadian Pacific Railway Company
Pipe Crossing Agreement With Canadian Pacific	OD 50545	Canadian Pacific Railway Company
Power Crossing Agreement With Canadian Pacific	OD 50655	Canadian Pacific Railway Company
Private Crossing Agreement with Canadian Pacific	OD 55466	Canadian Pacific Railway Company
Private Crossing Agreement with Canadian Pacific	OD 55467	Canadian Pacific Railway Company
Private Crossing Agreement with Canadian Pacific	OD 52703	Canadian Pacific Railway Company
PTTW - Little Cedar Lake Permit To Take Water	6028-A2LKAU	Ministry of Environment
PTTW - Theresa Lake Permit To Take Water	8024-A2LLFZ	Ministry of Environment
PTTW - WOC Cedar Creek Permit To Take Water	5533-A2LMPA	Ministry of Environment
PTTW - WOC C Zone and Sceptre Pit De-Watering Permit To Take Water	6881-A4DSY4	Ministry of Environment
PTTW - WOC Pit Storm Water Ponds Permit To Take Water	0716-8P4J5V	Ministry of Environment
Road Crossing Agreement With Canadian Pacific	OD 50170	Canadian Pacific Railway Company
Tailings Bridge - Legal Agreement Between MTO & Williams Operating Corp.	Tailings Bridge	Ministry of Transportation
Teck Corona Landfill COA	A71902-02	Ministry of Environment
Theresa Lake Dam Navigable Waters Approval	8200-85-54	Transport Canada
Water Treatment Plant	4-029-84-006	Ministry of Environment
Williams Mine Potable Water Directive	SDWS No.762001054	Thunder Bay District Health Unit
Williams Operating Corp. Landfill COA	A5825391	Ministry of Environment
Entrance Permit - DBOC Yellow Brick Road Entrance	EN-2016-61T-31	Ministry of Transportation
Entrance Permit - DBOC Mine Entrance	EN-2016-61T-32	Ministry of Transportation
Entrance Permit - DBOC Tailings Entrance	EN-2016-61T-33	Ministry of Transportation
Entrance Permit - WOC North Tailings Gate Entrance	EN-2016-61T-34	Ministry of Transportation
Entrance Permit - WOC Tailings Yellow Gate Entrance	EN-2016-61T-35	Ministry of Transportation
Entrance Permit - WOC A Zone Pit Hwy Entrance	EN-2016-61T-36	Ministry of Transportation
Entrance Permit - WOC Mine Entrance	EN-2016-61T-37	Ministry of Transportation
Encroachment Permit - WOC Moose Lake Water Line	EC-2016-61T-49	Ministry of Transportation
Encroachment Permit - WOC Pipeline Trestle Overpass	EC-2016-61T-50	Ministry of Transportation
Encroachment Permit - WOC Powerline HWY 17	EC-2016-61T-51	Ministry of Transportation
Encroachment Permit - DBOC Fibre Optic Line and 4160V	EC-2008-61T-24	Ministry of Transportation
Transport Canada - Standards Obstruction	Standard 621.19	Transport Canada



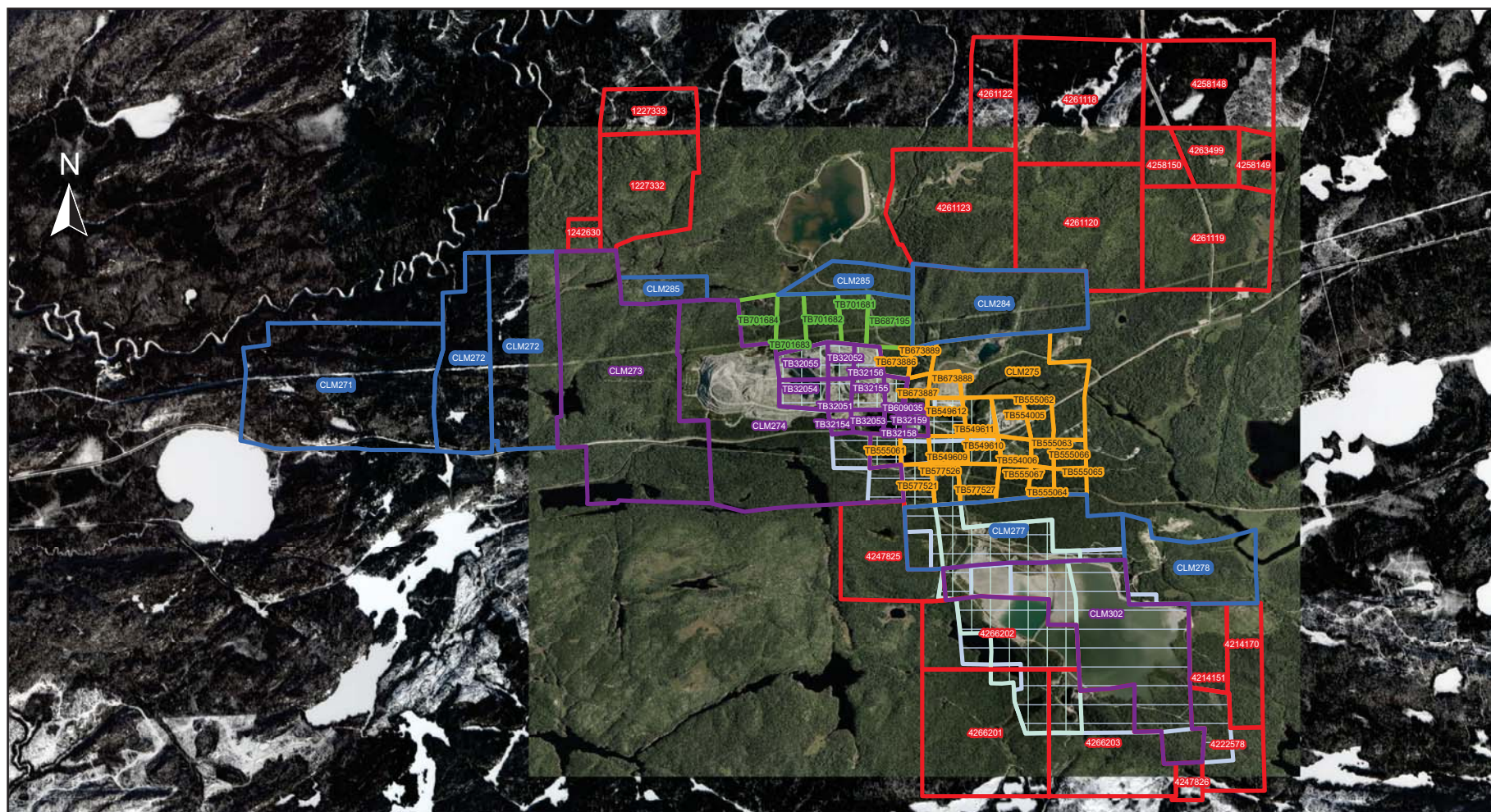


Figure 4-2

Legend:

- | | |
|---|---|
|  | <i>Williams Mine-BGI</i> |
|  | <i>BGI Acquired Leases</i> |
|  | <i>BGI Unpatented Claims</i> |
|  | <i>David Bell & Golden Giant</i> |
|  | <i>Williams Mine - Interlake Property</i> |
|  | <i>Teck Easement</i> |
|  | <i>Lac Easement</i> |

A horizontal number line with tick marks at 0, 1, 2, 3, and 4. The word "Kilometres" is written below the line. The segment between 1 and 3 is shaded black.

Provisional Map - This map was not prepared from a field survey and should not be relied on as a representation of legal land descriptions.

Information on this map may not be complete or up to date and its accuracy is not to be relied upon.

This map is intended only as a general representation of land status.

Barrick Gold Corporation

Hemlo Mine
Bomby Township, Ontario, Canada
Property Map

5 ACCESSIBILITY, CLIMATE, LOCAL RESOURCES, INFRASTRUCTURE AND PHYSIOGRAPHY

ACCESSIBILITY

The Hemlo Mine has excellent transportation and communication links as it is located adjacent to the Trans-Canada Highway #17 approximately 35 km east of the town of Marathon. The main Canadian Pacific Railway line is located just south of the mine sites. A major electrical transmission corridor runs just to the north of the properties.

CLIMATE

The average recorded temperature in the Marathon area over the last 36 years varies from a low of -13 °C in February to a high of 14 °C in August. Meanwhile the average high and low temperatures during the same period were -8 °C to -19 °C in February and 18 °C to 10 °C in August. The average precipitation over the 39 years on record ranges from a low of 49 mm in February to a high of 92 mm in September.

LOCAL RESOURCES

There is a skilled workforce in the neighbouring towns of Marathon, Manitouwadge, White River, and the Biigtigong Nishnaabeg and Pic Mobert First Nations communities. The Hemlo Mine employs approximately 700 people, including contractors and temporary employees. The average age of the workforce at Hemlo is 46 years old. The total turnover of employees during 2016 averaged 8.5%, largely due to a voluntary retirement program. The Williams Mine is non-unionized.

INFRASTRUCTURE

The mines in the Hemlo area have been active year round since the start of production in 1985. At the present, only the Williams Pit and Underground mines are in production. The David Bell and Golden Giant Mines have been closed. All surface infrastructure at these sites has been demolished, with the exception of the administration building at David Bell. The Williams Mine site consists of the following major infrastructure:

- A wash bay consisting of two bays for the 777 open pit haul trucks.
- A three bay garage near the open pit to service the pit equipment.
- A 10,000 tpd carbon-in-pulp (CIP) gold mill with electrowinning and refining capacity.
- A 431 hectare tailings management facility with plans to change the deposition method so it has capacity to operate for the LOM.
- A hoistroom and headframe (208 ft high) with a 1,300 m deep shaft.
- A 4.6 m diameter production hoist with 22 t capacity skips. A 3.7 m service hoist with cage and counterweight.
- A 100 m³/hr paste backfill plant.
- A service building housing the maintenance facilities, offices, and mine dry.
- A four metre by five metre decline and a series of ramp connected levels.
- A 115 kV power line from the Ontario power grid feeding two 33 MVA main transformers.

PHYSIOGRAPHY

The natural site topography is variable, with elevations ranging from 310 MASL to 360 MASL. The Canadian Shield terrain is comprised of weathered rocky ranges, boreal forested uplands (partially logged), a few wetlands, creeks, and lakes. Glacial overburden is typically less than five metres thick.

According to the Ontario's Ecological Land Classification, the area falls into Site Region 3W (a total area of 8.9 M ha) which breaks down into the following percentage of total area and includes ownership by the crown, parks, and others:

- Water 17.1%;
- Wetland 0.80%;
- Field/Agriculture 0.00%;
- Other 0.70%;
- Tree Bog 3.40%; and
- Productive Forest 78.1%.

Cedar Creek flows to the east of the Williams property and is located between the Williams and Golden Giant properties.

6 HISTORY

A summary of the history of the Hemlo Property is provided below and taken from an internal report entitled “Barrick Hemlo Reserves and Resources Summary End 2010” by Rodney Barber, P.Geo., dated December 31, 2010. Prior estimates of quantities and grades are provided for information only, and are not to be relied on as historical estimates.

- Peter Moses prospected the area in 1944 from which samples returned assays up to 14.23 g/t Au. Harry Ollmann formed a partnership with Dr. J.K. Williams of Maryland and eleven claims were staked in 1945 known as the Ollmann-Williams property.
- In the spring of 1946, consulting geologist Trevor Page staked additional adjoining claims south and east of the property which are referred to as the Lake Superior property. Page continued mapping, chip and channel sampling, and X-ray drilling on both properties, with fifteen holes completed on the Ollmann-Williams property and thirteen holes completed on the Lake Superior property in 1947.
- The eleven claims held by Ollmann and Williams were patented and when Mr. Ollmann died in December 1947 the claims were placed in trust under Williams’ name. Exploration continued and in 1949 drilling had indicated a zone with a strike length of 275 m and a width of two metres and an average grade of 8.78 g/t Au. This strike length was later reduced to 91 m and estimated to contain 28,527 t at 7.54 g/t Au.
- Lake Superior Mining Corporation Limited (Lake Superior) was optioned by Teck-Hughes Gold Mining Limited in 1951, the latter completing six diamond drill holes and increased the size of the No. 1 zone (also called the Main Zone) to 69,524 t grading 9.26 g/t Au, but the zone was uneconomic and no work was done until 1957, when Teck Exploration Company Limited completed some shallow drilling of what is now referred to as the Lower or Footwall Zone.
- Lake Superior optioned the ground to Cusco Mines Ltd., in 1958 with more diamond drilling completed on the Main Zone. Dimensions of the Main Zone revised to 168 m long by three metres wide and 91 m deep and estimated to contain 63,397 t at 7.45 g/t Au.
- Lake Superior’s claims lapsed and the area received only intermittent staking during the 1960s. In 1973, the former Lake Superior property was staked by Ardel Explorations Ltd. (Ardel), and three diamond drill holes completed. The deposit “reserve” estimate was further revised to 136,050 t at 7.23 g/t Au, but viewed as uneconomic and the ground dropped by Ardel and subsequently picked up by Cypress Resources Ltd., who also dropped the claims. The former Lake Superior property remained open from 1978 until late 1979.
- R.G. Newman staked claims that adjoined the western boundary of the Williams patented claim group in 1976 and anomalous gold values in soil samples were identified and appeared to be associated with quartz-feldspar porphyry rocks.

- In December 1979, prospectors Donald McKinnon and John Larche staked a large block of claims surrounding the 11 patented claims comprising the Williams block. Corona Resources Ltd. (later named International Corona Resources Ltd.) optioned a portion of these claims and after raising \$1.7 million, began line cutting and geophysical surveys in September 1980. A diamond drill program under the direction of David Bell was started in January 1981. Corona's drill program was designed to more completely explore the original mineralized zone (now known as the West Zone) that was discovered and drilled by T. Page in the late 1940s and early 1950s. The program amounted to 65 short holes and expanded the reserves to 340,000 t at 6.0 g/t Au. Corona had started negotiations with Lola Williams (the widow of J.K. Williams) for an agreement on the Williams property. During this time an additional ten holes were drilled east of a major diabase which intruded the ore package to evaluate the geology along strike, which David Bell had interpreted as an interface between mafic and felsic volcanic rocks and a favorable site for gold deposition. In November 1981, the tenth hole of this step-out drilling program (or the now famous hole 76) intersected the east arm of the Main ore zone and returned 3.3 m of mineralization grading 11 g/t Au.
- In May 1981, while Corona was negotiating with Lola Williams for the Williams property, representatives of Corona and Long Lac Minerals (Lac Minerals) exchanged information with the intent of formulating a joint-venture agreement. Lac Minerals were convinced that the geological environment at Hemlo was very similar to that at their Silverstack (Doyon) and Bousquet operations in Quebec. Lac Minerals also initiated negotiations with Lola Williams and in July 1981, she accepted Lac's offer.
- In October 1981, Corona alleged that Lac Minerals was in breach of fiduciary duty and launched a lawsuit over the ownership of the Williams claims. The ensuing three year court battle would become one of the best known legal disputes in Canadian mining history.
- Corona, who needed financial support for their legal dispute with Lac Minerals and for the development of the newly discovered zone (the East Zone), entered into a joint venture agreement with Teck Corporation in November 1981.
- R. Hughes and F. Lang optioned the remainder of Larche and McKinnon's claims in March 1981 and placed the claims located east of the Williams block and north of the Corona option into a company called Goliath Gold Mines Ltd. (Goliath). Claims located west of the Williams block were placed into the holdings of Golden Sceptre Resources Ltd. (Golden Sceptre). Drilling on the Golden Sceptre property began in August 1982, as a follow up to earlier trenching. David Bell supervised the Golden Sceptre and Goliath exploration projects in addition to his involvement with the Corona property.
- In late 1982, Lac Minerals announced the discovery on the Williams claim group, which they estimated to contain 1.6 M t grading 6.0 g/t Au.
- The continuity of the Main Zone was apparent by this time and drills working on the Golden Sceptre property were quickly moved and began drilling targets on the Goliath property. In September 1982, Goliath announced that a drill hole had intersected a down-dip extension of the Main zone and cored 29.9 m at 8.78 g/t Au. Goliath Gold Mines and Golden Sceptre Resources signed an agreement with Noranda Exploration Company Ltd. (Noranda), for development of their Goliath and Golden Sceptre properties. This gave Noranda a controlling interest in what would become the second largest mine in the Hemlo camp. By mid-1983, the potential size of the Hemlo deposit

was clear and Noranda, Teck-Corona, and Lac Minerals were all contemplating the development of their properties.

- The north dipping Main zone crosses onto the Goliath property about 300 m below surface and this made it difficult to position a shaft without penetrating the orebody at depth or requiring extensive and unfavorable hanging wall development. Thus, Teck-Corona agreed to option one quarter of a mining claim to Noranda in order to provide Noranda with a favorable site to collar their shaft. In return, Noranda would be required to mine 500 st of ore per day from the Quarter claim. Mining costs and recovered gold would be shared equally between the two parties. At the time of this agreement (known as the Quarter claim agreement), the Main zone was not perceived as a continuous zone but was interpreted as a number of westward plunging ore shoots.
- With regard to the legal dispute between Corona and Lac, the Supreme Court of Ontario decided in favour of Corona in March 1986. Operation of the Page-Williams continued under a court appointed committee pending an appeal by Lac Minerals. Lac Minerals lost the appeal to the Ontario Court of Appeal in October 1987, but was granted the right to appeal the provincial court ruling to the Supreme Court of Canada. However, the appeal by Lac Minerals was unsuccessful and the Supreme Court of Canada awarded the Page-Williams mine to International Corona Resources and Teck Corporation. The name of the mine was shortened to the Williams Mine and it is now managed by the Williams Operating Corporation.
- Homestake Mining Corporation purchased the assets of International Corona Resources in 1991 and Homestake was purchased by Barrick in 1999.
- In January of 1987, Golden Sceptre Resources Ltd., Goliath Gold Mines Ltd., Noranda Minerals Inc., and a consortium of minor shareholders amalgamated their holdings and formed Hemlo Gold Mines Inc. (Hemlo Gold). Noranda, as the major shareholder, continued to have a strong influence over the management of Hemlo gold. As a result of corporate restructuring in 1992, Noranda Minerals Inc., transferred all its gold assets to Hemlo Gold in exchange for a larger equity interest in the company and appointed a board of directors responsible for managing Hemlo Gold as an autonomous company within the Noranda Group.
- Ownership changed to Battle Mountain Canada Ltd., in 1996 and then to Newmont Canada Ltd., in 2001.
- In 1998, Williams acquired the surface and mineral rights of the Sceptre claims from Battle Mountain Canada to the 9450 elevation of the Williams Mine grid. In 1999 Williams also acquired the surface and mining rights on the Horizon claims from Battle Mountain Canada to the 10150 elevation of the Williams Mine grid. These acquisitions would permit pit expansion to the west, and allow evaluation of underground mining of the down dip extension of the C Zone pit. Both of these claim groups were part of lease 274. In addition to these two exchanges and as part of the same 1998 agreement, the David Bell Mine agreed to transfer the upper quarter claim and the M3 and M4 blocks of the C Zone to Battle Mountain Canada. Battle Mountain Canada also agreed to transfer Block 5 east and the Upper Block 5 to David Bell. Both of these latter exchanges were completed to facilitate mining for the parties involved.
- In 2002, Williams acquired the surface and mineral rights on lease 106623 from Newmont Canada Ltd. This acquisition allowed Williams to mine C Zone mineralization

above the 9450 elevation as well as the down dip extension of the C Zone mineralization on the Interlake property. In August, 2008 Newmont and WOC entered into an agreement to allow WOC to extend its underground mining operations on the WOC property through a 60 m restricted area (Boundary Pillar).

- The Williams and David Bell Mine are currently 100% owned by Barrick Gold Inc., after it purchased Teck Cominco Ltd.'s 50% interest in April 2009. The mining claims at the Williams Mine are subject to three NSR royalties totalling a net effective rate of 2.7%. The Interlake claims are subject to a 3% NSR royalty payable to Franco-Nevada Corporation. At the David Bell Mine, royalty holders have a 3% NSR.
- With the acquisition of the Newmont's Golden Giant Mine in September 2010, Barrick now owns all three Hemlo mines. The Golden Giant Mine had been dormant since 2006, but mining restarted in Q3, 2010. Golden Giant Mine claims have a royalty of 3% on the first 50,000 ounces and 3.5% on ounces thereafter payable to Newmont Canada for ore mined from this area. An additional 3% royalty will be payable to McKinnon and Larche on all Shaft Pillar and Quarter Claim ounces.
- In May 2014, production ceased from the David Bell Mine and David Bell Extension, i.e., the remnant mining at the Golden Giant Mine. Therefore the Williams Underground and Pit are the only remaining mines in production in the Hemlo camp.
- In March 2015, Barrick Gold Inc. acquired certain lands to the west and north of the Williams Mine, as well as claims underlying the Molson Lake TMF. Barrick acquired the following properties: CLM 271, CLM 272, CLM 284, CLM 277, CLM 278, and a southern portion of CLM 285. Barrick also acquired mineral rights on portions of CLM 273, CLM 274, the Sceptre claim, and Horizon claims that it did not already own. Subsequent to this transaction, no royalty is owed on the Sceptre claims and a 2.5% Gross Revenue Royalty is owed on any future mineral production from the remainder of these claims.

PAST PRODUCTION

Production from the Williams Mine began in mid-1985 from the A Zone open pit located at the east end of the property. This was soon augmented by underground ore from the same area to sustain an initial 3,000 tpd mining rate. The completion of the main shaft, the B Zone infrastructure and a mill expansion program in 1988 facilitated an increase in throughput to 6,300 tpd. Increased production from the C Zone pit brought the mill to a throughput of 10,000 tpd to the end of 2006. During 2007 the mill decreased to approximately 8,000 tpd production. Starting in 2009, the mill implemented a series of step-wise improvements that increased the daily throughput to approximately 9,500 tpd.

The Williams Mine has been active from 1985 until the present. The David Bell Mine was active from 1985 until 2014. The Golden Giant Mine was active from 1985 to 2004, and again

from 2010 to 2013. Production figures from the mines over that period of time are summarized in Tables 6-1, 6-2 and 6-3.

TABLE 6-1 WILLIAMS MINE PRODUCTION DATA
Barrick Gold Corporation - Hemlo Mine

Year	U/G			Pit			Total Yearly Production		
	Tonnes (000)	Feed Oz (000)	Feed Grade (g/t)	Tonnes (000)	Feed Oz (000)	Feed Grade (g/t)	Production Tonnes (000)	Feed Au Oz (000)	Feed Grade (g/t)
1985	156	23.1	4.60	-	-	0	156	23.1	4.60
1986	1,230	240.1	6.07	-	-	0	1,230	240.1	6.07
1987	1,224	297.2	7.55	-	-	0	1,224	297.2	7.55
1988	1,542	398.5	8.04	-	-	0	1,542	398.5	8.04
1989	1,923	520.1	8.42	-	-	0	1,923	520.1	8.42
1990	2,281	618.3	8.43	-	-	0	2,281	618.3	8.43
1991	2,153	543.4	7.85	32	2.5	2.43	2,185	545.9	7.77
1992	2,171	511.4	7.33	128	9.4	2.27	2,300	520.8	7.04
1993	2,177	504.7	7.21	143	11.6	2.52	2,320	516.2	6.92
1994	2,144	452.9	6.57	159	14.0	2.74	2,303	466.8	6.31
1995	2,162	407.8	5.87	204	18.3	2.80	2,355	426.2	5.60
1996	2,141	414.6	6.02	202	16.6	2.55	2,343	431.2	5.72
1997	2,091	400.4	5.96	318	23.5	2.3	2,409	423.9	5.47
1998	2,087	382.9	5.71	381	29.3	2.39	2,468	412.2	5.20
1999	2,058	421.4	6.37	374	23.6	1.96	2,432	445.0	5.69
2000	1,929	394.4	6.36	568	41.6	2.28	2,497	436.1	5.43
2001	1,975	412.9	6.50	1,063	65.7	1.92	3,038	478.7	4.90
2002	1,910	360.1	5.86	1,119	67.5	1.88	3,029	427.6	4.39
2003	1,796	358.2	6.20	1,401	82.2	1.83	3,197	440.4	4.28
2004	1,886	337.2	5.56	1,397	69.9	1.56	3,284	407.0	3.86
2005	1,811	315.0	5.41	1,329	63.7	1.49	3,140	378.7	3.75
2006	1,740	273.9	4.90	1,280	59.5	1.45	3,019	333.4	3.44
2007	1,157	176.3	4.74	1,591	98.6	1.75	2,748	265.9	3.01
2008	883	122.2	4.31	1,620	63.4	1.22	2,502	185.6	2.31
2009	826	142.6	5.37	1,481	72.2	1.52	2,307	214.8	2.90
2010	853	128.0	4.67	2,018	92.7	1.43	2,379	220.9	2.89
2011	756	137.1	5.64	2,216	79.0	1.11	2,972	216.0	2.26
2012	783	115.4	4.58	2,195	75.9	1.08	2,978	191.3	2.00
2013	850	112.7	4.12	2,118	77.4	1.14	2,968	190.1	1.99
2014	921	133.9	4.52	1,923	70.7	1.14	2,844	204.6	2.24
2015	1,004	150.7	4.67	2,116	79.9	1.17	3,120	230.7	2.30
2016	955	141.2	4.91	2,450	93.5	1.27	3,405	249.8	2.29
Total	49,573	9,957	6.25	29,825	1,399	1.46	79,399	11,357	4.45

TABLE 6-2 DAVID BELL MINE PRODUCTION DATA
Barrick Gold Corporation - Hemlo Mine

Year	Total Yearly Production		
	Production Tonnes	Feed Ounces	Feed Grade (g/t)
1985	153,381	34,439	6.98
1986	230,913	67,167	9.05
1987	361,730	155,189	13.34
1988	375,393	225,858	18.63
1989	425,150	323,242	23.65
1990	466,238	327,492	21.85
1991	470,197	295,473	19.55
1992	469,298	220,436	14.61
1993	491,654	225,523	14.27
1994	464,590	204,251	13.67
1995	441,509	168,734	11.89
1996	387,210	203,240	16.33
1997	429,379	187,865	13.61
1998	425,180	166,203	12.16
1999	443,967	169,078	11.85
2000	448,492	182,512	12.66
2001	455,004	163,064	11.14
2002	429,699	140,593	10.17
2003	379,243	123,757	10.15
2004	378,670	119,474	9.81
2005	363,515	112,353	9.60
2006	335,327	101,617	9.42
2007	287,900	92,694	10.01
2008	272,908	89,281	10.18
2009	219,146	72,775	10.33
2010	125,738	30,128	7.86
2011	65,300	18,009	8.58
2012	39,289	10,490	8.30
2013	15,921	3,492	6.81
Total	9,851,971	4,233,429	13.365

TABLE 6-3 GOLDEN GIANT MINE RESERVES AND PRODUCTION DATA
Barrick Gold Corporation - Hemlo Mine

Year	Total Yearly Production		
	Production Tonnes	Feed Ounces	Feed grade (g/t)
1984	-	-	-
1985	324,853	114,574	10.97
1986	733,413	272,111	11.54
1987	884,979	383,827	13.49
1988	927,100	352,914	11.84
1989	1,021,874	392,573	11.95
1990	1,185,364	460,792	12.09
1991	1,154,677	459,295	12.37
1992	1,071,930	478,282	13.88
1993	1,103,292	435,486	12.28
1994	1,107,348	465,639	13.08
1995	631,613	247,743	12.20
1996	1,087,970	377,205	10.78
1997	1,066,884	384,375	11.21
1998	1,029,229	374,816	11.33
1999	1,088,603	378,624	10.82
2000	1,070,456	339,858	9.88
2001	988,055	282,374	8.89
2002	997,556	279,996	8.73
2003	653,225	247,001	11.76
2004	470,689	160,169	10.58
2005			
2006			
2007			
2008			
2009			
2010	5,987	2,711	14.08
2011	10,288	2,325	7.03
2012	66,276	17,324	8.13
2013	125,658	25,418	6.29
Total	18,807,319	6,935,432	11.47

Note. The mine was shut down from 2005 to 2009.

7 GEOLOGICAL SETTING AND MINERALIZATION

REGIONAL GEOLOGY

The Hemlo Mine lies near the eastern half of the Schreiber-Hemlo greenstone belt, within the Wawa Subprovince of the Archean Superior Structural Province of Ontario. Figure 7-1 illustrates the Regional Geology.

The Wawa Subprovince has an east-west extent of at least 600 km, being truncated by the Kapuskasing structural zone in the east and extending an unknown distance beneath poorly exposed areas and Phanerozoic cover in northern Minnesota. The southern extent of the Subprovince is obscured by the waters of Lake Superior and unconformably overlying Proterozoic strata of the Animikie Basin.

The Subprovince comprises isolated, arcuate to linear greenstone belts with intervening masses of granitoid rock. The stratigraphy characterizing the greenstone belts is highly variable but typically consists of one or more mafic or ultramafic to felsic volcanic cycles. Subordinate clastic metasedimentary rocks are generally either a lateral facies equivalent of the intermediate to felsic portion of the volcanic sequence or overlie the metavolcanic rocks. One distinctive assemblage (Shebandowan), characterized by shoshonitic metavolcanic rocks, coarse conglomerate and subordinate wacke and mudstone, is interpreted to unconformably overlie the lower assemblages. Many assemblages are interpreted to be tectonically juxtaposed. Granitoid rocks internal to the greenstone belts include diverse syn-volcanic to post-tectonic stocks. The bounding granitoid complexes are poorly understood but are dominated by foliated to gneissic tonalitic to granodioritic rocks cut by relatively younger, more massive, granodioritic to granitic plutons.

The earliest structures interpreted or inferred in some of the larger greenstone belts are thrust faults and related small to large scale recumbent fold structures. These structures are overprinted by more upright, belt-parallel folds. Late, transcurrent deformation along ductile shears and faults is widespread. Mineral assemblages within the greenstone belts are indicative of greenschist to mid-amphibolite grade metamorphism with higher grades generally restricted to smaller greenstone belts.

The Schreiber–Hemlo greenstone belt of the Wawa subprovince, a volcano-sedimentary-plutonic terrane, is bounded to the east by the high-grade Kapuskasing structural zone and to the north by the metasedimentary-dominated Quetico subprovince. The geology of the eastern half of the Schreiber–Hemlo greenstone belt is designated as the Hemlo greenstone belt (HGB). Massive to pillowed, tholeiitic basalt flows and felsic to intermediate, calc-alkalic pyroclastic rocks with related sedimentary deposits dominate the western part of the belt, whereas turbiditic wacke–mudstone and minor conglomerate deposits dominate the eastern part. Granitoid plutons core and flank a large portion of the greenstone belt.

The minimum age of mafic volcanism is best constrained by crosscutting apophyses of the Dotted Lake pluton (ca. 2697 Ma). Contact relationships with the Pukaskwa batholith (ca. 2719 Ma) are unclear. Felsic calc-alkalic volcanism took place from ca. 2698 to ca. 2692 Ma, and intermediate volcanism appears to be ca. 2689 Ma. Sedimentation of turbiditic wacke–mudstone in the HGB occurred after ca. 2693 Ma for volcanoclastic deposits and possibly as late as ca. 2685 Ma for wacke.

The presence of felsic volcanic rocks dated at 2695 ± 2 Ma (near Heron Bay) and 2772 ± 2 Ma (at the deposit) has been cited as evidence that a fault in the deposit area, designated the Hemlo fault zone (HFZ), represents a major structural discontinuity between terranes having different ages.

Of the four main episodes of granitic magmatism in the HGB, only two (ca. 2720 to ca. 2718 Ma and ca. 2690 to 2684 Ma) are represented in the immediate area of the deposit. The latter episode represents the main magmatic event(s) and includes the Heron Bay, Cedar Lake, Cedar Creek, and Botham plutons and stocks and part of the Pukaskwa and Black-Pic batholiths. This episode may have occurred as late as ca. 2680 for the Cedar Lake pluton. The most strongly deformed granitoids occur in the Pukaskwa batholith, flanking the southern margin of the HGB. Although some parts of the Black-Pic batholith are also deformed, particularly along the contact with the Gowan Lake pluton, there is insufficient exposure along its southern contact with the HGB for characterization of strain. Most of the youngest plutons (e.g., Gowan Lake, Musher Lake, Bremner) are spatially associated with the northern margin of either the HGB or the sedimentary units surrounding the Cedar Lake pluton.

Supracrustal rocks in the HGB typically display moderately to strongly developed fabrics. A preliminary regional structural history of the HGB recognized two main stages: (i) a penetrative

regional foliation (D1R), which formed during a medium-grade metamorphic event; and (ii) a later regional event (D2R) involving the development of a second foliation and folding of the earlier fabric, greenstone belt boundaries, and some internal plutons.

Metamorphic grade increases from low in the western part of the HGB to medium in the central and eastern parts of the belt, with an increase also toward the batholiths and toward the “central axis” of the belt, particularly between the Pukaskwa batholith and the Cedar Lake pluton. Mesoproterozoic, rift-related magmatism, ca. 1.1 Ga, is reflected in the Port Coldwell alkalic complex and a variety of lamprophyre and alkalic intrusions, which occur throughout the HGB.

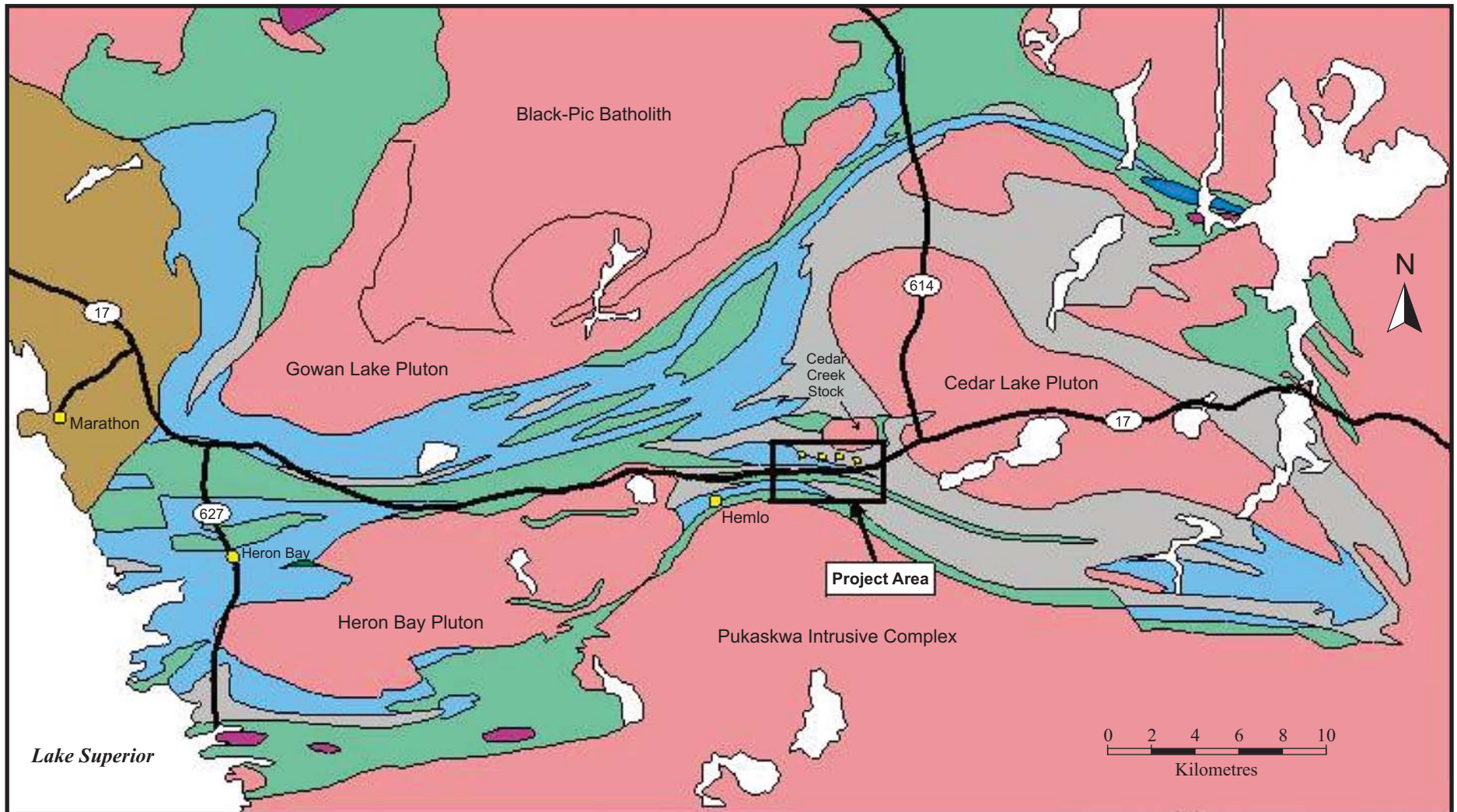
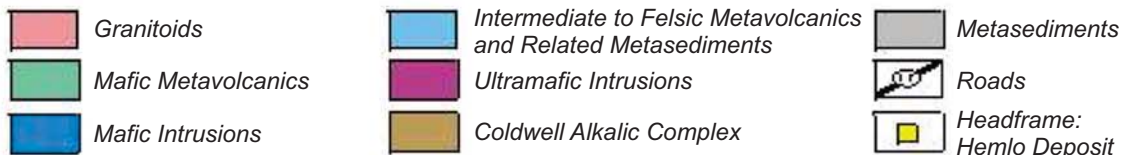


Figure 7-1



Barrick Gold Corporation

Hemlo Mine

Bomby Township, Ontario, Canada

**Regional Geology of the
Heron Bay**

LOCAL GEOLOGY

The description of local geology is taken from Muir (2003). Figure 7-2 illustrates the Local Geology.

The deposit is centred in the south-central part of the HGB, between the Pukaskwa batholith and the Cedar Creek stock, at and near a change in structural trend from westerly to west-northwesterly. In this area, volcano-sedimentary lithotectonic units generally dip to the north or north-northeast and are openly to isoclinally folded and transposed within high-strain zones. The most prominent structure is a dislocated, S-shaped, F2 fold pair delineated by a distinctive unit of felsic volcanic-subvolcanic rocks termed the Moose Lake volcanic complex (MLVC).

VOLCANIC ROCKS

Primary features in mafic volcanic flows are rarely preserved, but may include variably deformed plagioclase phenocrysts, varioles, and (or) pillows. A contiguous, west-striking mafic unit, which is parallel to and lies largely within the HFZ, thins eastward from 200 m to 40 m and consists of moderately to intensely strained, massive to pillowed flows and gneissic amphibolite. Locally, variolitic, pillowed flows on the south side of the HFZ are separated from medium- to coarse-grained amphibolite and finer grained gneissic amphibolite on the north side.

Calc-alkalic, felsic volcanic rocks make up several bodies, which together form the MLVC. The MLVC locally forms the footwall, hanging wall, or host unit to the deposit. The MLVC is a texturally diverse, composite unit consisting of massive and fragmental, felsic, quartz-plagioclase-phyric rocks and is inferred to have a U-Pb age of ca. 2694 Ma. The massive variety locally contains numerous lapilli-sized fragments and (or) displays discrete, non-S2-parallel primary layering inferred to be flow banding. Contacts between texturally distinct parts of the massive variety are sharp to abruptly gradational and cannot be traced for more than several metres. Contacts between massive and fragmental varieties are locally common and range from sharp and straight to bulbous or gradational. The MLVC is interpreted to comprise volcanic flows and sub volcanic intrusions coeval with pyroclastic deposits.

SEDIMENTARY ROCKS

Sedimentary rocks are subdivided into volcanoclastic and epiclastic units. The volcanoclastic units comprise tuffaceous conglomerate, tuffaceous sandstone, and tuffaceous mudstone and

have a white to light grey matrix. With the general exception of the tuffaceous mudstone, these subunits contain various amounts of discrete quartz and (or) plagioclase crystals (“phenocrysts”), and quartz-plagioclase-phyric clasts. The coarsest and (or) thickest packages of finer grained volcanoclastic units are generally proximal to the MLVC. The volcanoclastic rocks are interpreted as being derived from reworked detritus of the MLVC.

Epiclastic units comprise sandstone, mudstone, and conglomerate, which incorporate non-volcanic and volcanic detritus, and have a medium to dark grey matrix. The conglomerate is typically polymictic, and the sandstone mostly wacke. The mudstone is generally brownish grey, but rare, thin, dark grey to black mudstone occurs locally. Wacke and brownish-grey mudstone commonly occur in couplets and display characteristics consistent with turbidites, such as graded and massive beds, parallel-laminated beds, and rip-up clasts. Rare examples of scour channels and cross-bedding have also been reported in the wacke. The volcanoclastic and epiclastic units appear to be locally interbedded, suggesting that felsic volcanism, reworking of volcanic detritus, and sedimentation of distally derived material were broadly contemporaneous.

INTRUSIVE ROCKS

Early granitoid intrusions (ca. 2719 Ma) include tonalite to granodiorite within the northern margin of the Pukaskwa batholith. These rocks display a moderately developed to well-developed “proto-mylonitic” fabric, which locally shows subtle fold structures. Major plutonism occurred from ca. 2690 to ca. 2684 Ma. This stage involved emplacement of granodiorite to form the Heron Bay and Cedar Lake plutons and the Cedar Creek and Botham stocks, all of which occur internally to the HGB. Later plutons, from ca. 2679 to ca. 2677 Ma, chiefly involved emplacement of tonalite, monzodiorite, and diorite to form the Fourbay Lake, Musher Lake, Gowan Lake, and Bremner plutons, most of which occur externally or adjacent to the HGB.

Abundant, compositionally diverse dikes in the deposit area are a reflection of its complicated magmatic history. Most Neoarchean dikes have no chilled margins, indicating elevated ambient temperatures during intrusion. These dikes range from mafic-ultramafic to felsic in composition, and both predate and postdate alteration and gold mineralization at the deposit, although no consistent detailed sequence of emplacement, with respect to type of dike, could be established.

The mafic-ultramafic dikes lie mostly within the HFZ. Felsic (predominant) to intermediate (subordinate) dikes are most numerous in unit- or layer-parallel zones of volcano-sedimentary rocks between the Cedar Creek fault zone (CCFZ) and the HFZ and within about 250 m and 50 m of the Pukaskwa batholith and Cedar Lake pluton, respectively. Felsic dikes, many of which are plagioclase porphyritic, form the most abundant and widespread type in the map area. They tend to be the youngest dikes in that they typically crosscut mafic and intermediate dikes. Pegmatite and aplite dikes are restricted to the batholiths, the margins of plutons and stocks internal to the HGB, and locally the immediately adjacent volcano-sedimentary rocks. Minor deformed pegmatite locally occurs within 500 m of the HFZ on the footwall (i.e., south) side. Some felsic, plagioclase-porphyritic dikes appear to be related to marginal phases of the Cedar Lake – Heron Bay – Cedar Creek intrusions, given their similarities in texture, mineralogy, and geochemistry, and the presence of layer-parallel to sharply crosscutting dikes emanating from the east end of the Heron Bay pluton.

STRUCTURAL GEOLOGY

The four most detailed deformation histories for the deposit area reflect different perspectives. All studies infer a complex history of strain but differ in emphasis, namely detailed surface mapping, drill core section logging (Golden Giant Mine) with some surface mapping, underground mapping, and surface and underground mapping. These models range from progressive dextral shear, accounting for all structural elements, to polyphase models.

The style of strain varies from weak to intense across the map area, with the ambient strain being moderate to strong. There is an overall increase in strain and degree of recrystallization southward from outside the strain aureoles of the Cedar Creek stock and Cedar Lake pluton toward the Pukaskwa batholith. At least six generations of structural elements are recognized, two of which account for most of the strain. Although some generations of structures are likely products of a progressive strain event, they are designated here as D₁–D₆ based on overprinting relationships.

Collectively, primary features such as pillows, flow banding, and graded bedding (S₀) are recognizable locally in the area. They are preserved most commonly in the relatively low strain area north and northeast of the CCFZ and near Botham Lake, but also occur locally within lithons ranging from less than one metre up to several metres thick in the more highly strained area between the Hemlo and Cedar Creek fault zones.

In the vicinity of the Hemlo deposit, at least four generations of structures produced by at least two deformation events were identified: (1) an early phase resulting in small-scale F_1 folds and possible low-angle normal or thrust faults; (2) a major, regional, second phase producing small- to large-scale, tight to isoclinal, generally northwest- to north-northwest-plunging F_2 folds, along with a penetrative axial planar schistosity and differentiated layering (S_2), possibly associated with sinistral shearing and mylonitization; (3) dextral shear which locally resulted in s-c-c' mylonitic rocks and small- to medium-scale, generally northeast- to east-plunging, F_3 folds with axial planar schistosity and crenulation cleavage (S_3); and (4) small scale F_4 faults.

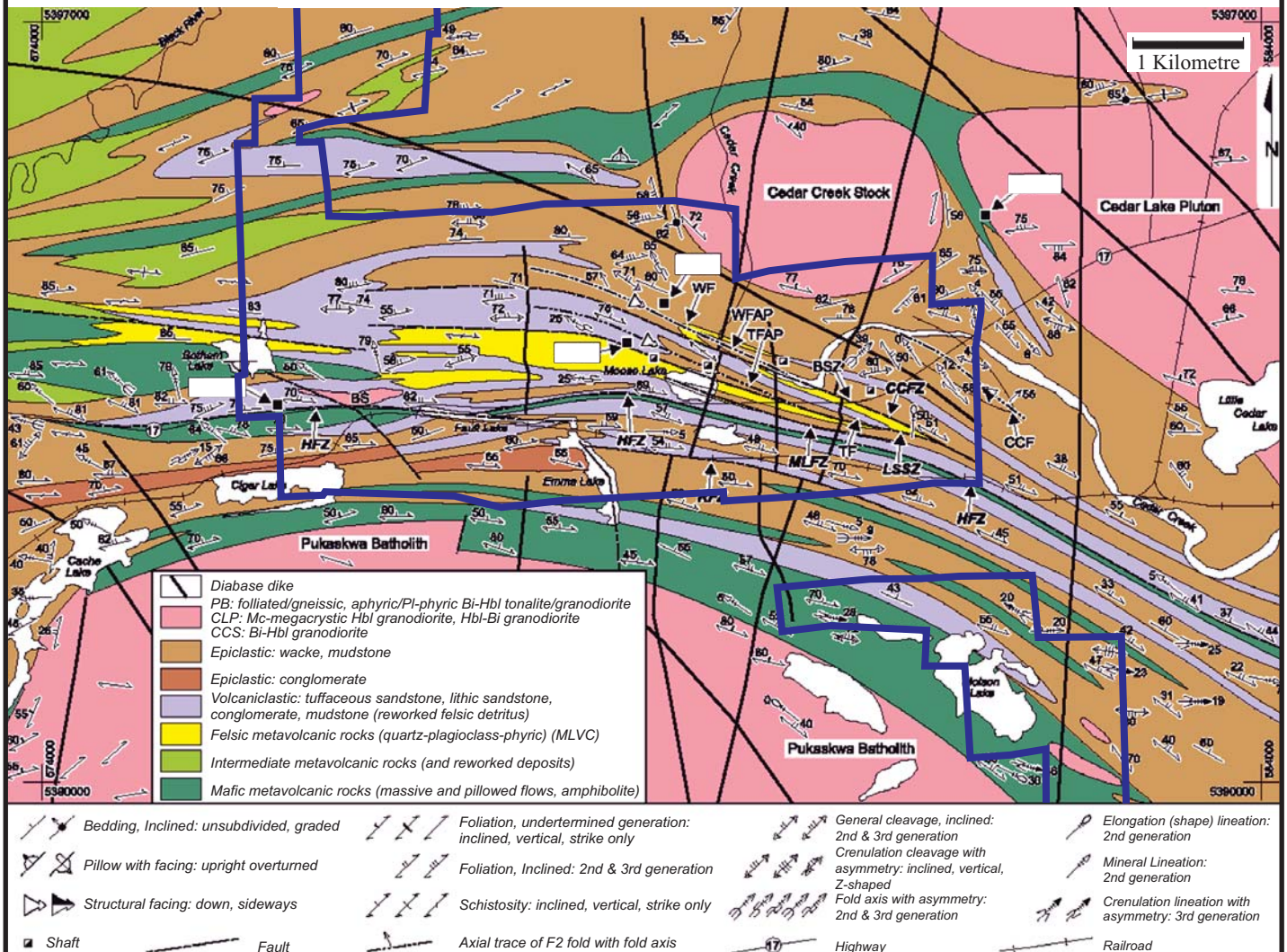


Figure 7-2

Property Outline

Barrick Gold Corporation

Hemlo Mine
Bomby Township, Ontario, Canada
Local Geology

PROPERTY GEOLOGY

The following description of the property geology is taken from Barber (2010).

WILLIAMS MINE

The ore zones at Williams consist of three principal rock types: (1) feldspathic rock, (2) muscovite schist and (3) biotite fragmental rock. These are locally enriched in Mo, Au, As, Hg, Sb, Ba, Ti, and V, which are typical mineralogical indicators of ore.

The Mineral Reserves at the Williams Mine are grouped into two main areas: the B and C Zones. The B Zone is further divided into the main zone and footwall zone. The A Zone has been mined out. The A and B Zones are geologically the same continuous zone, with the B Zone being the down-plunge westerly extension of the A Zone, which sub-cropped at surface. The C Zone represents multiple sub-parallel lenses of irregular, generally narrow, gold mineralization. C Zone mineralization is stratigraphically different from the main zone and occurs in two broad geological domains, the porphyritic felsic metavolcanic rocks and the intermediate to felsic volcanoclastic sedimentary unit.

WILLIAMS MINE B ZONE GEOLOGY

The B Zone main zone is the westerly downward extension of the A Zone extending to a depth of at least 1,300 m below surface and has a strike length of approximately 1,200 m. The mineralization lies at or near the contact between overlying metasedimentary rocks and underlying felsic porphyritic rocks and is composed mainly of a fine grained, quartz-feldspar groundmass with gold occurring as finely disseminated particles within the groundmass as well as associated with pyrite grains. The zone dips at 60° to 70° to the northeast and plunges to the northwest at approximately 45°. The average thickness of ore in the well-delineated areas is 25 m to 30 m. The maximum thickness of ore is 45 m and is located adjacent to the Golden Giant boundary. Figure 7-3 is a section through the B Zone at 9800E. Moving west from that property boundary the ore generally declines in width and grade. The western portion of B Zone Mineral Reserves, from approximately 45 stope west, incorporates two ore grade zones within the typical host lithology. Internal waste pillars of at least 10 m horizontal thickness separate these zones. This mineralization represents the fringe of the Hemlo deposit and as such, continuity of economic grade and thickness becomes increasingly erratic.

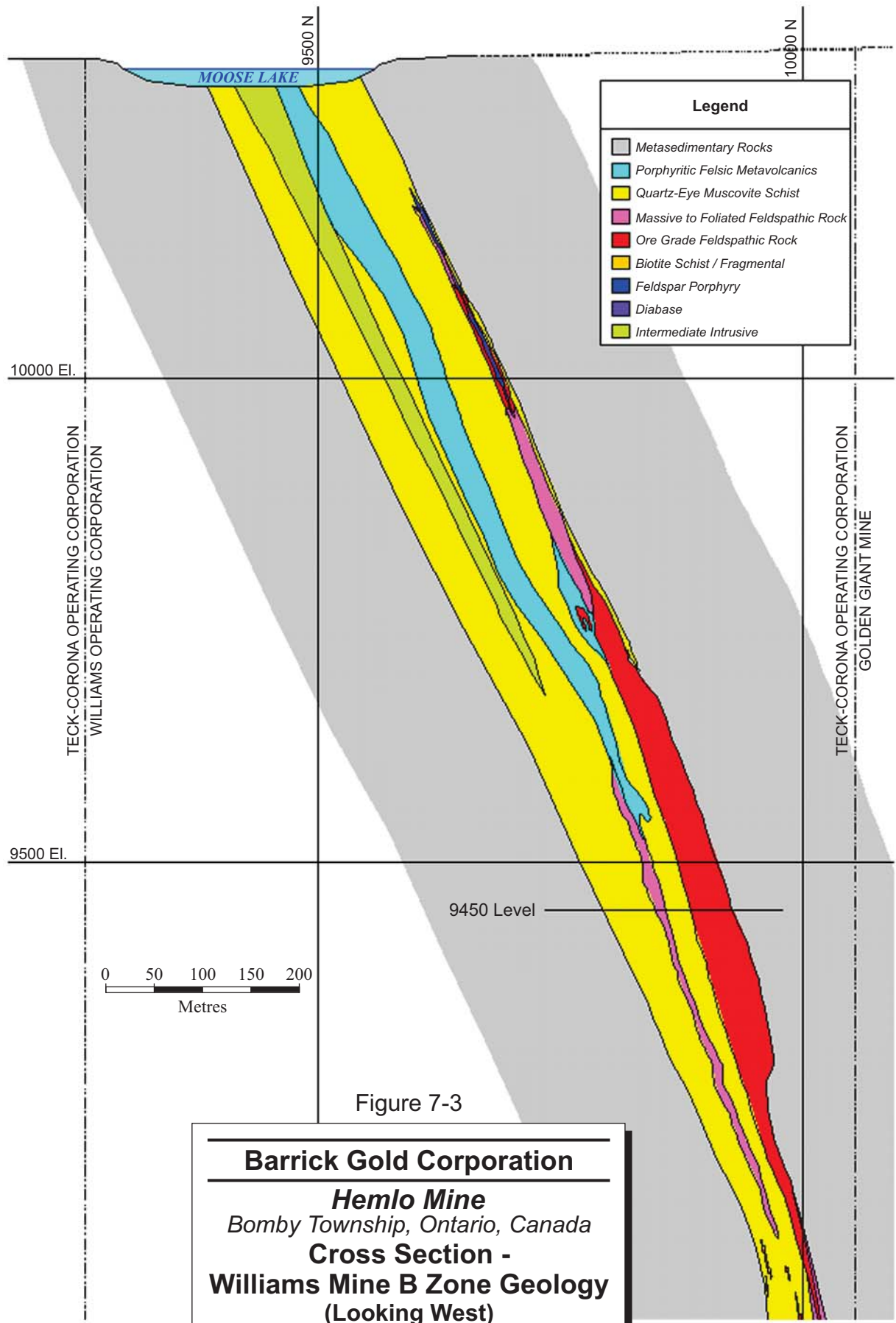


Figure 7-3

Barrick Gold Corporation
Hemlo Mine
 Bomby Township, Ontario, Canada
Cross Section -
Williams Mine B Zone Geology
 (Looking West)

April 2017

Source: Barrick Gold Corp., 2015.

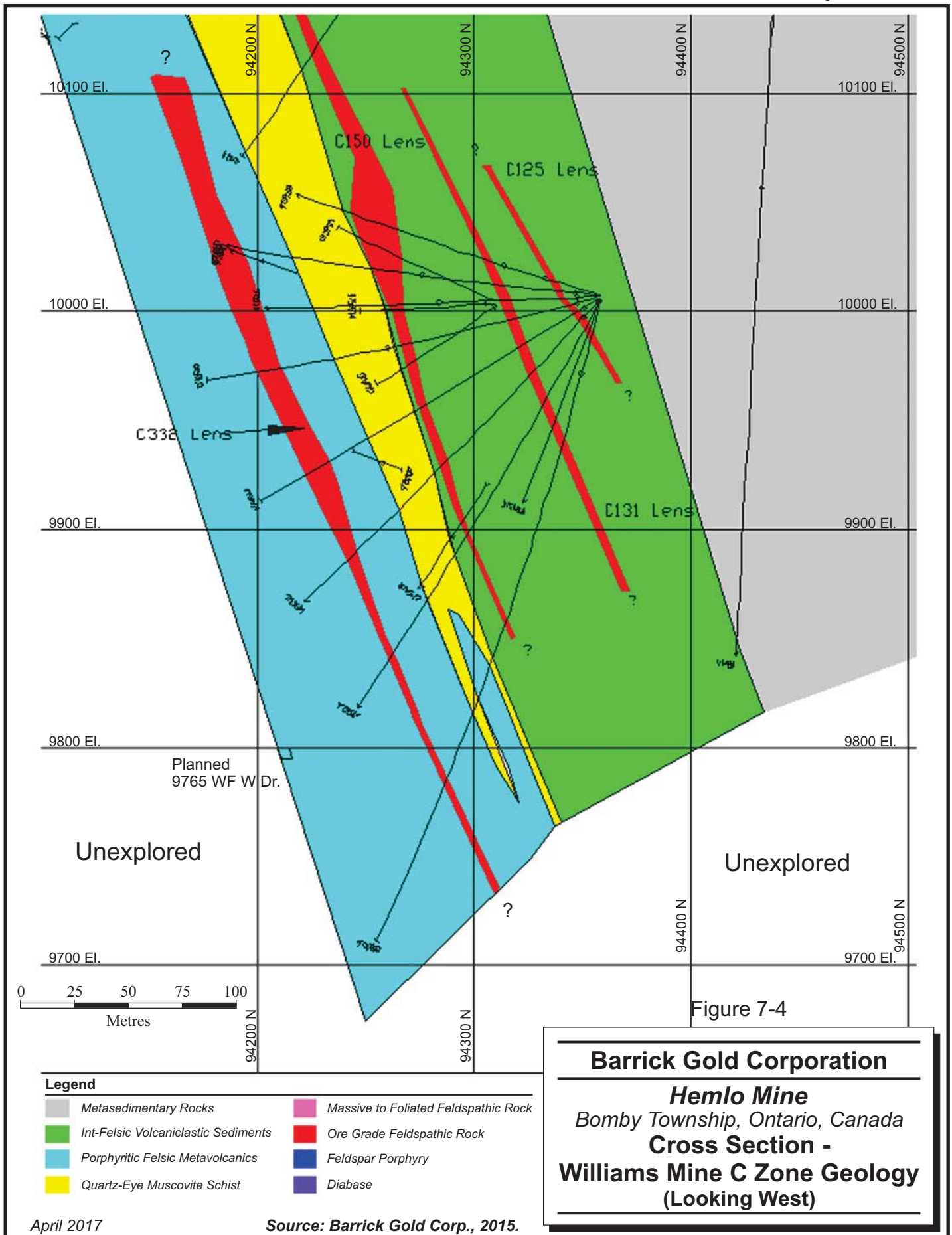
The Footwall zone is a series of lower grade mineralized lenses located 30 m to 80 m into the footwall of the B Zone main zone. It diverges from the main zone above the 9200 elevation and westward from the Golden Giant property boundary. For the level designations, these vary from mine to mine. The Williams Mine is approximately 360 MASL and 10,000 was added to the surface elevation to avoid having negative numbers as mining went deeper. At the Golden Giant Mine, 5,000 was added to the surface elevation for the same reason, while at the David Bell Mine, a simple level designation of 1, 2, 3, etc., increasing with depth, was used.

WILLIAMS MINE C ZONE GEOLOGY

The general C Zone stratigraphy from south to north comprises lower metasedimentary rocks, porphyritic felsic metavolcanics (Moose Lake Porphyry), quartz eye muscovite schist, intermediate to felsic volcanoclastic sediments (fragmental unit), and the upper metasedimentary rock sequence. The Williams Mine C zone geology is illustrated in the typical cross section in Figure 7-4.

Lower and upper denote the relative structural positions of the metasedimentary rock units as the younging directions are unclear. All of the major rock units are highly deformed with multiple events of deformation. Structural geology is complex. Rocks in the deposit area exhibit high strain. At the deposit scale, rocks in the area are tightly isoclinally folded. Most of the orebodies occur on one or more limbs of these folds. Local drag folding can be seen in the ore. Occasional transverse faults offset ore and wall rock units up to a few metres, and there is some shearing along major contacts. Regional metamorphism is up to amphibolite grade. The deposit has also been cut by a number of north-south trending diabase and lamprophyre dykes which post-date mineralization.

The 300-series (320, 332, and 333 lenses) of mineralized lenses are located in the porphyritic felsic metavolcanic rocks. Overall the unit can best be described as a variably sheared, quartz-feldspar porphyritic felsic rock defined by quartz eyes and/or feldspar phenocrysts and/or fragments. Highly variable alteration and deformation result in a highly variable appearance. The ore extends from the pit and economic intervals pinch out below 9450 elevation. The grade in the 300 lenses is more consistent than in the 100 lenses.



The 100-series (120, 125, 130, 140, 150 and 170 lenses) of mineralized lenses occur in the intermediate to felsic volcanoclastic sedimentary rocks (fragmental unit). There are at least two distinctive units—intermediate tuff/volcanoclastic sediments and felsic lapilli tuff with gradations between them. The intermediate tuff/volcanoclastic sedimentary unit is thickly bedded (two metres to 10 m) and composed of 40% to 60% biotite with generally fewer and smaller fragments than the felsic lapilli-tuff which consists of primarily coarse buff or light grey stretched feldspathic fragments (up to football sized) within a usually feldspathized matrix. The ore lenses within these units have some very high grade intersections which are less consistent on plan and in section than in the 300 lenses. The 100-series lenses extend onto the Interlake property with some very high grades especially in the 151 lens quartz vein system.

DAVID BELL MINE

There are three mineralized zones in the David Bell Mine. The A Zone is located within the Moose Lake Formation. It is a mixed fragmental-sedimentary unit. The A Zone is located on the contact of quartz-sericite porphyry schist and quartz-biotite hanging wall sediments. The C Zone is located within the Rule Lake Formation, about 80 m south of and parallel to the A Zone. The C Zone is on the footwall side of the quartz-sericite schist, in a sedimentary package similar to the hanging wall sedimentary rocks. The Highway Zone, originally called the Corona West Zone, is located approximately 500 m west of the Crown Pillar and approximately 100 m below surface. Mineralogy consists of Au and MoS₂ enrichment usually in a feldspathized or sericitic altered fragmental unit. Vanadium mica (roscoelite) and barite are prominent in the mineralized horizons as well as, to a lesser extent, realgar, stibnite, cinnabar, and orpiment.

GOLDEN GIANT MINE

The Main zone, which has produced most of Golden Giant's ore, lies on the north side of the Moose Lake Porphyry (MLP). The main lens of the Hemlo deposit is typically bounded on the hanging wall side by felsic-intermediate fragmental and sedimentary units and on the footwall side by felsic quartz-sericite schists. The lower zone lies on the south side of the MLP, about 70 m in the footwall of the Main Zone. It consists of one zone cut by a nearly concordant barren feldspar-porphyry sill.

The Hemlo deposit is crosscut by diabase dykes, some of which comprise significant volumes.

MINERALIZATION

The following is taken from Barber (2010).

The most significant mineralizing event (Au-Mo-K event) introduced Au, S, Mo, Zn, As, Sb, Hg, Tl, and W and the alteration elements K, Si, Fe, and V during the early stages of the major regional deformation event (G_2) and prior to peak metamorphism. Pervasive potassium-dominated alteration resulted in a core of feldspathization (microcline-quartz) coincident with the ore zone and an outer halo of muscovitization (muscovite-quartz). Significant pyritization, with biotitization and silicification are associated with alteration. Subsequent to the Au-Mo-K event, remobilization of Au-Sb-Si resulted in quartz veins with gold and stibnite, an Au-Ca event resulted in redistribution of gold with calc-silicate alteration assemblages, and an Au-As-Hg event precipitated low temperature sulphide minerals such as realgar, orpiment, and cinnabar from retrograde fluids. Molybdenite and green vanadiferous mica are the best visual mineralogical indicators of gold content in the Hemlo deposit.

Enrichment of Au, Mo, and lithophile elements suggests that magmatic fluids transported the metals, however, the source of the fluids has not been identified. The fluids were channelled along the feldspar quartz porphyry-metasediment interface and mafic fragmental in the restraining bend of a regional sinistral, ductile shear zone. The barite horizon contributed to competency contrast and may have triggered gold precipitation.

K, Si, Fe, and V enrichment and Ca, Mg, and Na depletion produced distinct alteration haloes both across and along strike from the deposit. The K enrichment produced a strong radiometric anomaly. Associated pyritization resulted in subtle but distinct very low frequency electromagnetic (VLF-EM) and induced polarization (IP) anomalies detected in both airborne and ground surveys. As, Sb, and Hg haloes are locally present as a result of metal redistribution during late alteration events.

The Hemlo deposit has several distinct mineralogical characteristics. High Hg content is a distinguishing feature of Hemlo native gold. Microcline related to alteration and mineralization has high Ba and muscovite has high Ba and V. Rutile in the mineralized zone is enriched in V, W, and Sb.

The Hemlo deposit is approximately three kilometres long by two kilometres deep and averages 20 m wide with a very small surface expression.

Mineral Reserves at the Williams Mine are currently grouped into two main areas: the B Zone and C Zone. The A Zone has been mined out. The A and B Zones are geologically the same continuous zone, with the B Zone being the down-plunge westerly extension of the A Zone, which sub-cropped at surface. The B Zone is further divided into the main zone and footwall zone. The C Zone represents multiple sub-parallel lenses of irregular, generally narrow, gold mineralization. C Zone mineralization is stratigraphically different from the main zone and occurs in two broad geological domains, the porphyritic felsic metavolcanics (Moose Lake porphyry) and the intermediate to felsic volcanoclastic sediment unit. Interlake is the down dip extension of the C Zone.

The B Zone extends to a depth of at least 1,300 m below surface and has a strike length of approximately 1,200 m. Main Zone mineralization occurs at or near the contact with the hanging wall sediments in a strongly feldspathized rock unit. The zone dips at 60° to 70° to the northeast and plunges to the northwest at approximately 45°. The horizontal thickness of the mineralization varies from three metres to 45 m. The maximum thickness of ore is located adjacent to the Golden Giant boundary. Moving west from that property boundary, the mineralization generally declines in width and grade. The western portion of B Zone Mineral Reserves includes two or three mineralized zones within the typical host lithology. Internal waste pillars of at least 10 m horizontal thickness separate these zones. This mineralization represents the fringe of the Hemlo deposit and as such, continuity of grade and thickness becomes increasingly erratic.

The B Zone Footwall (BZFW) zone is a series of lower grade mineralized lenses 20 m to 50 m south of the B Zone main zone. It diverges from the main zone above the 9200 elevation and westward from the former Newmont property boundary which is part of the Hemlo Property following the acquisition in 2014. The BZFW Zone is hosted in a series of structures sub-parallel to the main B Zone mineralization. Mineralization is hosted within a muscovite schist unit which can be traced from the David Bell Mine in the east part of the Hemlo area, through the Golden Giant Mine, to the western rim of the Williams Pit. Four of the seven FW lenses occur in mineable widths from two metres to nine metres over a strike length of up to 150 m and 125 m vertically. FW mineralization is characterized by rapid pinch-outs (within 20 m), but

where present, the mineralization is often high grade (10 g/t Au to 30 g/t Au). Associated minerals are molybdenite, pyrite, realgar, orpiment, arsenopyrite, tourmaline, and green mica.

The C Zone mineralization is approximately 400 m to 600 m west of the B Zone and distinctly different from the B Zone. Several zones or lenses are typical; most of which are thin, low to medium grade mineralization. These extend eastward roughly 700 m from the western boundary of the Williams property and vertically from surface to a depth of at least 1,300 m. The Interlake resource is the down dip extension of the C Zone mineralization.

The C Zone mineralization strikes at approximately 100° azimuth, dips 68° to the North, and plunges at 45° to 60° to the west. The higher-grade cores of the lenses are mined by underground methods where ore body widths are generally less than 10 m. Mineralization in the C Zone generally occurs along contacts between the fragmental units or as lenses that parallel the foliation. Individual lenses display pinch and swell features and large variances in grade can occur within each lens. Pre-existing geologic structures appear to have exerted the strongest controls on mineralization. The influence of later deformational events is not well understood. All of the major rock units are highly deformed with multiple events of deformation. Regional metamorphism is up to amphibolite grade. A muscovite schist, which is a key marker, pinches out at depth, between the porphyry and the fragmental unit.

The 300-series of lenses are found within the porphyritic felsic metavolcanics (Moose Lake Porphyry). The lens numbers increase to the south with the 320, 332, and 333 being the economic lenses underground. The C332/C333 zone displays the greatest continuity of all of the zones and has been traced 500 m along strike and over 900 m down dip.

The 100-series are contained in the intermediate to felsic volcaniclastic sediments (fragmental) unit. The 130, 131, 140, and 150 lenses make economic grade in various areas underground. The Interlake mineralization is the down dip extension of the 100-series of lenses.

8 DEPOSIT TYPES

Muir (2002) interprets the Hemlo deposit to be an atypical, mesozonal-orogenic, disseminated-replacement-stockwork deposit, broadly synchronous with D_2 and “middle” stage granitoid plutonism, prior to or synchronous with peak regional metamorphism, and involving magmatic $\pm$ metamorphic fluids. The following is taken from Muir (2002).

The deposit was emplaced after development of F_2 folds defined by lithologic layering; synchronous with incipient development of D_2 high strain zones during mid- D_2 ; and in association with the development of a restraining bend in the Hemlo greenstone belt during sinistral transpression.

Much of the mineralization is confined to high strain zones and spatially associated with the contact between felsic volcanic rocks and sedimentary rocks. Localization of hydrothermal fluid flow was aided by competency contrast at this contact, strain softening in the developing high-strain zones and formation of the restraining bend with induced dilation. The Hemlo deposit is spatially associated with and replaces felsic volcanic rocks of the Moose Lake Volcanic Complex (MLVC) reworked volcanoclastic rocks and epiclastic rocks, in increasing proportions, respectively. The deposit has undergone considerable progressive D_2 ductile strain, including mylonitization, with a sinistral component, interpreted to reflect sinistral transpression. The mineralized zones are structurally controlled by D_2 elements at a variety of scales, being broadly tabular and parallel to subparallel to S_2 and S_{2M} (mylonitic) fabrics. The deposit was further modified slightly by a D_3 , dextral transpressional overprint, in which the D_2 high-strain zones localized much of the D_3 strain. Numerous types of dikes crosscut the deposit. Strain and metamorphism have modified some characteristics of the deposit. More than one stage of gold remobilization has occurred.

The restraining bend marks general changes in many characteristics of the alteration and mineralization between what is termed the Main Segment and the West Segment. The Main Segment is characterized largely by potassium metasomatism and replacement-style mineralization, whereas the West Segment is noted largely for sodium (+ calcium) and potassium metasomatism, as well as fracture- controlled mineralization. Mineralization consists dominantly of Au and Mo, with anomalous Sb, As, Hg, V, and Ba. Barite is common

in parts of the deposit. Its origin is contentious, but is considered to be part of the hydrothermal system.

Mineralization is poorly constrained between 2694 and 2677 Ma and predates or is nearly synchronous with regional peak metamorphism, which culminated during late-D₂ and is poorly constrained between 2688 and 2675 Ma. Muir (2002) estimates ages for mineralization and peak metamorphism are approximately 2689 to 2684 Ma and 2684 Ma, respectively, based on inferred, broadly contemporaneous plutonism, mineralization, metamorphism, and crosscutting relationships.

Muir (2002) favours emplacement at a depth from six kilometres to 12 km, with syntectonic (mid-D₂), magmatic ± metamorphic hydrothermal system, which may have originated from an unexposed pluton. The fluids are considered to have utilized an existing planar D₂ crustal anisotropy, enhanced, in part, by competency contrast, to focus the fluids. An inferred magmatic association reflects the considerable potassic alteration and molybdenum content of mineralization. The siting of significant alteration and/or mineralization in post-regional-folding, complex high-strain zones is the cumulative result of many variables, particularly those operating in the upper crust. The direct role of deep-seated metamorphism in generating magmas and mineralizing fluids during Archean crustal-scale evolution is likely.

9 EXPLORATION

EXPLORATION POTENTIAL

Despite 30 years of mining in the Hemlo camp, significant exploration potential still exists within reach of the current mine workings. Ongoing projects include a data compilation in the lower B Zone area as well as diamond drill programs to explore mineralization at depth in the C Zone and B Zone, and to the west of C Zone. Data acquired with the acquisition of additional land to the west of the current workings and wide-spaced drilling in 2016 suggests significant exploration potential could exist in this area.

The 2016 Diamond Drilling Program met with a high rate of success. Known mineralization was extended to the west and down dip and in most cases, remains open along strike and at depth. Near surface, a program tested the strike and down-dip extension of higher grade mineralization exposed in the Pit, in order to assess the viability of mining this mineralization by underground methods.

As new drilling augments the recently acquired historical data, and the understanding of the Hemlo deposit progresses, it is likely that new exploration targets will emerge.

Exploration advantages include:

- Relatively low drilling cost on a per ounce basis.
- A history of resource to reserve conversion.
- Property acquisition to the west has removed underground and open pit mining constraints (see Figure 9-1).
- A Geological Compilation Study completed in 2016 established the first model of the entire Hemlo deposit using all available data from the Williams, Golden Giant, and David Bell mines.
- Good exploration potential identified to the west and at depth (see Figure 9-2).

In RPA's opinion, exploration success has the potential to significantly extend the mine life at Hemlo. Additional resources, together with some debottlenecking of current operations, may

provide an opportunity for modest increases in production throughput, and more significant increases in gold production.



Figure 9-1

Barrick Gold Corporation

Hemlo Mine
 Bomby Township, Ontario, Canada
Hemlo Mining Claims

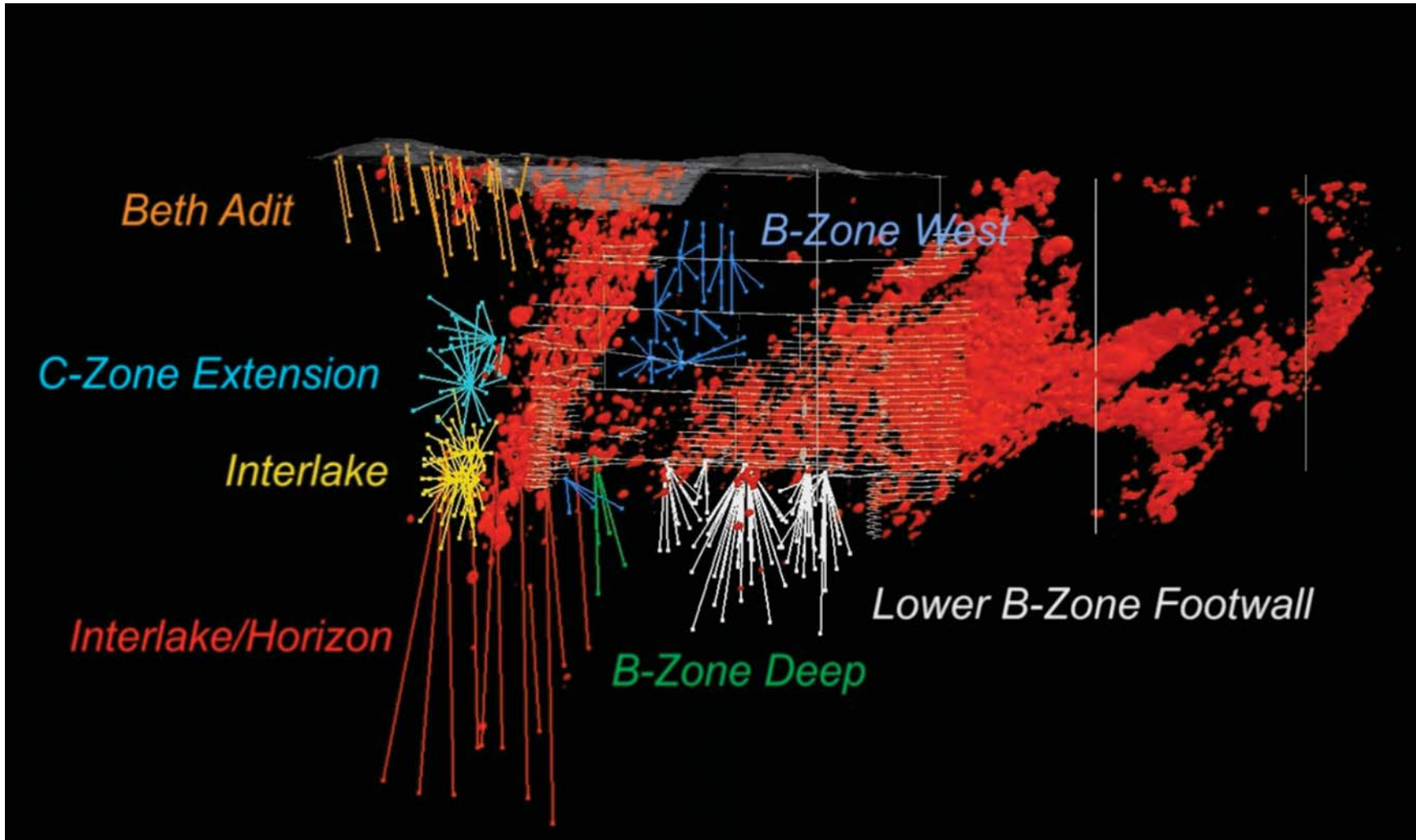


Figure 9-2

Barrick Gold Corporation

Hemlo Mine

Bomby Township, Ontario, Canada

**2017 Exploration - Long Section
(Looking North)**

10 DRILLING

Diamond drilling on the Hemlo Mine property was conducted in phases by several companies from 1947 to 2016. By the end of December 2016, a total of approximately 10,845 diamond drill holes have been completed at the Williams Mine; 2,999 holes in B Zone, 3,398 holes in C Zone, 566 holes in the Interlake zone, 1,308 in the Golden Giant zone, 2,094 holes at David Bell, and service and geotechnical holes.

The diamond drill holes from surface and underground at the Williams and David Bell mines are summarized in Table 10-1.

Currently, all diamond drill core is photographed and logged noting lithology, alteration, structure, and mineralization. Geotechnical logging includes core recovery and rock quality designation (RQD). All the observations are recorded on a laptop computer using Acquire software.

Exploration holes are cut prior to sampling. Approximately one-third of the underground definition holes are cut while half are sampled whole.

Samples are taken, nominally at one metre intervals, however samples may be as short as 30 centimetres, as dictated by lithology and mineralization. Samples do not cross major lithological contacts.

TABLE 10-1 SURFACE AND UNDERGROUND DIAMOND DRILL HOLES
Barrick Gold Corporation – Hemlo Mine

Year	Company	Williams B Zone		Williams CZone/SC/HZ		Interlake		Golden Giant		David Bell		Geotech		Totals	
		No. of Holes	Footage	No. of Holes	Footage	No. of Holes	Footage	No. of Holes	Footage	No. of Holes	Footage	No. of Holes	Footage	No. of Holes	Footage
1982	Lac Minerals	23	2,796	1	171									24	2,967
1982	Golden Sceptre			21	2,244									21	2,244
1983	Lac Minerals	106	52,852	142	12,333									248	65,184
1983	Noranda			32	9,178									32	9,178
1983	Teck/Noranda					6	11,131							6	11,131
1984	Lac Minerals	27	15,868	60	11,698									87	27,566
1984	Noranda			128	19,539									128	19,539
1984	Teck/Noranda					4	9,398							4	9,398
1985	Lac Minerals			84	5,795									84	5,795
1985	Teck/Noranda					2	3,423							2	3,423
1986	Lac Minerals	70	6,981											70	6,981
1986	Noranda			7	6,750									7	6,750
1987	Lac Minerals			141	18,961									141	18,961
1987	Teck/Noranda					4	10,773							4	10,773
1987	Noranda			5	1,944									5	1,944
1988	Lac Minerals	134	10,773	37	3,992									171	14,765
1989	Teck/Corona	96	8,789	6	3,268									102	12,057
1990	Teck/Corona	191	23,847											191	23,847
1991	Teck/Homestake	107	14,986											107	14,986
1991	Hemlo Gold			54	5,026									54	5,026
1992	Teck/Homestake	101	9,527	20	5,577									121	15,104
1992	Noranda			11	11,697									11	11,697
1993	Teck/Homestake	164	14,378	5	2,337									169	16,715
1994	Teck/Homestake	135	14,332	14	3,436									149	17,767
1995	Teck/Homestake	116	8,680	64	12,338									180	21,018

Year	Company	Williams B Zone		Williams CZone/SC/HZ		Interlake		Golden Giant		David Bell		Geotech		Totals	
		No. of Holes	Footage	No. of Holes	Footage	No. of Holes	Footage	No. of Holes	Footage	No. of Holes	Footage	No. of Holes	Footage	No. of Holes	Footage
1996	Teck/Homestake	182	19,258	28	5,282									210	24,540
1997	Teck/Homestake	110	13,192	7	1,986									117	15,179
1997	Franco Nevada					21	10,008							21	10,008
1997	Noranda			4	5,443									4	5,443
1998	Teck\Homestake	104	9,286	42	6,694									146	15,980
1998	Franco Nevada					19	10,055							19	10,055
1998	Noranda			6	4,429									6	4,429
1999	Barrick/Teck	47	5,799	77	10,688									124	16,487
2000	Barrick/Teck	27	3,455	77	16,417									104	19,872
2001	Barrick/Teck	91	5,656	84	11,731									175	17,387
2002	Barrick/Teck	136	10,445	76	11,562									212	22,008
2003	Barrick/Teck	113	8,464	127	17,440									240	25,904
2004	Barrick/Teck	119	11,811	221	28,289									340	40,100
2005	Barrick/Teck	65	3,765	257	31,999									322	35,764
2006	Barrick/Teck	149	16,288	262	29,889							34	2,089	445	48,266
2007	Barrick/Teck	177	14,177	109	15,736	37	11,327					24	1,794	347	43,035
2008	Barrick/Teck	9	1,683	177	21,621	104	15,777					60	4,910	350	43,992
2009	Barrick/Teck	40	5,796	111	13,115	77	18,898					37	3,489	265	41,298
2010	Barrick	54	5,849	291	39,347	36	9,870					11	1,936	392	57,001
pre 2011	various owners							1165	131,382	2,088	377,583			3,253	508,966
2011	Barrick	73	18,943	164	19,856	52	8,364	86	5,143			28	1,388	403	53,695
2012	Barrick	15	6,840	83	15,705	83	10,952	49	2,690	6	273			236	36,460
2013	Barrick	27	6,388	58	8,651			8	334			32	1,003	125	16,375
2014	Barrick	45	5,332	19	3,326	29	2707					108	1000	202	12,365
2015	Barrick	82	12,119	99	18,368	37	4,809					46	926	276	37,113
2016	Barrick	64	8,719	183	41,291	47	8,637					100	1,715	394	60,362
TOTAL		2,999	377,074	3,398	515,925	566	146,243	1,308	139,549	2,094	377,856	480	20,250	10,845	1,576,900

11 SAMPLE PREPARATION, ANALYSES AND SECURITY

SAMPLING METHOD AND APPROACH

DIAMOND DRILL CORE

All core samples are tagged, bagged in sample bags, and forwarded to the sample preparation laboratory. Most holes are selectively sampled on one metre intervals, however, core sample lengths vary from 0.3 m to 1.1 m, to honour geological units and obvious geological features. Historically, whole core was sampled in most cases. Current practice is that most exploration holes and selected definition holes are sampled and sawn into equal halves. One half is placed in a sample bag and the other half remains in the core box, which is stored in a secure location on site.

UNDERGROUND SAMPLES

Chip samples are taken in the Alimak raises and at the undercut and overcut of the stopes.

Underground test holes are drilled and sampled by miners for all Alimak stopes in the C Zone prior to cable bolting. Test holes are drilled at 20 m spacing up the raise in the footwall and at 40 m spacing in the hangingwall. Test holes are sampled every 1.2 m down the hole. Drill cuttings are collected in sample bags inserted in a customized sampling device constructed for the collection of drill cuttings. The sampling device is placed underneath the drill rods at the rock interface to collect the drill cuttings that are flushed out of the hole with the water from the drill. Residual water in the sample bag is drained. Drill cuttings are collected to a depth of six metres in the footwall and 4.8 m in the hangingwall. The test hole number, interval between samples, date and name of the sampler are recorded on the sample tag. One part of the sample tag is inserted into the sample bag and sealed with a tie wrap. Barrick has documented the underground Alimak test hole sampling procedure.

BLASTHOLE SAMPLES

Cuttings from each blast hole are sampled and analyzed for grade. Only every fourth blast hole is sampled in geological units which have consistently proven to be uneconomic. In and around ore units, every blast hole is sampled and used for ore control.

SAMPLE PREPARATION

A complete analytical laboratory is located on the Williams mine site adjacent to the core facility. Both mine and definition diamond drill core samples are assayed at this facility. The onsite laboratory is not certified. Currently, most exploration samples are shipped to an external lab. Samples are placed into large bags (10 to 12 at a time) and the large bags are sealed. The samples are placed in a crate, taken to the warehouse and shipped to the laboratory via transport truck. WOC currently sends samples to Activation Laboratories Ltd. (Actlabs) in Thunder Bay, Ontario. Actlabs is an ISO/IEC 17025 accredited laboratory (Certificate number 266) with CAN-P-1579 (Mineral Analysis). Results from Actlabs are emailed and imported into the database. A paper certificate is mailed at a later date.

Check assaying may be conducted at the Williams laboratory or at ALS Chemex in Vancouver. The ALS Chemex laboratory in Vancouver is independent and accredited in accordance with ISO/IEC 17025:2005.

ASSAYING

Samples are oven dried for 24 hours then crushed to minus 3 mm. They are assayed by fire assay with an atomic absorption finish. The assay data is put into the database electronically as it is received from the laboratory.

A minimum of five percent of the sample pulps are sent to an additional secondary laboratory for check assays.

SAMPLE SECURITY

The Williams Mine has its own analytical laboratory which is located adjacent to the core facility on the secure mine site. The core is handled only by authorized mine personnel.

QUALITY ASSURANCE AND QUALITY CONTROL

The quality assurance/quality control (QA/QC) protocols include the insertion of certified reference material (CRMs or standards) into the sample stream. Standards are inserted at the rate of one standard for every 20 samples and one blank is inserted for every 40 samples. One duplicate is assayed out of every 40 samples.

CERTIFIED REFERENCE MATERIAL

Certified standard samples, prepared by CDN Resources are used. Standards are inserted at the rate of one in 20 samples. The results are reviewed on a per sample basis and the sample is re-assayed if the results are greater than two standard deviations from the expected result. The entire tray is re-assayed if the result is greater than three standard deviations from the expected result.

An assay on a standard that is more than two standard deviations from the mean concentration triggers a re-assay of the sample. An assay on a standard that is over three standard deviations from mean concentration triggers re-assaying of the entire batch of samples. There is an automatic re-assay of any sample that is greater than four grams per tonne.

BLANK SAMPLES

Hemlo uses commercially supplied blank samples purchased from Mines Assay Supply in Kirkland Lake, Ontario. The blanks are inserted into the sample stream at a ratio of 1:40. The results are reviewed by Hemlo and an investigation is initiated if the result is greater than 0.01 g/t Au. The whole batch is rejected if a blank returns a value over 0.2 g/t Au.

DUPLICATE SAMPLES

One out of 60 samples is assayed in duplicate. Results that are within plus or minus two standard deviations are acceptable results. If the results are in excess of two standard deviations, the sample set is re-assayed.

RPA OPINION OF SAMPLE PREPARATION, ANALYSIS, AND SECURITY

In RPA's opinion, the sample preparation, analysis, and security meets industry best practice standards.

12 DATA VERIFICATION

STANDARDS

Certified standard samples, prepared by CDN Resources are used. Standards are inserted at the rate of one in 20 samples. The results are reviewed on a per sample basis and the sample is re-assayed if the results are greater than two standard deviations from the expected result. The entire tray is re-assayed if the result is greater than three standard deviations from the expected result.

GRADE DISTRIBUTION OF CHECK ASSAY VALUES

The Hemlo Mine uses commercially supplied blank samples purchased from Mines Assay Supply in Kirkland Lake, Ontario. The blanks are inserted into the sample stream at a ratio of 1:40. The results are reviewed and an investigation is initiated if the result is greater than 0.01 g/t Au. The whole batch is rejected if a blank returns a value over 0.2 g/t Au.

DUPLICATE SAMPLES

One out of 60 samples is assayed in duplicate. Results that are within plus or minus two standard deviations are acceptable results. If the results are in excess of two standard deviations, the sample set is re-assayed.

RPA DATA VALIDATION

The Hemlo Mine is a producing mine with a long history of gold production. No independent samples were required by RPA to demonstrate the existence of gold-bearing mineralization.

RPA OPINION OF QA/QC

In RPA's opinion, the database is appropriate to be used for the estimation of Mineral Resources and Mineral Reserves.

13 MINERAL PROCESSING AND METALLURGICAL TESTING

HISTORICAL METALLURGICAL TESTING AND DATA

Predicting metallurgical performance at Hemlo is based on a combination of historical operating performance and sampling and test work. Predictive models are challenged by the fact that the ore is sourced from different areas and that each of these areas has different geological and mineralogical characteristics. Primary variables used in modelling include mill throughput rate, which influences grinding circuit product size distribution, and the gold head grade of the feed to the process plant.

Characterization studies of the open pit ore have been on-going since 2001 with the most comprehensive work completed during 2010 and 2011. Two holes each were drilled in the 100- and 300-series of lenses and samples were taken to represent the entire pit. Samples were taken at varying depths and used to evaluate the grinding characteristics of the ore. Tests included JK Drop Weight Tests, SAG Mill Comminution (SMC) tests, Rotary Breakage Tests (RBT), Bond Ball Mill Work Index (BWi) tests, Bond Rod Mill Work Index (RWi) tests, and Crusher Work Index (CWi) tests. The comminution tests indicated that the 300-series samples were more resistant to impact breakage than the 100-series samples.

Samples were also taken from 10 m intervals of the drill core to conduct grind-recovery tests. The gold grades ranged from 0.5 g/t Au to 2.5 g/t Au and particle sizes tested ranged from 70% to 100% passing 75 µm.

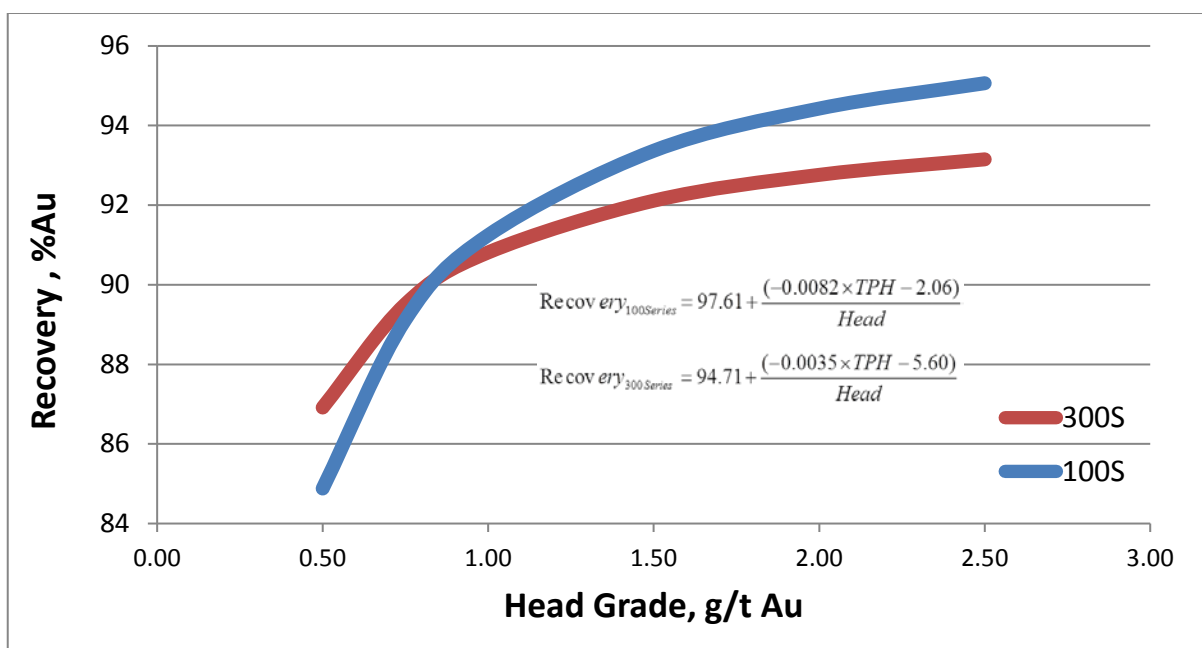
Results indicated that gangue minerals from the 100-series samples included muscovite, biotite, calcite, and feldspar and the samples contained 0% to 2% pyrite. The gangue mineral in the 300 series samples only included feldspar, however, the pyrite concentration was 5% to 8%.

Models to predict recovery based on grind size were developed by Barrick. No correlations were observed between depth and recovery, however, subsequent plant trials resulted in higher recoveries than those predicted using the laboratory data. Therefore, scale-up coefficients were developed to provide recovery estimates that were consistent with the plant

performance (Yalcin, 2012). Relationships between plant throughput and the particle size distribution of the ground ore were also developed. This made the development of correlations between plant throughput and recovery possible.

Further improvements to the recovery estimates were made using data from bulk tests and production campaigns including secondary crushing of the ore fed to the mill. Correlations developed from 2012 plant data were higher than the estimates made using 2010 data. The most recent estimates based on laboratory data that was scaled up to plant data are shown in Figure 13-1.

FIGURE 13-1 RELATIONSHIP BETWEEN GOLD HEAD GRADE AND GOLD RECOVERY



The gold recovery relationships are:

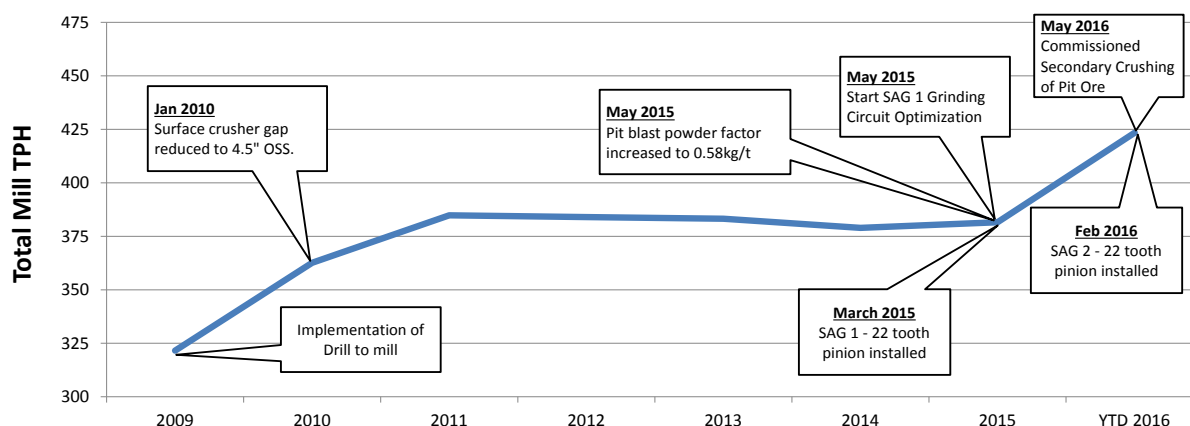
$$Au \text{ Recovery } 100 \text{ series, \%} = 97.61 + \frac{-0.0082 \times \frac{t}{h} - 2.06}{Au \text{ Head Grade, g/t}}$$

$$Au \text{ Recovery } 300 \text{ series, \%} = 94.71 + \frac{-0.0035 \times \frac{t}{h} - 5.06}{Au \text{ Head Grade, g/t}}$$

Tinney (2016) reported that additional laboratory and plant scale testing programs have determined that process plant throughput is SAG mill limited by the 300-series ore. A series

of step-wise improvements have been made to the plant since 2009, including increased SAG mill power utilization and reduction of the SAG mill feed size, resulting in higher and higher throughput as shown in Figure 13-2.

FIGURE 13-2 IMPROVEMENTS TO PLANT THROUGHPUT



Improvements shown in Figure 13-2 include:

- Increased blast fragmentation
- Reduction in the crusher closed size setting and improvements to the mantle control
- Increased grinding power
- Coarser particle sizes
- SAG mill stability improvements

Metallurgical testing programs have been completed for numerous reasons over the years, including:

- Design of a secondary crushing circuit to further increase plant capacity
- Evaluation of the impact of pebble crushing
- Development of a cyanide detoxification process for CIP tailings
- Evaluation of various tailings preparation and deposition options that were used to optimize the design and capacity of the TSF
- Development of a flotation process that can be used to de-sulphurize the CIP tailings and reduce the costs for mine closure activities

FUTURE METALLURGICAL TESTING

Wiputri (2016) prepared a scope of work to evaluate the characteristics of mineralization from new areas underground. The data will initially be used as an input to an internal study to evaluate extensions to the underground mine at Williams. The work is proposed for 2017.

Objectives include:

- Determine the variability in grindability
- Determine the variability in gold recovery
- Determine the optimum grind size
- Compare the results from carbon-in-leach (CIL) and carbon-in-pulp
- Assess the impact of coarse gold particles on the plant operation

PLANT OPERATING DATA

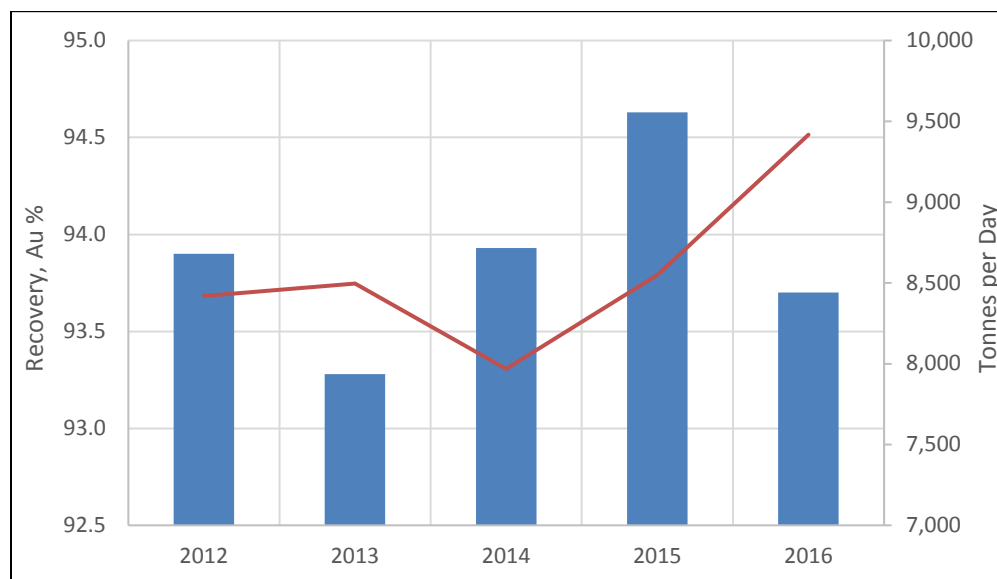
Table 13-1 summarizes the plant operating data from 2013 through 2016.

TABLE 13-1 PLANT OPERATING DATA
Barrick Gold Corporation – Hemlo Mine

Description	Units	2016	2015	2014	2013
Tonnes Milled	Mmt	3.408	3.120	2.844	2.968
Tonnes per Operating Hour	dmt/h	422	382	378	382
Tonnes per Calendar Day	tpcd	9,417	8,549	7,968	8,496
Total Production	oz	234,668	218,560	192,197	177,343
Gravity Production	oz	0	0	0	1,843
Recovery	%	93.74	94.64	93.99	93.26
Head Grade	g/t	2.28	2.30	2.24	1.99
Tails Grade	g/t	0.14	0.12	0.13	0.13

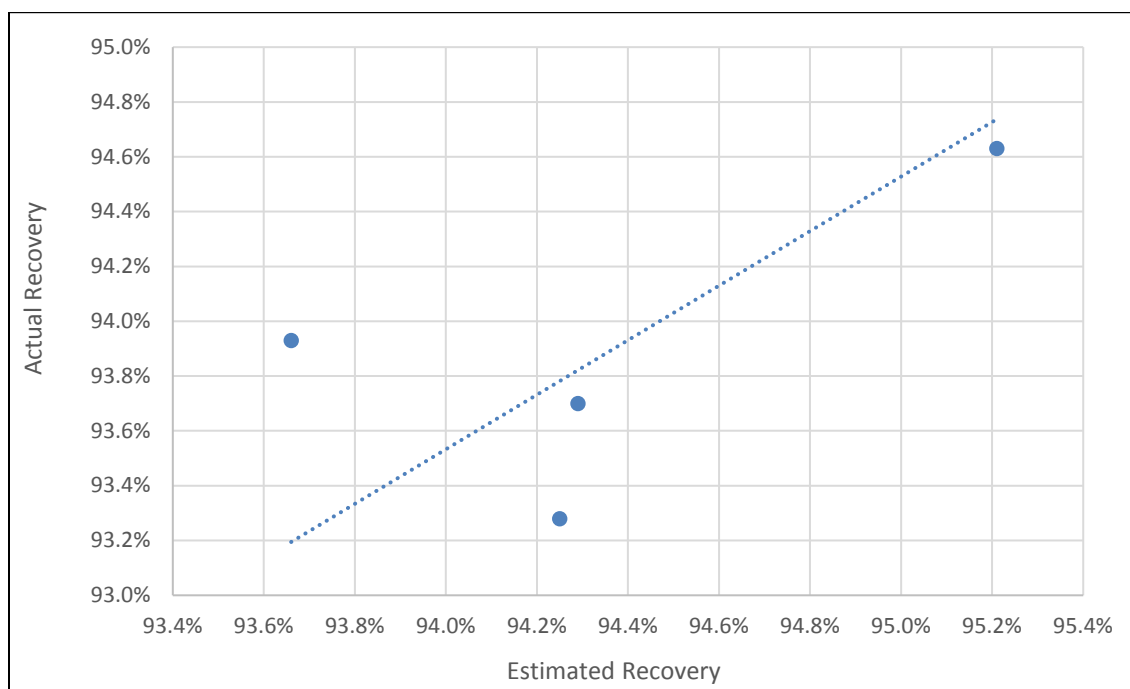
Figure 13-3 compares the gold recovery and the plant throughput for a similar timespan.

FIGURE 13-3 GOLD RECOVERY AND PLANT THROUGHPUT



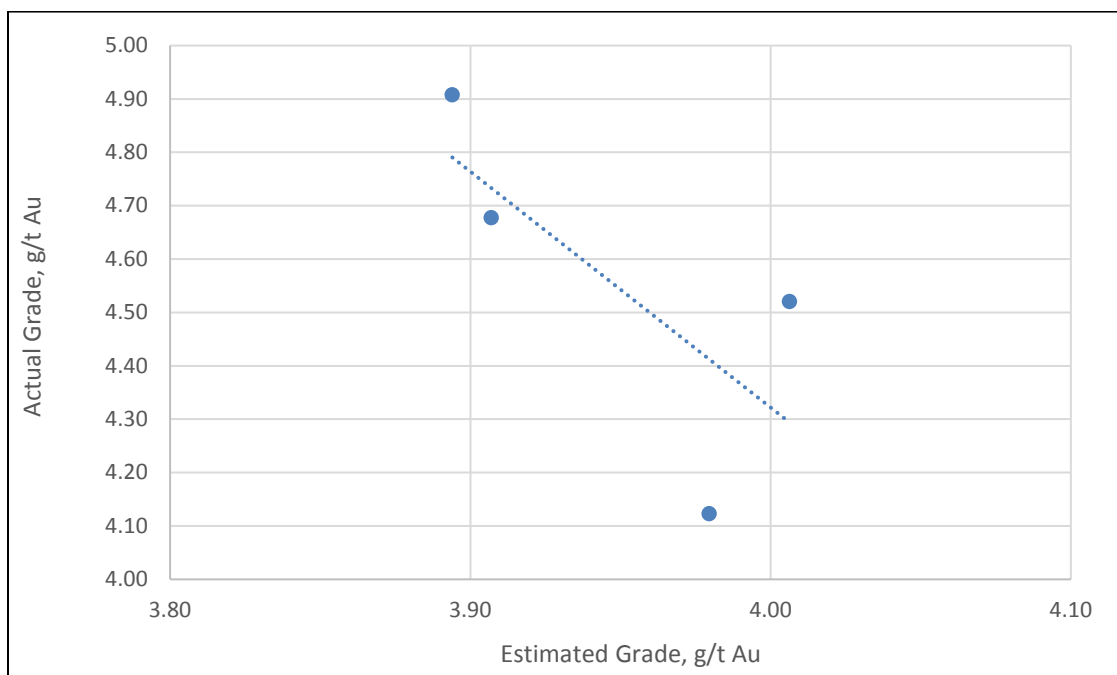
RPA considers evaluation of plant operating data to be the best indication of the accuracy of plant recovery estimates. Figure 13-4 compares budgeted (or estimated) gold recovery with actual gold recovery for 2013 through 2016.

FIGURE 13-4 ESTIMATED GOLD RECOVERY COMPARED TO ACTUAL GOLD RECOVERY



The recovery estimates differ from the actual recovery by less than 1%, which indicates that the estimates are very accurate. It is also worthwhile to compare the estimated head grades of the plant feed to the actual head grades since recovery is correlated to gold head grade. Figure 13-5 provides the data from the underground mine and Figure 13-6 compares the data from the open pit mine.

FIGURE 13-5 COMPARISON OF ESTIMATED GOLD FEED GRADE AND ACTUAL GOLD FEED GRADE FOR WILLIAMS UNDERGROUND

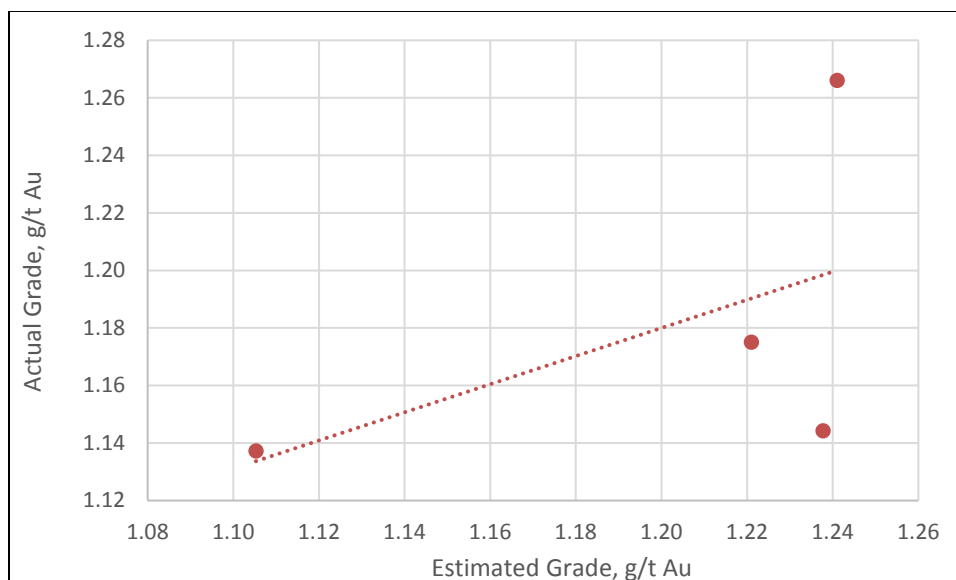


For underground ore, the difference between the estimated feed grade and the actual feed grade varies from 3.6% to 26% between 2013 and 2016, as summarized in Table 13-2.

**TABLE 13-2 ESTIMATED HEAD GRADE COMPARED TO ACTUAL HEAD GRADE FROM WILLIAMS UNDERGROUND
Barrick Gold Corporation – Hemlo Mine**

	Budget	Actual	% Difference
2013	3.980	4.123	-3.6%
2014	4.006	4.521	-12.8%
2015	3.907	4.678	-19.7%
2016	3.894	4.908	-26.0%

FIGURE 13-6 COMPARISON OF ESTIMATED GOLD FEED GRADE AND ACTUAL GOLD FEED GRADE FOR WILLIAMS OPEN PIT



The actual feed grades are less than 8% different from the estimated feed grades for the open pit ore which RPA considers reasonable.

RPA is of the opinion that the metallurgical estimates are based upon comprehensive testing programs that meet industry standards and the recovery estimates are appropriate, however, if the grade of the ore that will be milled from the underground mine cannot be estimated accurately, it is likely that the recovery and production estimates will become less accurate.

RPA is not aware of any processing factors or deleterious elements that could have a significant effect on potential economic extraction at the Hemlo Mine.

14 MINERAL RESOURCE ESTIMATE

SUMMARY

Table 14-1 summarizes the Mineral Resources exclusive of Mineral Reserves as of December 31, 2016, based on a gold price of US\$1,500 per ounce. Measured and Indicated Mineral Resources total 58.9 million tonnes averaging 0.91 g/t Au and contain 1.72 million ounces of gold. In addition, the Inferred Mineral Resources total 7.8 million tonnes averaging 1.94 g/t Au and contain 0.48 million ounces of gold.

TABLE 14-1 SUMMARY OF MINERAL RESOURCES - DECEMBER 31, 2016
Barrick Gold Corporation - Hemlo Mine

Classification	Tonnes (000 t)	Grade (g/t Au)	Contained Gold (000 oz Au)
Measured			
Williams Open Pit			
Williams Underground	126	2.72	11
Total Measured	126	2.72	11
Indicated			
Williams Open Pit	56,484	0.80	1,449
Williams Underground	2,287	3.53	259
Total Indicated	58,771	0.90	1,709
Measured + Indicated			
Williams Open Pit	56,484	0.80	1,449
Williams Underground	2,413	3.49	270
Total Measured + Indicated	58,897	0.91	1,720
Inferred			
Williams Open Pit	5,118	0.62	102
Williams Underground	2,648	4.49	382
Total Inferred	7,765	1.94	484

Notes:

1. CIM definitions were followed for Mineral Resources.
2. Mineral Resources are estimated using a gold price of US\$1,500 per ounce, and a US\$/C\$ exchange rate of 1.30:1.
3. Mineral Resources are estimated at gold cut-off grades that vary by mining method and material type, averaging 0.36 g/t Au for open pit and 1.95 g/t Au for underground.
4. Mineral Resources are exclusive of Mineral Reserves.
5. Mineral Resources that are not Mineral Reserves do not have demonstrated economic viability
6. Numbers may not add due to rounding.

The current Mineral Resources are based on resource models updated in 2016 for the Williams C Zone, Interlake, B Zone West, B Zone Footwall, Lower B Zone, and Beth Adit areas. Mineral Resources are based on grade models updated in 2015 for the C Zone open pit area. The modelling work was completed by Barrick Geological Staff under the supervision of Rodney Barber, P.Geo, Hemlo Geology Superintendent.

RPA reviewed the resource assumptions, input parameters, geological interpretation, and block modelling procedures and is of the opinion that the Mineral Resource estimate is appropriate for the style of mineralization and that the resource model is reasonable and acceptable to support the 2016 year-end Mineral Resource and Mineral Reserve estimates.

RPA is not aware of any environmental, permitting, legal, title, taxation, socio-economic, marketing, political, or other issues that could materially affect the Mineral Resource estimate.

DATABASE

The database used for resource estimation consisted of 10,701 drill holes containing 982,326 m of drilling. Assay data was uploaded directly from laboratory electronic files to avoid manual data entry errors.

COMPOSITES

Drill holes were composited to three separate composite databases in one-, two- or four- metre lengths. The composite length used for estimation depends on the block model domain type and sub-block size. Less than one percent of the assays have a length greater than 1.4 m with 89% at or less than one metre. This indicates that additional data points are not being created at the compositing stage. Composite intervals were stopped and re-started at geological boundaries.

CUT-OFF GRADES

Due to the complex processing strategy in use at the Hemlo Mine where operating costs vary by multiple ore types, multiple cut-off grade values are used. Separate Mineral Resource and Mineral Reserve cut-off grades, based on the Mineral Resource and Mineral Reserve gold

prices, are used to report the Mineral Resources and Mineral Reserves. The cut-off grade is based on process cost, G&A cost (split between open pit and underground), ore type, and recovery. All blocks that fail to meet the cut-off grade are classified as waste. The inputs used to determine the Mineral Resource cut-off grade of 0.36 g/t Au are shown in Table 14-2. The principal difference between the Mineral Reserve and Mineral Resource cut-off grade calculation is the gold price of \$1,000/oz Au and \$1,500/oz Au, respectively.

**TABLE 14-2 OPEN PIT RESOURCE CUT-OFF GRADE CALCULATION
PARAMETERS
Barrick Gold Corporation - Hemlo Mine**

Description	Units	Value
Mining cost	\$/t mined	4.70
Processing cost	\$/t milled	16.37
General and Administrative costs	\$/t milled	3.89
Currency Exchange Rate	USD/CDN	1.30
Gold Metal Recovery	%	89.16
Gold Selling Price	US\$/oz	1,500
Shipping and Refining Cost	\$/oz	2.77
Royalty	%	2.09
OP Resource Cut-off Grade	g/t Au	0.36

The underground resource cut-off grade was determined using the parameters shown in Table 14-3. The same gold prices assumptions used in the open pit calculation were used for underground. The differences in metallurgical recoveries used in all cut-off grade calculations for open pit and underground Mineral Resources and Mineral Reserves is a result of the grade recovery relationship which is described in more detail in Section 13 of this report.

**TABLE 14-3 UNDERGROUND RESOURCE CUT-OFF GRADE CALCULATION
PARAMETERS
Barrick Gold Corporation - Hemlo Mine**

Description	Units	Value
Mining cost	\$/t mined	87.18
Processing cost	\$/t milled	10.99
General and Administrative costs	\$/t milled	13.22
Currency Exchange Rate	USD/CDN	1.30
Gold Metal Recovery	%	93.86
Gold Selling Price	US\$/oz	1,500
Shipping and Refining Cost	\$/oz	2.77
Royalty	%	2.55
UG Resource Cut-off Grade	g/t Au	1.95

CAPPING

Barrick performed statistical analysis using its CumHist package on each of the mineralized domains, as well as the waste domain. Grade caps were chosen such that the coefficient of variation of the data was less than 2.0. No more than 10% of the length-weighted gold grades were capped. A summary of the Tier 3 open pit grade caps used is shown in Table 14-4, and examples of the CumHist are provided in Figures 14-1 and 14-2.

**TABLE 14-4 ASSAY CAPS AND RESOURCE CLASSIFICATION FOR TIER 3 PIT
Barrick Gold Corporation - Hemlo Mine**

Domain	Grade cap (g/t Au)
101	30
102	30
103	30
104	30
120	45
125	80
130	70
131	100
140	120
150	90
151	40
325	30
332	80
333	80
340	40
345	20
2000 (C Zone)	120
1000 (C Zone)	30
2000 (Sceptre)	55
1000 (Sceptre)	15
Diabase	0
0 (Waste)	10

FIGURE 14-1 GOLD HISTOGRAM

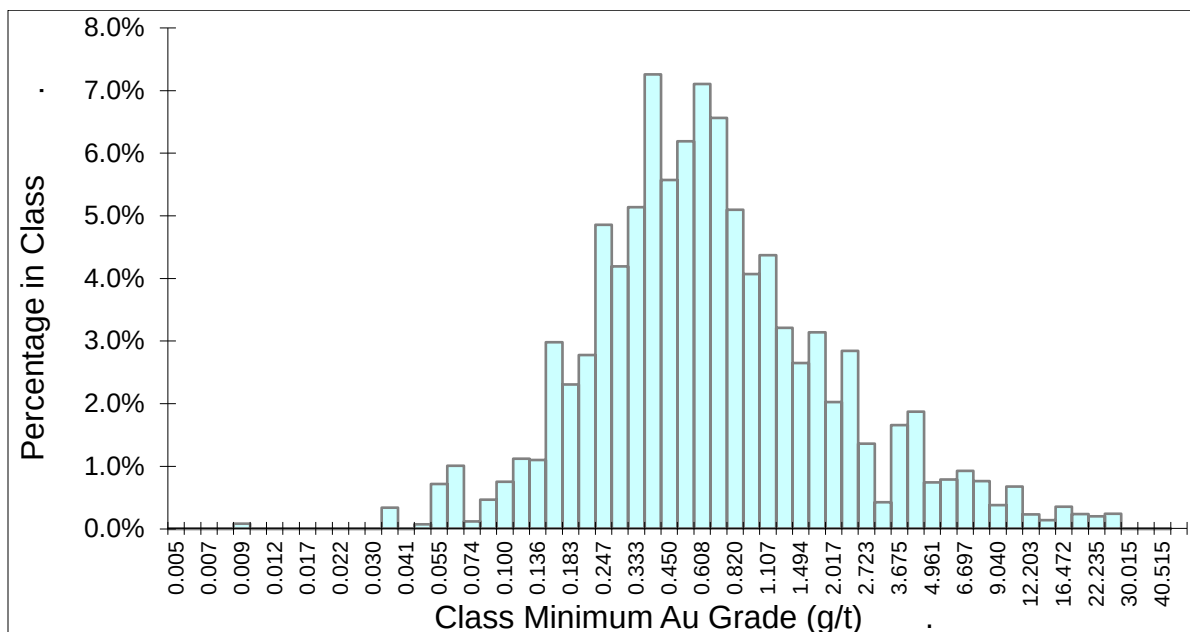
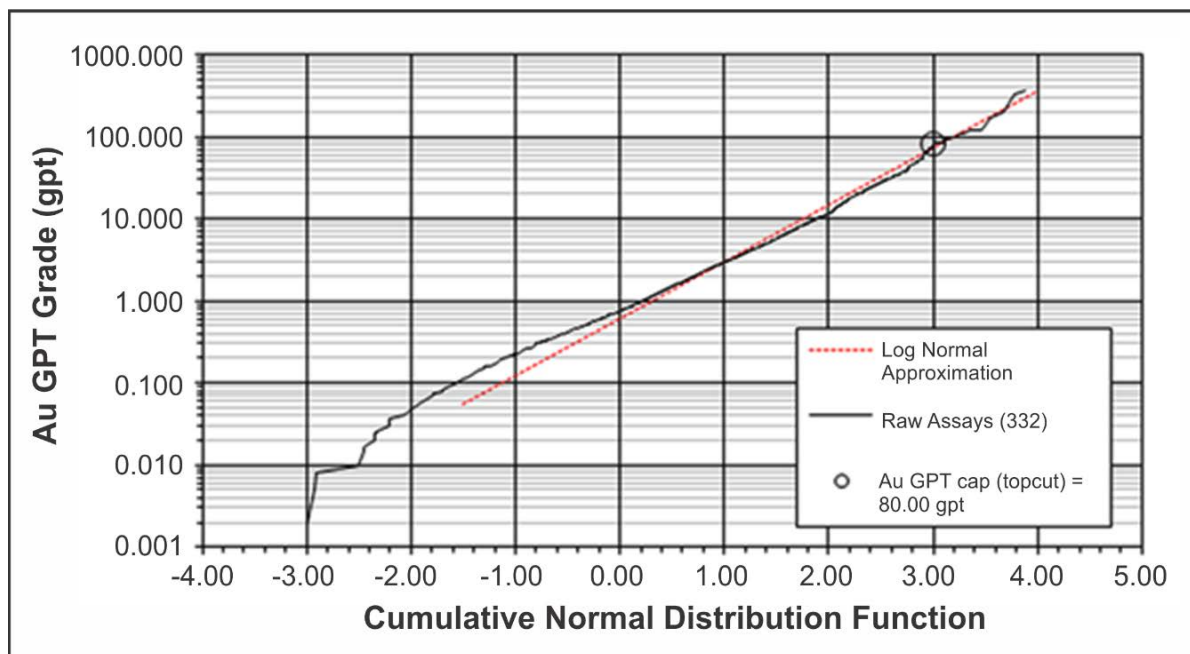


FIGURE 14-2 CUMULATIVE NORMAL DISTRIBUTION FUNCTION



VARIOGRAPHY

Omni-directional variography was used to analyse the individual ore lenses, high grade indicators, low grade indicators, and the waste domain. Barrick determined that zonal composites of capped gold grades were the best indicators for the search criteria for the grade estimation at the Hemlo Mine.

DENSITY DATA

Previously mined stopes and underground development as well as stopes planned to be mined were removed from the block model. The density of these areas was set as 1.9 to account for the assumption that these areas will be paste-filled.

Cone penetration tests performed on the tailings indicate that the tailings in the facility are generally greater than the assumed density of 1.4 t/m³.

A study was undertaken in 2007 and 2008 to determine the average density at Williams Mine. A total of 3,071 samples were first weighed in air and then placed in a container of water and the volume of water displaced (in mL) was measured. The following equation was used to derive the bulk density:

$$\frac{\text{Weight of sample (g)}}{\text{Volume of water displaced (mL)}} = \text{Density}$$

Average densities were tabulated for each resource area and for the mine as a whole. For the purposes of estimating Mineral Resources, a global bulk density of 2.7 tonnes per cubic meter was applied to the open pit block model.

GEOLOGICAL MODELS

Geological wireframes were constructed to constrain block model grade estimations as well as to provide mine planning parameters. All wireframes were constructed in Vulcan™ software. Wireframes models were updated for geological features based on additional drilling for each resource area.

The higher grade portions of the mineralization were modelled in section and plan using Vulcan software to produce 3-dimensional ore shapes. These correlate to zones mined underground. Because of the sporadic nature of the mineralization, high-grade and low-grade indicators were generated for the open pit block model to further constrain assays above 2.0 g/t Au (2000 Domain) and the assays above 0.2 g/t Au (1000 Domain) respectively.

OPEN PIT BLOCK MODEL

Parameters of the block model are shown in Table 14-5.

TABLE 14-5 OPEN PIT BLOCK MODEL PARAMETERS
Barrick Gold Corporation - Hemlo Mine

Blocks	2015 Model
Sub-block size (metres)	1.25 by 1.0 by 1.0 2.5 by 2.0 by 5.0 5.0 by 4.0 by 10.0
Re-blocked block size (metres)	10 by 8 by 10
Origin	X= 77,300, Y = 93,600, Z = 9,788
Number of Blocks	Sub-block: 45,645,507 Re-block: 2,142,450
Composite size (metres)	Varies: 1.0, 2.0, 4.0
Samples used (Indicated material)	Varies: CZ 1 min 3 max, Sceptre 4 min 6 max, Waste Domain 2 min, 6 max
Min. No. holes used (Indicated material)	Varies: CZ: 1-2, Sceptre 3, Waste Domain 2
Max No. samples/hole (Indicated material)	Varies: CZ 1, Sceptre 2, Waste Domain 2

GRADE ESTIMATION

A series of six passes were used to estimate block grades in each domain. As a block was estimated, it was flagged with the corresponding pass number in order to classify it.

The first pass estimated only blocks that were pierced by a diamond drill hole (box search). The second pass estimated within a search ellipse based on 80% of the sill of the variogram. The third pass, known as a “donut pass” used the closest sample to estimate within one-half of the search range of the second pass and was intended to estimate blocks which were

located in proximity to a single drill hole which were not estimated in Pass 2. Usually, this involved only a small percentage of blocks. The fourth pass estimated grades within a search ellipse based upon 90% of the sill of the variogram. The fifth pass was another “donut” estimation which estimated grades based on a single drill hole. The last pass was intended to show “blue sky” potential. The search region was based on the ranges used in Pass 4 multiplied by 1.5.

CLASSIFICATION

The mineralized material within the block model was classified as Measured, Indicated, or Inferred Resource, based on the estimation passes. This was written to a variable in the block model called Categ. Measured Resource (Categ = 1) was considered to be material estimated during Pass 1. That is, only material within the immediate vicinity of a diamond drill hole was classified as Measured Resource. Indicated Resource (Categ = 2) included material estimated during Passes 2 and 3. Within the modelled lenses 130, 131, 140, 150, 332, 333, Indicated Resources included blocks estimated within a 40 m search radius. Material estimated during passes 4 and 5 were classified as Inferred Resource (Categ = 3), except within the modelled lenses, where a 55 m search radius was used. Blue sky potential (Categ = 4) was shown by material estimated by Pass 6. Blue sky material was not included as Mineral Resource, and was used only to provide possible future exploration targets.

Table 14-6 defines the resource classification for the different Hemlo resource domains by search dimensions.

TABLE 14-6 MEASURED, INDICATED, AND INFERRED CLASSIFICATION BY DOMAIN
Barrick Gold Corporation - Hemlo Mine

Domain	Measured	Indicated	Inferred
101	0.625X0.50X0.50	30X30X3	50X50X3
102	0.625X0.50X0.50	30X30X3	50X50X3
103	0.625X0.50X0.50	30X30X3	50X50X3
104	0.625X0.50X0.50	30X30X3	50X50X3
120	0.625X0.50X0.50	30X30X3	50X50X3
125	0.625X0.50X0.50	30X30X3	50X50X3
130	0.625X0.50X0.50	40X40X3	55X55X3
131	0.625X0.50X0.50	40X40X3	55X55X3
140	0.625X0.50X0.50	40X40X3	55X55X3
150	0.625X0.50X0.50	40X40X3	55X55X3
151	0.625X0.50X0.50	30X30X3	50X50X3
325	0.625X0.50X0.50	40X40X3	55X55X3
332	0.625X0.50X0.50	40X40X3	55X55X3
333	0.625X0.50X0.50	40X40X3	55X55X3
340	0.625X0.50X0.50	30X30X3	50X50X3
345	0.625X0.50X0.50	30X30X3	50X50X3
2000 (C Zone)	0.625X0.50X0.50	50X50X3	75X75X3
1000 (C Zone)	0.625X0.50X0.50	50X50X3	75X75X3
2000 (Sceptre)	0.625X0.50X0.50	50X50X3	75X75X3
1000 (Sceptre)	0.625X0.50X0.50	50X50X3	75X75X3
Diabase	N/A	N/A	N/A
0 (waste)	0.625X0.50X0.50	50X50X3	75X75X3

VALIDATION

Reconciliation between the block model and actual mill output is discussed in Section 15, below. In the open pit, more tonnage was mined than was shown in the Mineral Reserves, at a decreased grade, to return roughly the same ounces. Dilution around narrow ore lenses was indicated in the pit, as well as finding ore through ore control drilling in the northeast part of the pit.

In the underground, both tonnage and grade were higher than modelled. There is a tendency to mine larger stopes than originally planned. This meant that in many cases, two or more ore lenses were taken together with internal waste pillars. Conservative grade caps in the Mineral Reserve estimation, however, ensured that generally higher grades were realized upon production.

RPA VALIDATION

All of the files provided by Barrick were originally constructed in Vulcan™ software. The Mineral Resource audit carried out by RPA utilized Vulcan™ as well.

The End of Year 2016 block models were validated by RPA using the following methods:

- Visual inspections for all coding and estimation steps;
- Visual comparison with composite data;
- Check estimation and comparison to the end of year 2016 Resource and Reserve (R&R) spreadsheets provided by Barrick
 - o Small difference can be attributed to the calculations and roll up numbers from the R&R sheets. All checks are within 2%.

Ongoing model reconciliation of the model and production show that the approach and procedures used to create the Hemlo open pit and underground resource model are adequate for mine planning purposes and are in line with industry standards. The models are created and documented by a qualified person and follow standard industry practice. The model procedure is well documented through the master Mineral Resource and Mineral Reserve spreadsheets which include records of visual checks and economics. RPA is of the opinion that the Mineral Resources as stated by Barrick are appropriate for the style of mineralization and that the resource model is reasonable and acceptable to support the December 31, 2016 Mineral Resource and Mineral Reserve estimates.

15 MINERAL RESERVE ESTIMATE

SUMMARY

The Mineral Reserves as of December 31, 2016, are presented in Table 15-1.

TABLE 15-1 MINERAL RESERVES – DECEMBER 31, 2016
Barrick Gold Corporation - Hemlo Mine

Category	Tonnes (000 t)	Grade (g/t Au)	Contained Gold (000 oz Au)
Williams Open Pit			
Proven	6	1.07	0.2
Probable	19,119	1.17	718.6
Total Pit	19,125	1.17	718.8
Williams Underground			
Proven	972	3.76	117.4
Probable	5,645	4.13	750.2
Total Underground	6,617	4.08	867.6
Stock Pile			
Proven	40	1.25	1.6
Probable			
Total Stockpile	40	1.25	1.6
Total Williams Mine	25,782	1.92	1,587.9

Notes:

1. CIM definitions were followed for Mineral Reserves.
2. Mineral Reserves are estimated using an average long-term gold price of US\$1,000 per ounce and a US\$/C\$ exchange rate of 1:1.30.
3. Underground Mineral Reserves are estimated at an average break-even cut-off grade of 2.92 g/t Au and an average Incremental cut-off grade of 2.43 g/t Au.
4. Open Pit Mineral Reserves are estimated at an average cut-off grade of 0.43 g/t Au.
5. Totals may not add due to rounding.

RPA is not aware of any mining, metallurgical, infrastructure, permitting, and other relevant factors which could materially affect the Mineral Reserve estimates.

CUT-OFF GRADES

For the Williams open pit operation, two cut-off grades are typically used: one for the delineation of the economic pit limit (break-even cut-off grade) and the other for the definition of material as ore or waste within the confines of the pit limits (incremental cut-off grade).

The difference between the two cut-off grades is that the break-even cut-off grade calculation for pit optimization includes all costs that impact the final tonne mined, whereas the incremental cut-off grade is used for defining ore/waste within the optimized pit limit and does not include mining operating costs. Once material has been identified as belonging within the economic pit limits the mining cost must be incurred whether the material is ore or waste and can be excluded from the equation, assuming the haulage cost for ore and waste is the same from the pit crest.

The 2016 year-end open pit reserve break-even cut-off grade of 0.52 g/t Au is based on a mining cost of \$4.70 per tonne moved, \$16.37 per tonne processed, and G&A costs of \$3.89 per tonne processed (G&A costs are apportioned between open pit and underground). The incremental open pit cut-off grade used for reporting Mineral Reserves is 0.43 g/t Au.

The underground break-even and incremental cut-off grades were estimated at 2.92 g/t Au and 2.43 g/t Au, based on an operating cost of \$87.18 per tonne (based on 1.2 Mtpa or approximately 3,300 tpd). Processing and G&A costs for the break-even cut-off grade calculation are \$10.99/t and \$13.22/t, respectively. Only the variable portion of the processing cost (\$4.78/t) is used in the incremental cut-off grade calculation.

The mining extraction rate utilized for both longhole and Alimak stoping was 90% with the exception of “one shot blast” (i.e., blind stopes that are blasting without the possibility of re-entry) where an extraction rate of 60% was applied. For longhole stopes that are drilled off with “uppers”, based on 14 m high stopes, an extraction rate of 85% was used.

It should be noted that each stope is evaluated separately to determine economic viability. Factors such as distance to the nearest orepass, the amount of capital development required, geotechnical risk and other operational factors are considered to ensure that each stope that is included in the reserve category is cash-flow positive.

CONVERSION TO MINERAL RESERVES

For the conversion of underground resources to underground reserves, the output from Vulcan was imported to an MS Excel® spreadsheet and the appropriate dilution and extraction factors were applied. The underground dilution factors, for various zones and methods, are shown in Table 15-2.

TABLE 15-2 UNDERGROUND MINING DILUTION
Barrick Gold Corporation - Hemlo Mine

Category	Stope Width (m)	Dilution HW (m)	Dilution Grade (g/t Au)
Williams Mine			
Longhole/Alimak	<15.0	1.25	0.50
Longhole/Alimak	>15.0	1.50	0.50

The average dilution adds up to approximately 22%. There is no blending underground for grade control and blasthole cuttings are not assayed. Grade control is carried out mainly through diamond drill assays, plus chip and channel samples taken in the drifts.

For open pit, dilution and extraction are addressed in the block model by using whole blocks at a size of 10 m x 8 m x 10 m.

The Mineral Reserve estimate by area and zone is shown in Table 15-3.

TABLE 15-3 MINERAL RESERVES BY ZONE – DECEMBER 31, 2016
Barrick Gold Corporation - Hemlo Mine

Description	Proven			Probable			Proven & Probable		
	Tonnes (000)	Grade (g/t Au)	Oz (000)	Tonnes (000)	Grade (g/t Au)	Oz (000)	Tonnes (000)	Grade (g/t Au)	Oz (000)
Williams Mine									
Open Pit									
C Zone Pit Phase 5	-	-	-	4,766	1.18	181.4	4,766	1.18	181.4
C Zone Pit Phase 6	6	1.07	0.2	8,448	1.27	345.8	8,454	1.27	346.0
Sceptre Pit Phase 5	-	-	-	1,444	1.21	56.2	1,444	1.21	56.2
Sceptre Pit Phase 6	-	-	-	4,461	0.94	135.1	4,461	0.94	135.1
Sub-Total Pit	6	1.07	0.2	19,119	1.17	718.6	19,125	1.17	718.8
Underground									
B Zone	95	3.09	9.4	205	3.59	23.6	299	3.43	33.0
B Zone FW	-	-	-	88	3.59	10.2	88	3.59	10.2
B Zone Below 9160	-	-	-	1,118	3.86	138.8	1,118	3.86	138.8
C Zone Long Hole	209	3.88	26.1	1,214	4.05	158.1	1,424	4.03	184.3
C Zone Alimak	152	3.76	18.4	345	3.63	40.2	497	3.67	58.6
Interlake	516	3.83	63.4	1,813	4.12	240.3	2,329	4.06	303.7
Sceptre Upper	-	-	-	177	3.69	21.0	177	3.69	21.0
Beth Adit	-	-	-	157	4.14	20.9	157	4.14	20.9
Sceptre Lower	-	-	-	528	5.72	97.1	528	5.72	97.1
Sub-Total UG	972	3.76	117.4	5,645	4.13	750.2	6,617	4.08	867.6
Stockpiles									
ROM + Crush Stockpile	40	1.25	1.6	-	-	-	40	1.25	1.6
Low Grade Stockpile	-	-	-	-	-	-	-	-	-
Stockpiles Sub-Total	40	1.25	1.6	-	-	-	40	1.25	1.6
Total Williams Mine	1,018	3.64	119.2	24,764	1.85	1,468.7	25,782	1.92	1,587.9

Notes:

1. CIM definitions were followed for Mineral Reserves.
2. Mineral Reserves are estimated using an average long-term gold price of US\$1,000 per ounce and a US\$/C\$ exchange rate of 1:1.30.
3. Underground Mineral Reserves are estimated at an average break-even cut-off grade of 2.92 g/t Au and an average Incremental cut-off grade of 2.43 g/t Au.
4. Open Pit Mineral Reserves are estimated at an average cut-off grade of 0.43 g/t Au.
5. Totals may not add due to rounding.

RPA verified the Mineral Resource to Mineral Reserve conversions carried out by Hemlo engineering staff. This included stope dimensions, dilution factor (metres from HW), dilution grade, average stope widths, diluted tonnes, calculated percent dilution per stope, and mine recovery factors per stope.

RECONCILIATION

The mine reconciliation at year end 2016 is shown in Table 15-4. This compares the tonnage, grade, and ounces produced from the Mineral Reserves model, Grade Control model, and Mill (DOM), for the Williams Mine operation.

TABLE 15-4 WILLIAMS YEAR END RECONCILIATION 2016
Barrick Gold Corporation - Hemlo Mine

Williams Pit	Tonnes	Grade	Ounces
DOM	2,193,236	1.304	91,934
Grade Control	2,129,497	1.232	84,342
Mineral Reserves	2,057,894	1.334	88,244
Williams Underground	Tonnes	Grade	Ounces
DOM	912,715	4.711	138,249
Grade Control	914,924	4.016	118,140
Mineral Reserves	861,454	4.005	110,920
Williams Total	Tonnes	Grade	Ounces
DOM	3,105,951	2.305	230,183
Grade Control	3,044,421	2.069	202,482
Mineral Reserves	2,919,348	2.122	199,164
Williams Pit	Tonnes	Grade	Ounces
DOM vs Grade Control	3.0%	5.8%	9.0%
DOM vs Mineral Reserves	6.6%	-2.2%	4.2%
Grade Control vs Mineral Reserves	3.5%	-7.6%	-4.4%
Williams Underground	Tonnes	Grade	Ounces
DOM vs Grade Control	-0.2%	17.3%	17.0%
DOM vs Mineral Reserves	6.0%	17.6%	24.6%
Grade Control vs Mineral Reserves	6.2%	0.3%	6.5%
Williams Total	Tonnes	Grade	Ounces
DOM vs Grade Control	2.0%	11.4%	13.7%
DOM vs Mineral Reserves	6.4%	8.6%	15.6%
Grade Control vs Mineral Reserves	4.3%	-3.5%	1.7%

Reconciliation to 2016 production indicates that the open pit resource model is performing well within the normal range of variability (+/- 10%), and that the underground resource model is conservative. Overall, Mineral Reserve estimates are within 10% of production results for tonnes and grades.

RPA recommends a review of 2016 production versus underground resource model results within the same volume, to assess model performance independent of other estimating factors. Consider adjusting some model input parameters to bring the conservative estimate closer to production results.

16 MINING METHODS

Mining at Hemlo consists of a combination of open pit and underground operations at current process production rate of approximately 9,500 tpd. The present split between the open pit (including stockpile) and underground tonnage is 72% and 28%, respectively, during 2016, while the open pit (including stockpile) supplied 40% of the produced ounces and the underground, 60%.

The underground mine at Williams is accessed via a vertical shaft equipped to move personnel and material, and also ore and waste from the underground to the surface. The Williams Mine is equipped with a 4.6 m diameter double drum hoist with 22 t capacity skips and a 3.7 m diameter service hoist with cage and counterweight.

OPEN PIT

During 2012, a Feasibility Study was completed on the open pit expansion at the Williams Mine. The open pit expansion would have added approximately 42.4 million tonnes grading approximately 1.10 g/t Au to Mineral Reserves, and would have extended the mine life from 2017 until 2029. During 2013, low gold prices forced the pit expansion to be suspended. During 2014, plans were made to mine the most readily accessible material within this resource and resulted in a small lay-back on the south wall (Phase 5). Shortly thereafter, studies began for additional lay-backs on the north wall, which would allow access to deeper ore. A Pre-Feasibility Study on Phase 6 was completed in 2016, and Barrick will proceed with a Feasibility Study this year.

Production from open pit mining (and stockpiles) makes up 74% of the mill feed in the current LOM plan. Over the next few years of Phase 5 mining, production will average approximately 5,700 tpd or 2.1 million tonnes per year ore, with 12,000 tpd of waste for a yearly average of five million tonnes of ore and waste. Near term open pit production will come predominantly from the C Zone, which is the main source of ore from the pit.

Phase 6 is expected to utilize the open pit production capacity as Phase 5 winds down, averaging a rate of approximately 5,200 tpd or 1.9 million tonnes per year ore. Phase 6 has the potential to extend the life of the open pit to 2027.

The open pit presently has five (5) CAT777D and two (2) CAT777B haul trucks. Loading is done with two (2) CAT992 loaders; in addition, a CAT330 and CAT365 excavator are employed to clean the bench faces. Production drilling is carried out using 6.75 in. diameter holes for a bench height of 10 metres, with one metre of subgrade drilling. The drilling is carried out using Atlas Copco PV235 rotary type drills.

The open pit powder factors are 0.50 kg/t for ore and 0.47 kg/t for waste and the explosive is a blend of 70% emulsion and 30% ANFO. The use of the electronic type of detonators (iKons), for the harder 300-series ore, has improved fragmentation, through precision timing. Surface drilling patterns in ore have a burden of 3.5 metre burden and 4.0 metre spacing, whereas waste patterns have a 4.0 metre burden and 4.0 metre spacing in waste.

Significant improvements in throughput rates for the open pit have been obtained over the last five years due to improvements in blasting practices, including the use of RFID tagged “blasting balls” for blast-movement monitoring. The “Drill to Mill” initiative resulted in significant improvements to throughput.

The present open pit status is shown in Figure 16-1, with Phases 5 and 6 overlaid.

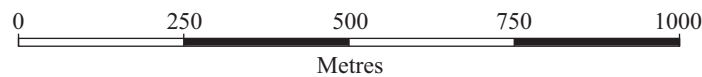
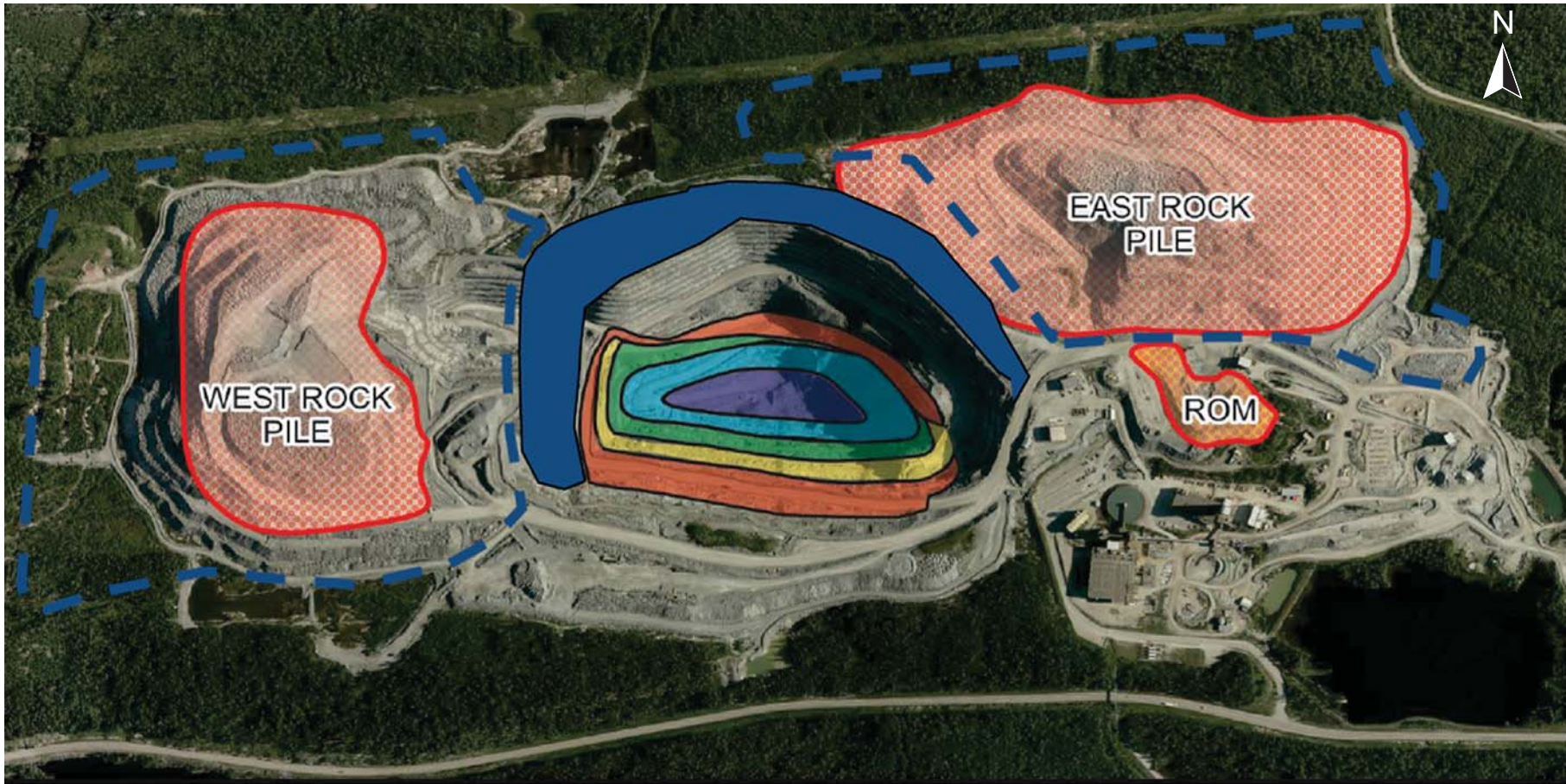


Figure 16-1

Barrick Gold Corporation

Hemlo Mine
 Bomby Township, Ontario, Canada
Williams Open Pit -
December 2016

MINE DESIGN

Mine models are developed using Maptek's Vulcan software, which is then output to Whittle 4X that employs the Lerchs-Grossmann pit optimization algorithm. Whittle produces a series of pit shells based on multiple gold prices. The design of the phases is based on taking the most profitable pit shells first. The phase designs are completed using Vulcan mine planning software. These are all well recognized software packages, and are commonly used for open pit mine optimization.

The pit design is based on 10 m bench heights. Slope angles vary based on pit location. Table 16-1 shows the general final pit slope design criteria discussed later in this chapter.

TABLE 16-1 OPEN PIT MINE DESIGN PARAMETERS
Barrick Gold Corporation - Hemlo Mine

Design Parameter	Value, Units
Haul Road Width	26 m
Haul Road Grade	10%
Mining Bench Height - Waste	10 m
Mining Bench Height - Ore	10 m
Nominal Catch Bench (Berm) Width (north, east, and west wall)	11 m
Nominal Catch Bench (Berm) Width (south wall)	8 m
Minimum Operating Width	26 m
Benching (berm) Cycle	Double
Bench Face Angle – North Wall	90°
Bench Face Angle – South Wall	65°
Nominal Pit Slope Design Angle – North Wall	61°
Nominal Pit Slope Design Angle – South Wall	49°
Pit Bottom Elevation	78 MASL
Pit Exit Elevation	357 MASL
Typical Sinking Rate	74 m/yr
In Pit Haul Distance	2,500 m
West Mine Rock Pile Elevation	450 MASL
East Mine Rock Pile Elevation	440 MASL
Mine Rock Pile Slope Angles	19° to 25°

GEOMECHANICS AND DEWATERING

The geotechnical analysis to develop the open pit slope design parameters for the Hemlo Mine was completed by Golder in 2011 and 2012. In October 2011, Golder conducted pit slope studies for the Williams open pit. The studies were comprehensive in nature and involved

assessments of the double benching (hanging wall) and triple benching (footwall) techniques, structural analyses, blasting practices, and proposed deepening of the pit by approximately 150 m, as well as a review of the pit slopes used in the final mine design.

Rock slopes in the Williams open pit have been documented to be dry, strong, competent, blocky meta-sedimentary and meta-volcanic rocks with foliation that dips steeply to the north. There are two major structures, three minor structures, and dykes that intersect the pit. Foliation changes dip between two of the dykes, and this controls the slope design on the south wall. To date, no significant movement in the pit walls has been recorded and the pit depth from the surface is currently at 210 m. Wall monitoring by surveyed prisms and MPBX cables was introduced in 2014, and a radar monitoring systems was recently installed.

Care is applied to the drilling and blasting to prevent damage to the highwalls. Presplit and trim drilling is utilized in the drilling and blasting procedures to reduce excessive vibration into the highwalls.

As previously noted, there are existing underground workings below the Williams open pit. There may be some localized instability created as the open pit mines through some of the existing underground workings. Any instability will be addressed by the operations department at the time of mining. Dewatering of the open pit will be required by dewatering wells or surface sumps, to minimize the influx of water to the underground.

AMEC provided a detailed design for the Mine Rock Pile (i.e., rock waste dumps) construction, which accounted for the handling of potentially acid generating (PAG) waste material.

OPEN PIT MINE EQUIPMENT

As previously noted, the Hemlo open pit mine is a typical truck and loader/hydraulic shovel operation. The fleet is shown in Table 16-2.

Mine mobile equipment production rates were reviewed with availability and utilization to determine whether mining production rates and costs are appropriate. In RPA's opinion, there are sufficient trucks, loaders, and support equipment in the current LOM design plan to meet the budgeted production requirements.

TABLE 16-2 OPEN PIT MINE EQUIPMENT FLEET
Barrick Gold Corporation - Hemlo Mine

Number of Units	Manufacturer	Model	Description
1	CAT	992G	Wheel Loader
1	CAT	992K	Wheel Loader
1	CAT	998K	Wheel Loader
2	CAT	777B	Haul Truck
5	CAT	777D	Haul Truck
1	CAT	D9T	Track Dozer
1	CAT	D8R	Track Dozer
1	CAT	D7G	Track Dozer
1	CAT	14H	Motor Grader
1	CAT	14M	Motor Grader
1	CAT	140G	Motor Grader
1	CAT	140M	Motor Grader
1	Atlas Copco	D65	Utility Drill
1	Ingersoll Rand	DM45	Production Drill
2	Atlas Copco	PV235	Production Drill
1	CAT	966	Utility Front End Loader
1	CAT	330BL	Track Excavator
1	CAT	365CL	Track Excavator
1	CAT	336	Track Excavator
1	CAT	CT660	Sander/ Aggregate Truck
1	CAT	773B	Water Truck
1	CAT	420E	Rubber Tire Back Hoe
1	International Harvester	Workstar	Plow Truck/ Sander/ Water Truck

UNDERGROUND MINE

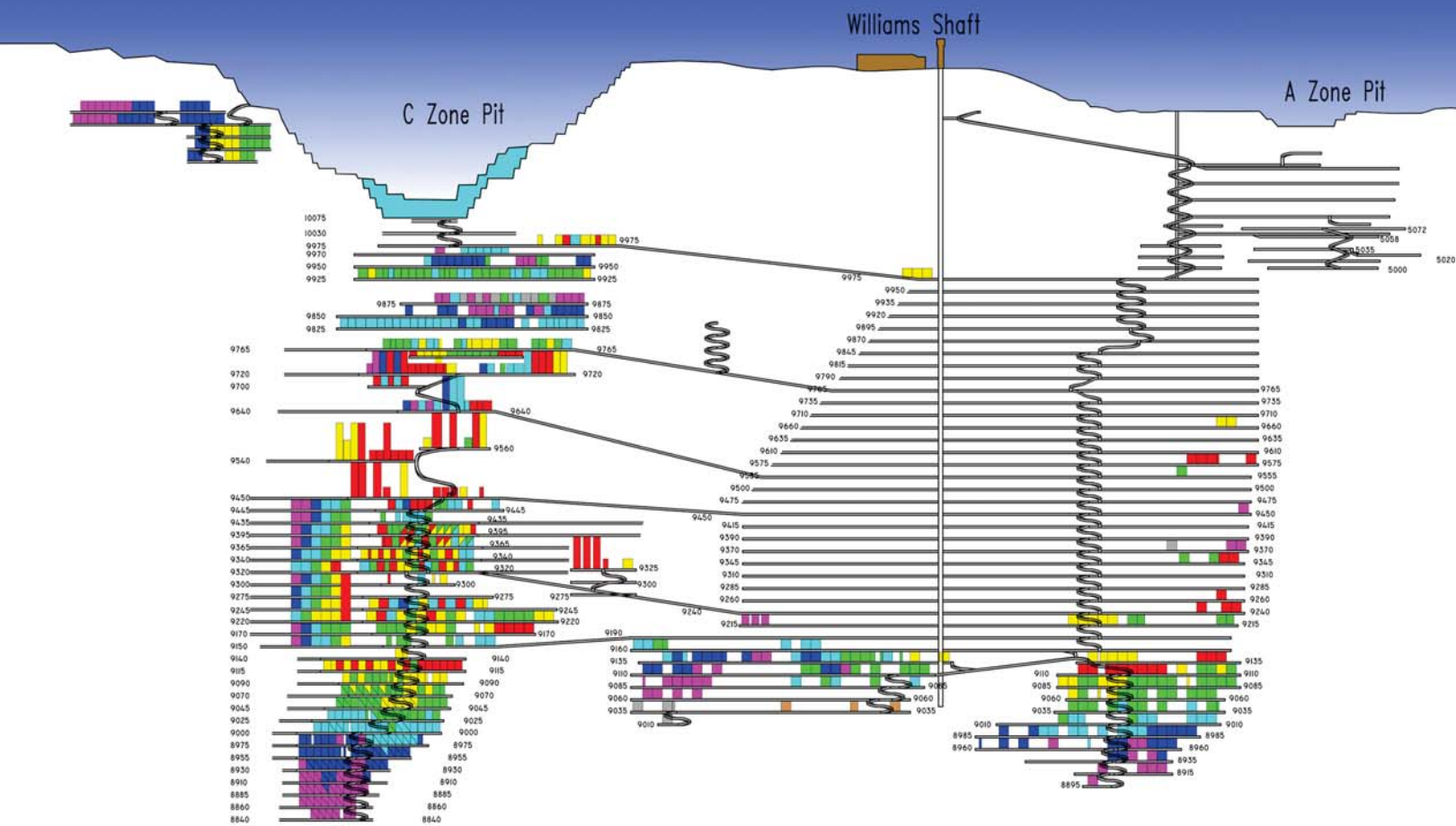
A long section illustrating the location of the underground Mineral Reserves is provided below, in Figure 16-2.

UNDERGROUND MINE DESIGN AND MINING METHOD

Two stoping methods are utilized at the Williams Mine, including the longhole and Alimak methods. Longhole “Longitudinal” stopes are typically 20 m to 25 m high and 15 m to 20 m along strike, with an average ore width of three to eight metres. Longhole “Transverse” stopes are between 15 m to 20 m along the strike, typically 25 m to 30 m high and stope widths ranging from three to thirty metres wide. Production drilling is performed with ITH (In-the Hole) drills using 4.5 in. diameter holes or Stopemaster drills using 3 in. diameter holes.

Alimak stoping, involves mining from the Alimak platform, after the slot raise has been driven in ore and the ground support installed. The Alimak method is a specialized method and carried out by a mining contractor (Manroc Mining). Generally speaking, the Alimak method includes driving a three metre by three metre raise to create an initial opening in the ore. The raise is driven to breakthrough on an upper level or it can be stopped without breakthrough. The hanging wall is then cable bolted, providing support in the middle of the raise opening. Once the support is completed, a longhole drill is set up on the Alimak platform and the stope is drilled off, along strike. If the stope length is 14 m, for example, the drill holes are drilled for seven metres from each side of the raise centre line. Once completed the longhole rings are loaded with explosives and blasted, either in a series of rings (i.e. two to four rings) or the stope can be blasted entirely in two shots using high precision electronic delays such as the iKon delays.

The inherent advantage with Alimak stopes is the unlimited stope height that can be achieved. At Williams, stope heights up to 180 m have been used. The Alimak technique is used mainly in the C Zone area between the 9450 level and the C Zone open pit. The Alimak stopes are typically 14 m to 15 m in length and a 1.5 m pillar is left between the stopes in order to comply with the ministry regulations regarding drilling near previously blasted ground. The Alimak now makes up approximately 30% of the mining tonnage, declining to 10% by 2020.



Mining Sequence:

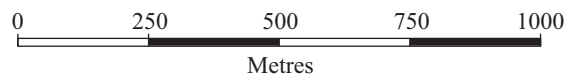


Figure 16-2

Barrick Gold Corporation

Hemlo Mine

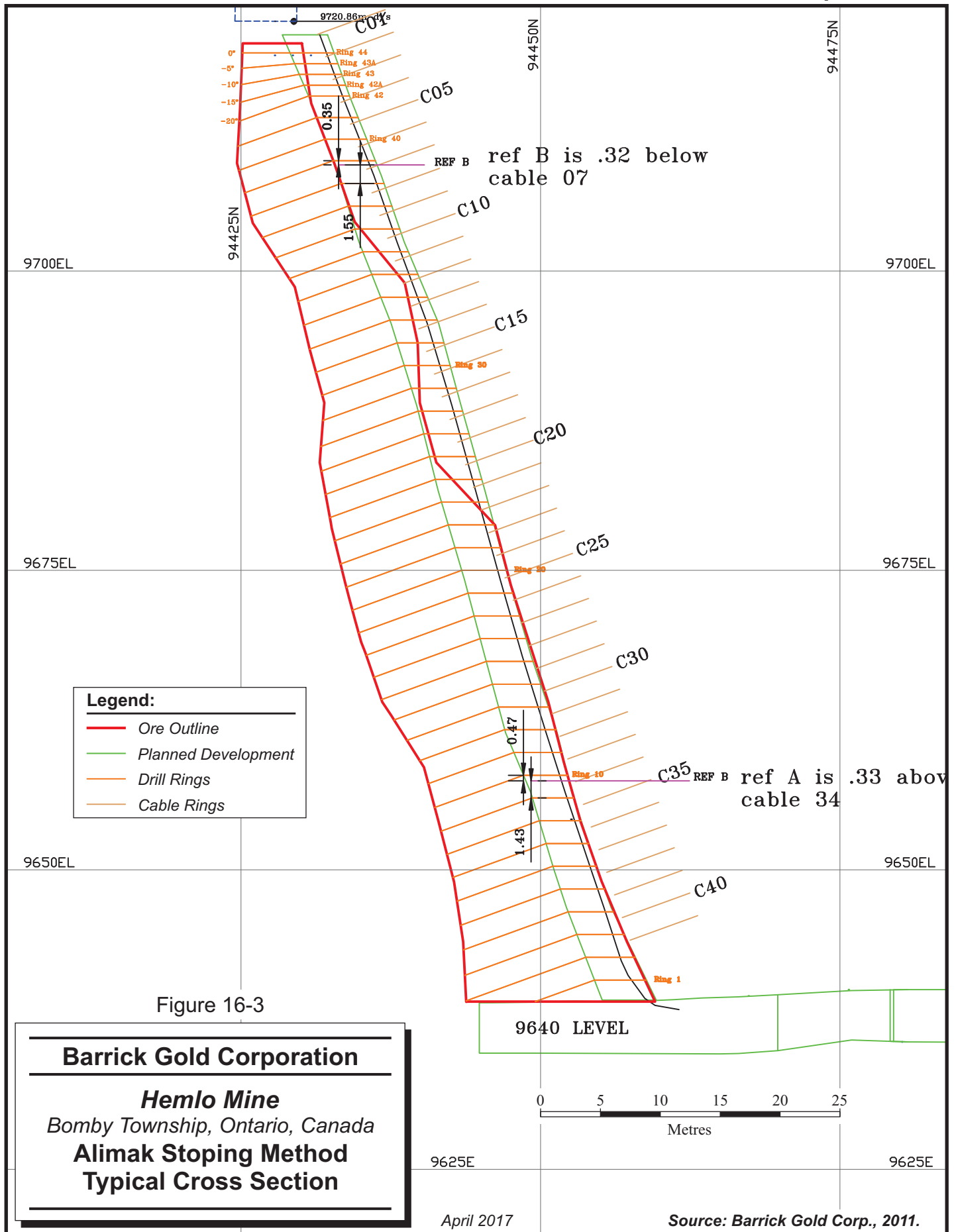
Bomby Township, Ontario, Canada

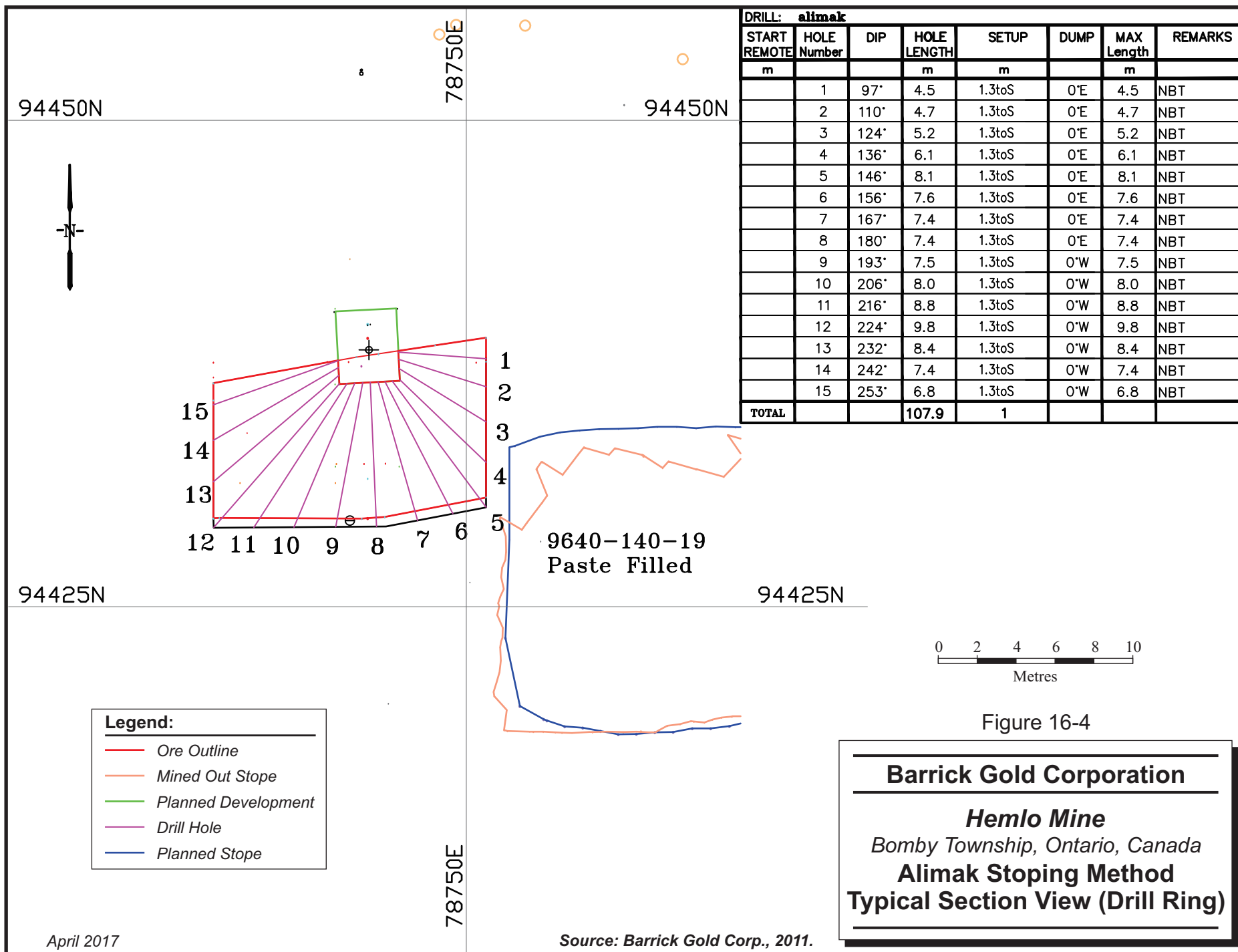
**Underground Long Section
(Looking North)**

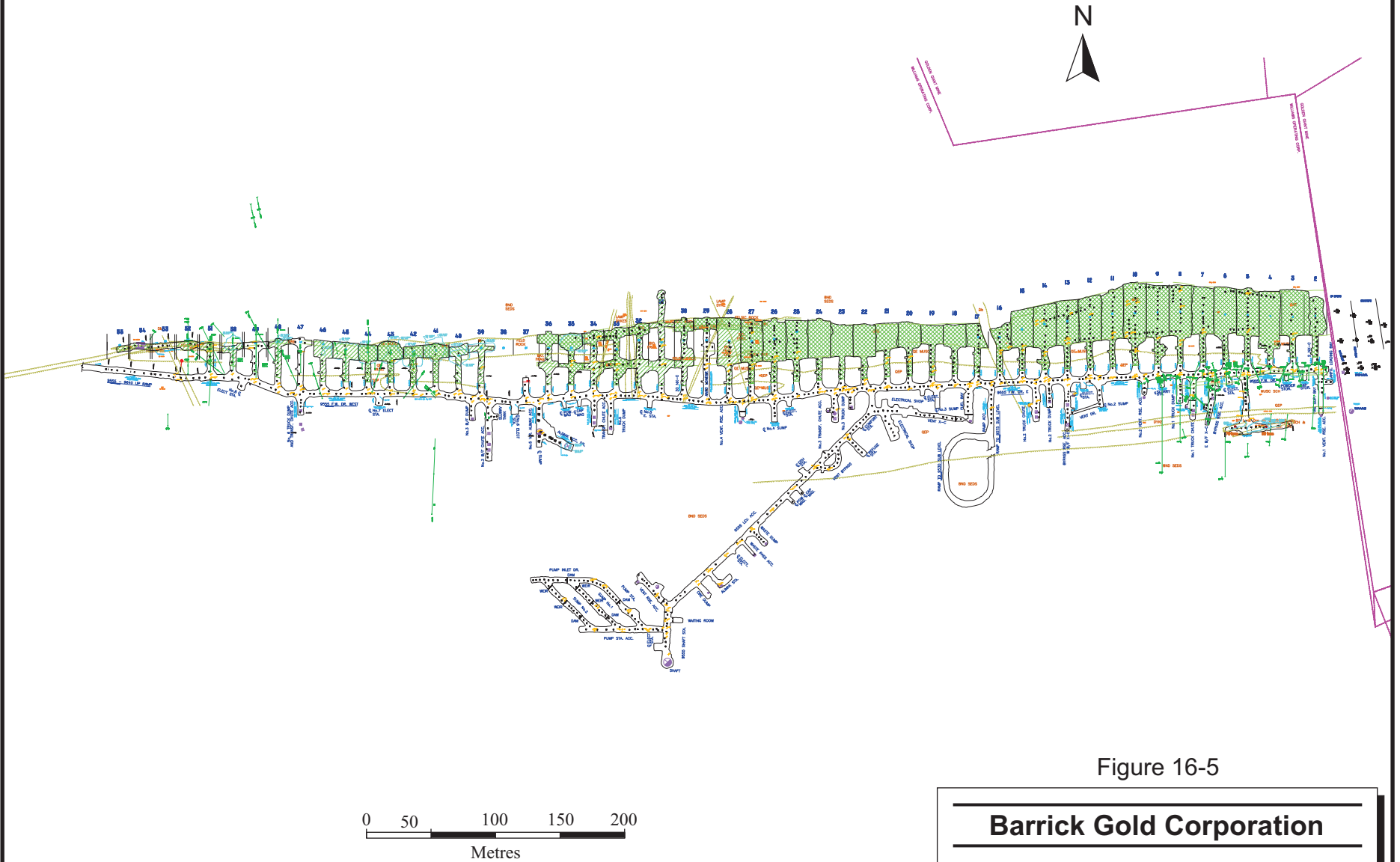
Alimak production drill holes are 2.5 in. in diameter with rings at 1.8 m intervals. Emulsion explosives are used to load the holes and Nonel delays are used unless there are poor ground conditions when the iKon electronic delays are utilized. The planned production drilling rate is 150 m per drill shift. When the stopes have been mucked (by Williams Mine employees) the stopes are typically backfilled with paste fill. The Alimak stoping method is shown in Figure 16-3 and 16-4. These show a cross section of the raise and stope outline and a plan view of a drill ring in the stope. The Transverse longhole stopes are typically 25 m in height and drilled with 3 in. diameter holes (for uppers) or 4.5 in. diameter holes for downholes. Burden on the rows drilled is two and a half to three metres.

Other types of blasthole stopes are used at Hemlo, including “Panelled Stopes”, where the stope is mined in two panels; a hanging wall panel, which is followed later by a footwall panel. This type of mining sequence reduces the back and wall hydraulic radii, which permits installation of sufficient cable bolts support to maintain stability. The interface between hanging wall and footwall may be vertical or inclined. Also, “Longitudinal Stopes” are used, where the ore zones are narrow (less than eight metres wide) and typically accessing two to four stopes from one cross cut. Mining then retreats towards the access crosscut. The stopes are then filled and hanging wall control is maintained by controlling the stope dimension on strike.

A typical level plan (9555 level) is shown in Figure 16-5. The main shaft, the inter-level ramp access and development in the footwall is shown as well as the stope dimensions and arrangement. A typical transverse stope longitudinal section and cross section is shown in Figures 16-6 and 16-7.



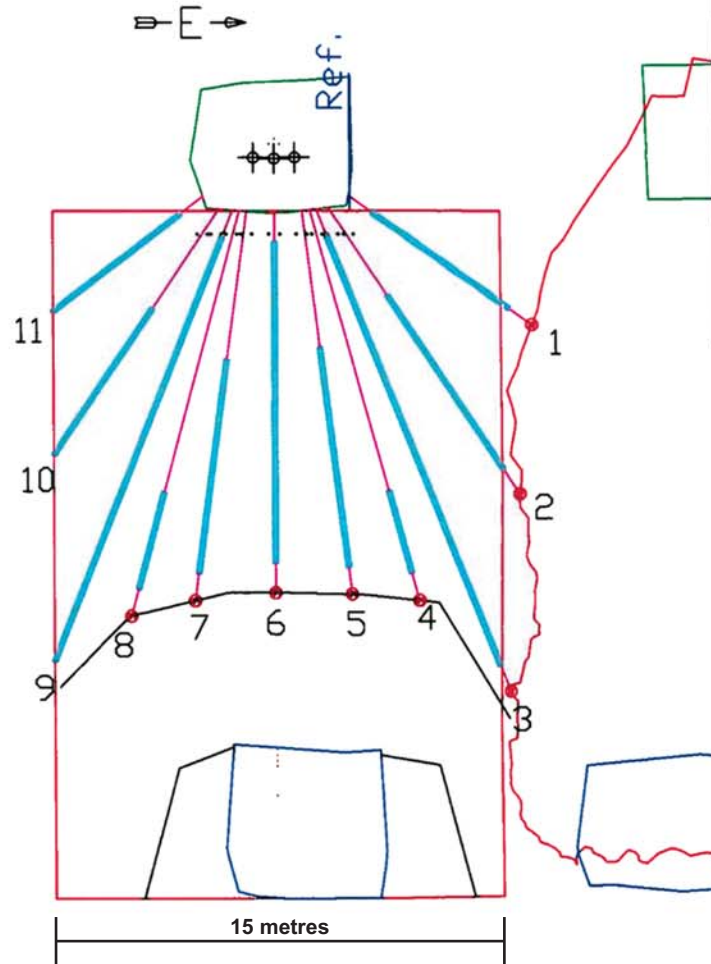




Barrick Gold Corporation

Hemlo Mine
Bomby Township, Ontario, Canada

**Typical Underground Level Plan
(9555 Level)**



LOADING INSTRUCTIONS						
HOLE	CAP	POWDER TYPE	kg	LOAD (m)		REMARKS
				From	To	
1		EMULSION	58.1	6.6	1.0	BF
2		EMULSION	72.6	10.6	3.6	BF
3		EMULSION	161.9	16.6	1.0	BF
4		EMULSION	29.1	12.7	9.9	BT
5		EMULSION	76.8	12.1	4.7	BT
6		EMULSION	112.1	11.8	1.0	BT
7		EMULSION	74.7	12.3	5.1	BT
8		EMULSION	34.2	13.2	9.9	BT
9		EMULSION	159.5	16.4	1.0	NBT
10		EMULSION	61.4	10.0	4.1	NBT
11		EMULSION	56.0	6.4	1.0	NBT
TOTAL			896			

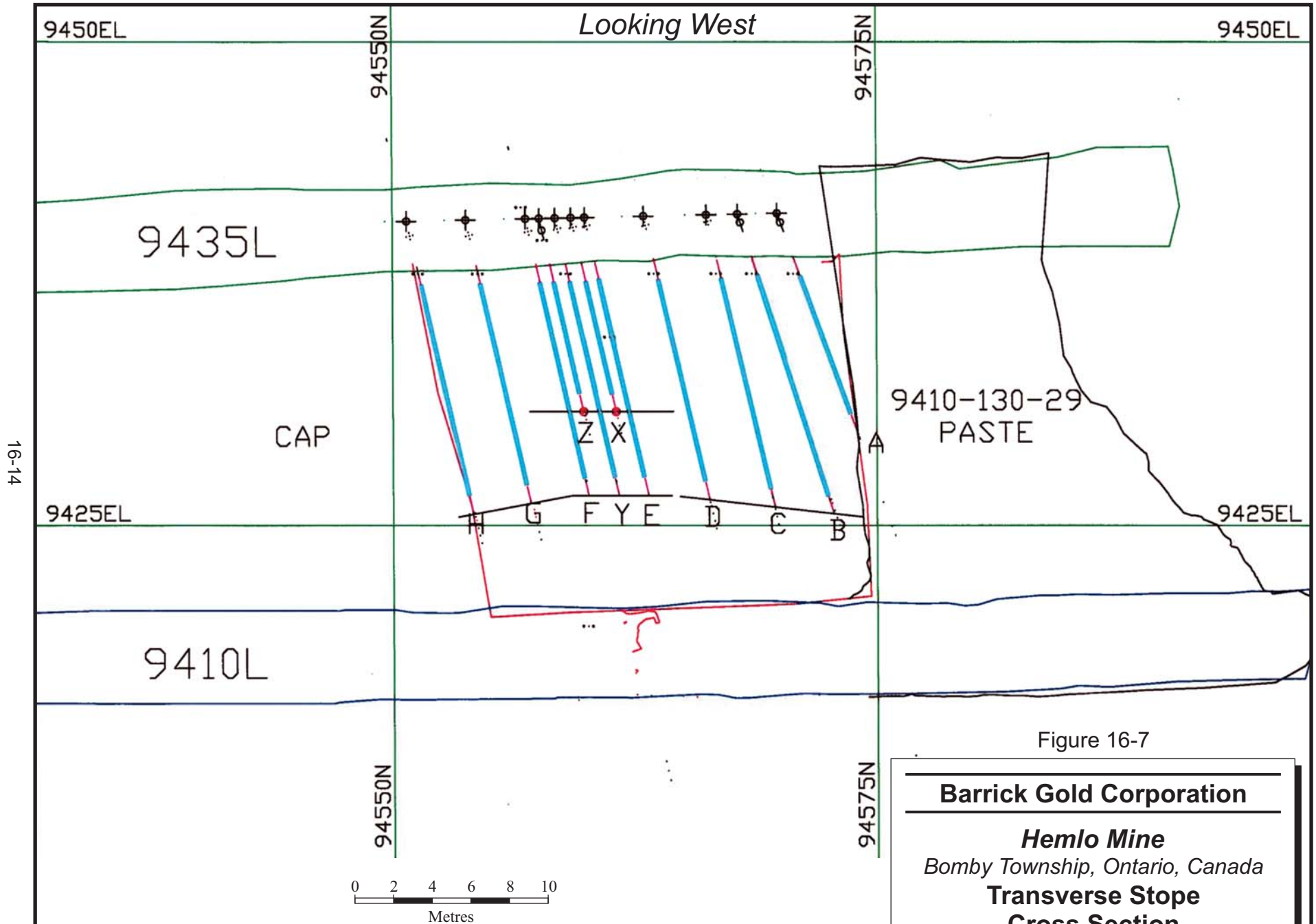
Figure 16-6

Barrick Gold Corporation

Hemlo Mine

Bomby Township, Ontario, Canada

**Typical Transverse Stope
Long Section**



Barrick Gold Corporation

Hemlo Mine
 Bomby Township, Ontario, Canada
**Transverse Stope
 Cross Section**

GEOMECHANICS AND GROUND SUPPORT

The Williams Mine has a comprehensive geomechanics and ground control/support programs in place. Ground support standards have been developed for a multitude of situations including the following:

- Main ramp and cross-cut support.
- Footwall and hanging wall drift support.
- Mining through cemented paste fill.
- Mining through or under cemented rock fill (CRF).
- Cable bolt support for various stope overcut configurations (transverse, parallel, panelled, longitudinal).
- Support in Alimak raises/stopes.

The support types include the following:

- Five and seven foot (3/4 in. dia.) resin rebar on 1.1 m x 1.1 m patterns for medium to permanent openings.
- Fiberglass resin rebar (7/8 in. dia.) eight feet and 12 feet long for Alimak development faces.
- Split Sets five feet long for wall rehabilitation in highly stressed fractured ground and 18 in. length used for utility hangers.
- Cable bolts six metres and nine metres “double bulge” in Alimak and longhole stopes and small to medium stopes (~20 m to 25 m), longitudinal mining, drawpoints, intersections.

Ground control at the Williams Mine also includes seismic monitoring. Response to a seismic event includes, reporting to the supervisor concerned, and for production blasting in a seismically active area, a re-entry protocol has been established which involves a number of checks before mining activities can resume.

Prior to adopting a particular mining method and sequence of mining, the proposed sequence is subjected to 3-D numerical modelling to assess the effects of redistribution of stresses in the adjoining area which may affect the stability of the stopes. Each of the proposed options is compared for the conditions of ground stability and the best option is selected.

The intact rock uniaxial compressive strength (UCS) for typical ore at Williams Mine is 150 MPa, while the average rock mass strength is estimated to be approximately 50 MPa. After the peak stress is exceeded, the rock fails and transfers to the “post-peak” regime. In the post-peak regime, the stresses in the rock mass decrease as it undergoes increasing damage and deformation. The effect of stress damage has been incorporated into the Mathew’s stability graph while designing the stope geometry and sequence of mining. The stress factor “A” in the stability graph is a linear function of boundary stress and intact rock strength. This factor remains constant at 0.14 after the rock mass strength is exceeded. The value of the resource dictated a mining plan aimed at complete extraction. Numerical modelling with three and two dimensional methods was initially used to examine various mining alternatives. Corresponding to the calculated modified stability number, the hydraulic radius (area/perimeter) for each wall of the stope is calculated to determine a suitable and most stable geometry.

The Williams Mine primary geomechanical consultant, Dr. W. F. Bawden, has conducted numerous site inspections and written many papers on the geomechanical properties of the Hemlo camp. His findings are that the hydraulic radius changes significantly from the B Zone, C Zone, and Interlake areas. The allowable hydraulic radii for the hanging wall of the B Zone and C Zone are respectively 4.25 and 2.75. Initial estimates give the Interlake an upper hydraulic radius bound of 4.0 and a lower bound of 2.5 for the hanging wall. It has been found that stope sequencing has a tremendous effect on the hanging wall failure rates in the Interlake area. These findings have now been incorporated into the mine plan.

BACKFILL

Approximately 30% of the mill tailings are used to prepare paste backfill for the Williams underground mine. The Williams backfill plant has a capacity of 100 m³/hr. A typical mixture includes 1.5% cement and 1.5% fly ash, up to 4% each. The paste fill is delivered at 75% to 80% solids.

MINE EQUIPMENT

The mine equipment used underground at the Williams Mine is shown in Table 16-3.

TABLE 16-3 UNDERGROUND MINE EQUIPMENT (WILLIAMS MINE)
Barrick Gold Corporation - Hemlo Mine

Description	Manufacturer	Model	No. Units
Compressors-Booster	Cubex	700	4
LHD- 9 Yd.	Elphinstone	R1700G	7
LHD- 7 Yd.	Elphinstone	R1600G	3
LHD- 7 Yd.	Tamrock	EJC-210	1
LHD- 2Yd.	EJC	JS 220	1
LHD- 2Yd.	RDH	MM250D	6
Truck- 26 T	EJC	JDT426	2
Truck- 30 T	Caterpillar	AD 30	6
Truck- 40 T	Toro	D 40	3
Jumbo Drill 2B	Tamrock	Minimatic	2
ITH Drill	Cubex	various	5
Production Drill	Boart	Stope Master	1
Bolter	RDH	RDH-200	4
Bolter	Boart	R712	1
Block Holer	MacLeans	BH3	1
Cable Bolter	Cubex	Gemini	1
Scissor Lift	Teledyne	M-60SL	4
Scissor Lift	Marcotte	Ramlift 6000	2
Scissor Lift	Getman	A-64	1
ANFO Loader	Getman	A-64	2
Boom Truck	MacLeans	MFM-977	1
Grader	Caterpillar	various	2
Mobile Rockbreaker	Teledyne	TM15HD/925X	1
Fork Lift	EJC	EJC-130	1
Backhoe/Man Basket	Kubota	R520S-FBH	1
Fuel/Lube Truck	MacLeans	MEM-966	1
Bulk Emulsion Truck	Marcotte	Truck	1
Concrete Hauler	Marcotte	M50	2
Shotcrete Boom	Normet	6050W	1
Tractor	various	various	47

UNDERGROUND INFRASTRUCTURE

DEWATERING

The underground mine water distribution system consists of Techno Sub pumps in up to 13 stages. The water is pumped from the mine workings and sent to the sedimentation basin on surface. Main sumps are located on the 9030, 9555, and 9975 levels off of the main shaft. There is very little discharge from the Hemlo Mine with the Actiflo system that permits the recycling of water for use in backfill placement.

VENTILATION

Ventilation of the underground workings is provided via surface fans that provide some 1.4 million cfm to adequately ventilate the mine equipment and underground installations. Fresh air downcasts via the main shaft and a separate ventilation raise in the B Zone and flows through the mine workings towards the A Zone Shaft and ramp, up to surface. There is a system of internal exhaust ventilation raises that permit adequate exhausting airways. Refuge stations are located strategically underground to ensure safe areas for personnel in case of emergency. A schematic of the ventilation system is shown in Figure 16-8. The Williams Mine active fan inventory includes 1,750 HP development fans, 1,775 HP production fans, and 6,035 HP system fans.

Six heating units of 16 MBtu/hr each provide for heating into the main shaft, with an additional two units at 15 MBtu/hr each at the No. 3 ventilation raise. Also a glycol heat recovery system is installed in the No. 5 ventilation raise.

MATERIAL HANDLING AND MAINTENANCE

Most of the mine materials are delivered via the main shaft. Levels are also interconnected with a ramp system that allows movement of equipment from one level to the next. The C Zone and Interlake Zone are accessed on several levels from the shaft and also internal ramps. Equipment servicing is carried out in underground garages located at strategic levels.

COMPRESSED AIR

The compressed air for the underground at Williams is supplied via three compressors, with two operating, each delivering 4,100 cfm of air at 100 psig.

COMMUNICATIONS

A Leaky Feeder system provides the communications system for the Williams Underground Mine.

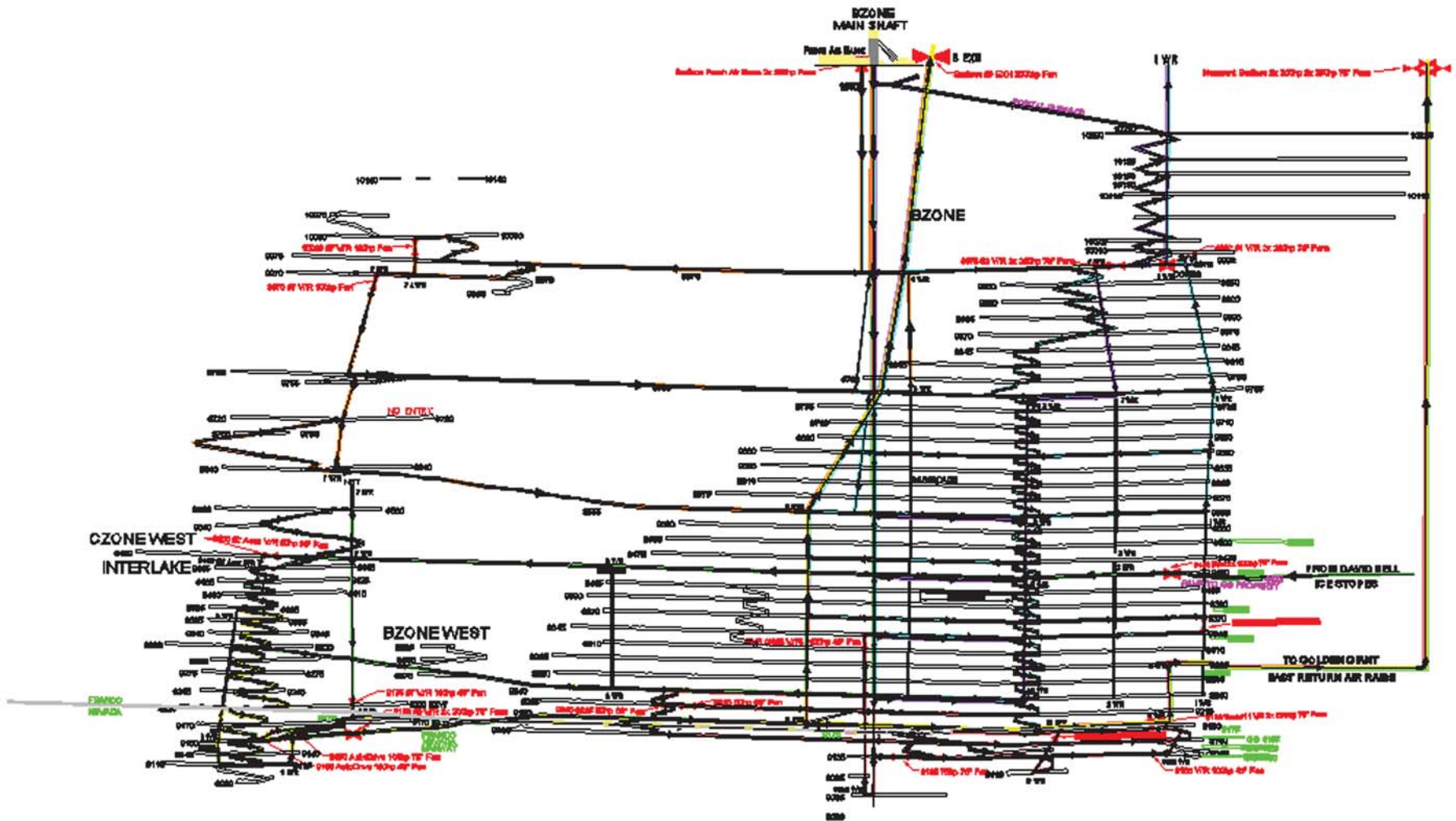


Figure 16-8

Barrick Gold Corporation

Hemlo Mine

Bomby Township, Ontario, Canada

Ventilation Schematic
(Looking North)

LIFE OF MINE PLAN

The LOM plan is made up of the Williams open pit and underground mines. RPA verified the economics of the LOM 2016 plan which includes only Proven and Probable Mineral Reserves.

Phase 5 in the open pit is expected to provide ore through to 2019, with stockpiled ore providing mill feed into 2020. A feasibility study is currently underway to confirm the economic viability of the Phase 6 extension, which involves a 60 m layback to the north wall. As shown below, Phase 6 could potentially extend the open pit life up to 2026.

The Williams underground has an expected mine life to the end of 2021, with production increasing from 3,300 tpd in 2017 to 3,800 tpd in 2021. A Pre-Feasibility study is currently underway to review the viability of an underground expansion, which would increase the production rate from a current target of 1.2 Mt per year to 1.9 Mt per year in 2021. An extensive diamond drilling program is currently underway to support the study. If successful, the underground expansion could potentially extend the mine life to 2026.

Both Phase 6 and the underground expansion are subject to permitting, social consultation, and corporate approvals before they will go ahead.

The LOM plan for the Hemlo Mine is shown in Table 16-4.

TABLE 16-4 2017 LOM PLAN FOR HEMLO OPERATIONS
Barrick Gold Corporation – Hemlo Mine

	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	TOTAL
Underground												
Tonnes Milled ('000s)	1,274	1,201	1,367	1,375	1,400							6,617
Head Grade (g/t Au)	3.55	4.30	4.35	4.14	4.04							4.08
Feed Ounces (koz)	145.6	166.0	191.3	182.8	181.9							867.6
Bullion Produced (koz)	137.0	157.0	181.0	172.7	171.8							819.7
Open Pit & Stockpile												
Tonnes Milled ('000s)	2,111	2,142	1,895	1,172	1,900	1,900	1,900	1,835	1,900	1,836	574	19,165
Head Grade (g/t Au)	1.33	1.03	1.13	0.99	1.46	1.36	1.09	0.94	1.02	1.23	1.24	1.17
Feed Ounces (koz)	90.2	71.3	69.1	37.3	89.0	83.0	66.7	55.6	62.5	72.8	22.8	720.3
Bullion Produced (koz)	82.9	64.7	63.3	34.6	83.6	76.1	60.6	51.3	57.5	67.4	20.8	662.8
Total												
Tonnes Milled ('000s)	3,385	3,342	3,263	2,546	3,300	1,900	1,900	1,835	1,900	1,836	574	25,782
Head Grade (g/t Au)	2.17	2.21	2.48	2.69	2.55	1.36	1.09	0.94	1.02	1.23	1.24	1.92
Feed Ounces (koz)	235.8	237.2	260.5	220.1	270.9	83.0	66.7	55.6	62.5	72.8	22.8	1,587.9
Bullion Produced (koz)	219.9	221.7	244.3	207.3	255.4	76.1	60.6	51.3	57.5	67.4	20.8	1,482.3
Overall Recovery	93.3%	93.4%	93.8%	94.2%	94.3%	91.6%	90.9%	92.1%	92.0%	92.5%	91.0%	93.3%

17 RECOVERY METHODS

The Williams Mill flowsheet is shown in Figure 17-1. The flowsheet consists of crushing, grinding, gravity concentration (not currently in use), cyanide leaching, carbon-in-pulp (CIP), carbon stripping, electrowinning, acid wash, reactivation, and refining.

Gold contained within the ore is exposed, or liberated, by grinding. The mill has two parallel grinding circuits, each consisting of a semi-autogenous grinding (SAG) mill, a ball mill, and a Knelson concentrator for gravity gold recovery. Knelson concentrates are upgraded on a gold table to produce a gravity concentrate.

Grinding is followed by thickening, aeration, and cyanide leaching to recover the gold from the ore. Gold is then extracted from the leach slurry using the CIP process followed by carbon stripping and Au electrowinning. Carbon is reactivated through acid washing followed by thermal regeneration in a kiln. The electrowinning cell sludge and gravity circuit concentrate are refined onsite by conventional mercury retorting and smelting in an electrical induction furnace to produce doré bullion.

Approximately 30% of the mill tailings are sent to the Williams paste plant to produce backfill for ground control underground. Surplus mill tailings are discharged to the TMF. Water is reclaimed from the TMF and returned to the mill for reuse. Excess water is treated through a seasonally operated effluent treatment plant prior to discharge to the environment.

The original mill capacity was 3,000 dry tpd and subsequently expanded to 6,000 tpd in 1988. Over the next ten years, throughput gradually increased to 7,000 tpd. In 1999, the David Bell Mill was shut down and all ore was trucked to Williams. This added another 1,250 tpd to Williams Mill. In 2001, the budgeted tonnage was increased to 10,000 tpd but, subsequently in 2006 tonnage targets dropped to less than 8,000 tpd. The current tonnage as of December 2016 month end reporting was 9,417 tonnes per calendar day.

The mill has two identical parallel grinding circuits. The grinding circuit can be operated in either the split milling mode or the blend milling mode. In the split milling mode, all underground ore is sent to the No. 2 circuit while open pit ore is sent primarily to the No. 1 circuit, with some overflow being bled into the No. 2 circuit. The underground ore tends to have significantly

higher gold grades relative to open pit ore. The purpose of split milling is to allow for a finer grind on the grind sensitive underground ore, while pushing throughput on No. 1 circuit. The blend milling mode is preferred and is current practice, whereby the underground and pit ores are mixed up to 50/50 ratio between the two coarse ore bins, allowing the bin levels to be maintained at 60%. Current operations from the Williams underground and open pit averaged 9,417 tpd in 2016.

Ore is delivered to the transfer tower and split or blended, as required, to two coarse ore bins. The SAG mills discharge onto screens. Oversize from the screens is returned to the SAG mill feed. Screen undersize flows to the cyclone feed pump box. From the pump box, slurry is pumped to the cyclone clusters. Overflow from the hydrocyclones in both grinding circuits is combined and fed to a linear screen. The cyclone underflow is ground in the ball mills. Slurry discharging from the linear screen is dewatered in a thickener.

The thickener underflow is pumped to the cyanide leach circuit which consists of eight leach tanks that operate in series. The slurry overflows by gravity from one tank to the next. The discharge from the final leach tank flows through a trash screen before advancing to the CIP circuit. The slurry flows by gravity through the six CIP tanks. Activated carbon is added to the sixth tank and is advanced counter-current to slurry flow.

Loaded carbon is removed from the first CIP tank and sent to carbon stripping and regeneration circuits. The gold-bearing solution from carbon stripping is pumped through electrowinning cells where the gold is recovered as a sludge. Electrowinning cell sludge is dried in a mercury retort, where mercury is also collected, and then refined to produce gold bullion. A portion of the stripped carbon is acid washed to remove inorganic contaminants, thermally regenerated to remove organic contaminants, and returned to the last tank in the CIP circuit.

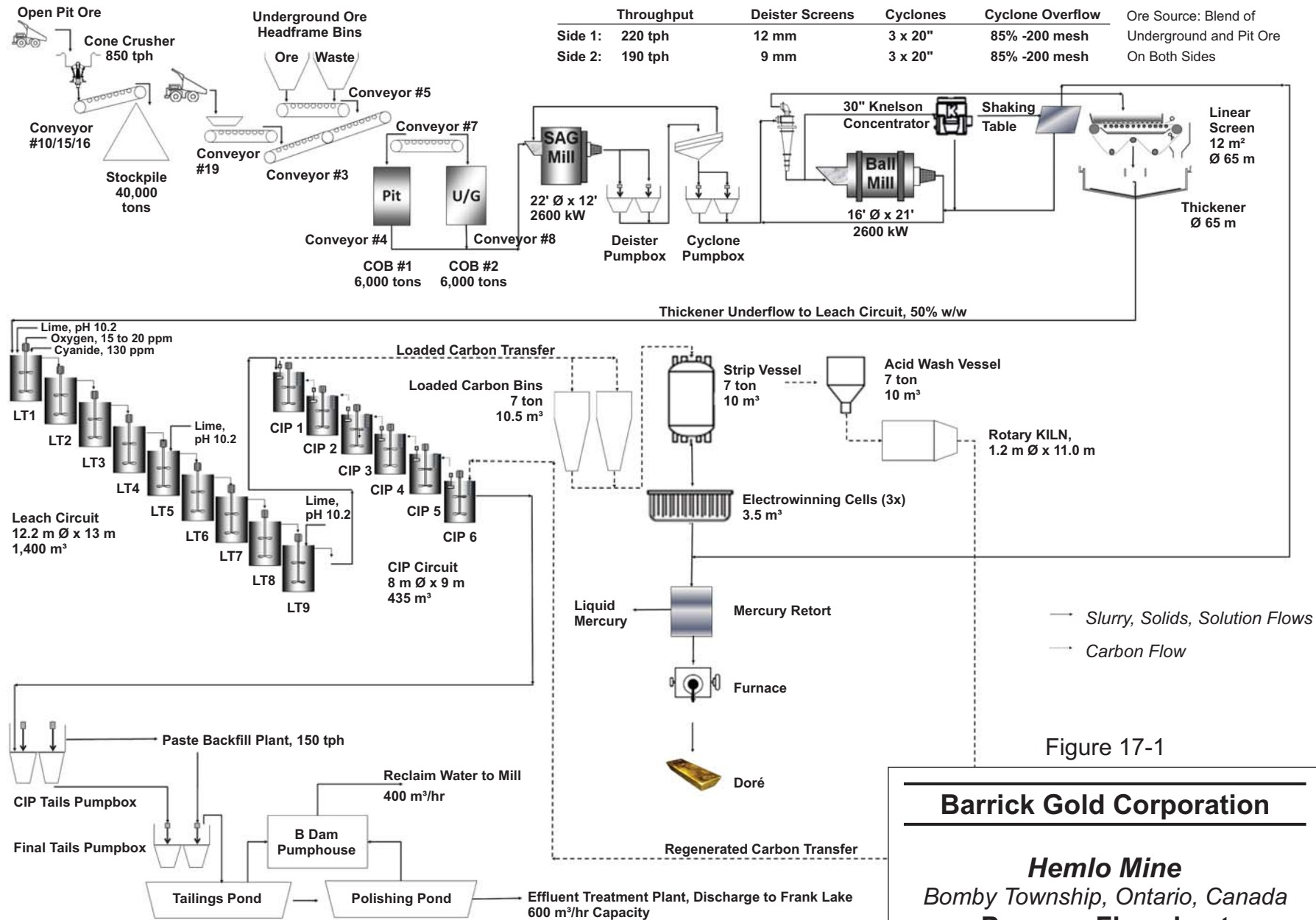


Figure 17-1

Barrick Gold Corporation

Hemlo Mine
Bomby Township, Ontario, Canada
Process Flowsheet

18 PROJECT INFRASTRUCTURE

The Hemlo underground mine has been in operation since 1985 and the open pit has been in operation since 1991. The Williams Mine infrastructure is shown in Figure 18-1.

SURFACE INFRASTRUCTURE

The Williams mine site has been operational for some 25 plus years with surface infrastructure that includes the following:

- C Zone open pit shop with three service bays
- Pit wash bay for large equipment
- A gyratory crusher with a capacity of 1,000 tph
- A 208 ft high headframe structure
- Hoist room facilities for a 15 ft diameter, double drum hoist and a 12 ft diameter service hoist with cage and counterweight
- A paste plant facility with a capacity of 100 m³/hr of paste fill
- A service building with offices, service facilities and warehousing
- An assay laboratory
- A mill facility to process open pit and underground ore
- A 30 m diameter thickener
- Two 6,000 t coarse ore bins
- Several laydown areas
- Miscellaneous conveyor ways
- Sedimentation pond and polishing ponds
- Propane tank farm

TAILINGS FACILITIES

Surplus mill tailings that are not used to produce paste backfill are sent via pipeline to the TMF which is approximately three kilometres away.

Initially, the David Bell and Williams tailings areas were separately managed. Today, these tailings areas are combined into a single facility that is managed by Williams.

The Williams TMF consists of the following structures and appurtenances:

- Williams pipeline highway overpass, drainage ponds, pipeline corridor, and pipelines: tailings to tailings pond, seasonal water transfer to the David Bell pond, tailings pond water reclaim, and polishing pond water reclaim
- Tailings deposition pond and impoundment with integrated landfill
- Polishing pond
- North Rim seepage dyke system
- Effluent Treatment Plant (ETP)
- Treated effluent pipeline to Frank Lake
- ETP feed pipeline and sludge pipeline to tailings pond
- Reclaim and water transfer pump house
- Dams A, B, D, E, F, G, H, and I
- Dam A emergency spillway and seepage pond
- Ground water monitoring stations
- Piezometric monitoring stations

Tailings from the Williams Mill paste plant are pumped to the TMF for solids deposition and impoundment. Water is reclaimed year-round from ponds to the Williams Mill to limit fresh water consumption.

Residual cyanide is treated passively by natural decomposition using sunlight and neutral pH. Total cyanide concentrations in the TMF range from approximately 0 ppm in the summer to 30 ppm in the winter. Water from the TMF is transferred to the polishing pond between mid-June and late-September, while the total cyanide concentration is low. Surplus water in the tailings management area is pumped seasonally from the Polishing pond water, treated to remove metals, and discharged to Frank Lake. The annual treatment season is April 1st through November 15th.

Seismic Stability Analyses were carried out during the design of the dams using an earthquake with a recurrence interval of 10,000 years. In performing the stability analysis, an earthquake time history for an area with a similar geologic setting was used and scaled to the peak acceleration for the recurrence interval as provided by the Geological Survey of Canada. The dams were designed with minimum factors of safety as recommended by the Canadian Dam Association (CDA) and Barrick.

The dams are zoned earth fill dams, constructed with a compacted glacial till core and grouted foundation bedrock. The basin was constructed around a former lake with perimeter dams constructed as required to close off topographic lows around the perimeter of the basin. All dams on the property are raised following centreline and modified upstream construction. Dam A has an emergency spillway that was designed to pass the runoff from the Probable Maximum Precipitation (PMP) storm event with a rise in pond level of less than 0.5 meters.

An external consultant carries out annual visual inspections. Routine visual monitoring of the pipeline corridor is carried out by mill operating staff three to four times per day and environmental staff on a weekly basis. Perimeter dams are inspected daily by mill operating staff. An annual review of the basin operation is also conducted by the external consultant, which includes projections of pond levels to allow scheduling of dam raises and water management activities.

In Canada, the CDA provides a rating system with respect to "Consequence Categories" that is recognized world-wide. Based on the CDA tables, the Williams perimeter dams are rated as having a "Low to High Consequence Category". The Ontario Ministry of Natural Resources (MNR) publishes Dam Safety Guidelines which follow a similar methodology. Using the MNR guidelines, the Hazard Potential are rated as "Low to Significant".

Pipeline, pipeline overpass, or pond containment failures are events with significant consequences to the environment and to the Hemlo Mine. The likelihood of each is mitigated to a low probability through the following controls:

- Access to the tailings area is restricted and controlled
- Differential alarms monitor flows into and out of the pipeline corridor for continuous leak detection for all pipelines

- Mill personnel inspect the pipeline corridors and perimeter dams each shift for a minimum of three inspections each day
- The overpass is inspected by a third-party engineering consultant every two years
- Secondary containment exists along critical sections of the pipeline corridor and is currently under review
- All containments are designed, constructed and inspected by a third-party engineering consultant

The capacity of the TMF was not sufficient to meet the LOM plan, so Barrick completed a study to evaluate different tailings deposition methods including slurry, thickened, and dry stack processing options. Since the tailings also contain sulphide minerals, such as pyrite, they are susceptible to acid rock drainage (ARD). Barrick also evaluated addition of a flotation plant to produce tailings that will be non-acid generating. Based on the study, thickened, de-sulphurized tailings was selected as the optimum option. Changing to these methods provides sufficient storage to accommodate the total production in the LOM plan and reduces closure costs by providing non-acid generating (NAG) cover material for the TMF.

POWER

The power supply at Williams is from the main Ontario Hydro grid and delivered at 115 kV to two 25 MVA main transformers. These main units in turn feed two, 20 MVA transformers, one with a 4.16 kV secondary and the second with a 13.8 kV secondary. These units then feed the various load centres throughout the mine surface plant and underground operations as well.

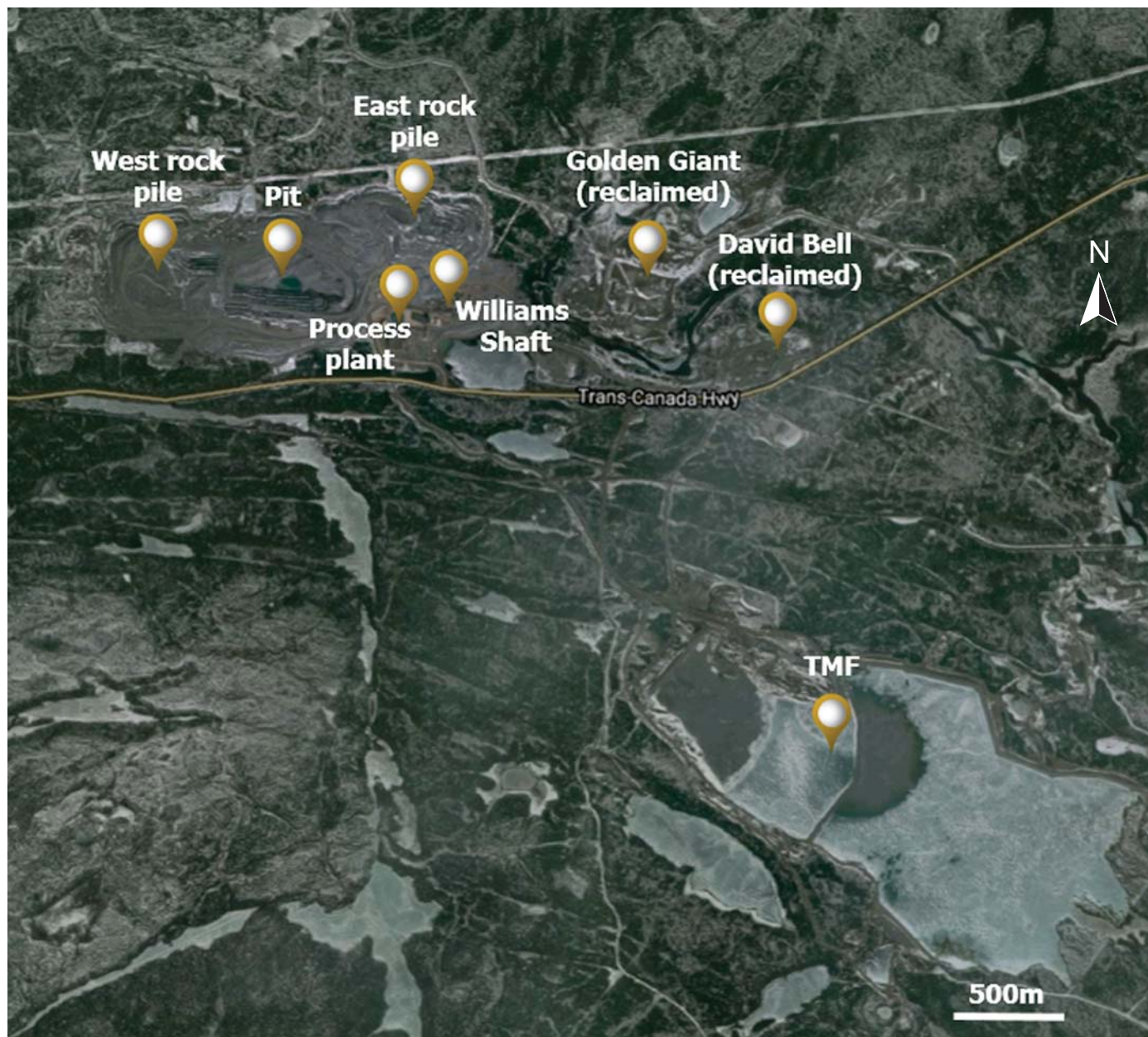


Figure 18-1

Barrick Gold Corporation

Hemlo Mine

Bomby Township, Ontario, Canada

Williams Mine Surface Plan

19 MARKET STUDIES AND CONTRACTS

MARKETS

The principal commodities produced at Hemlo are gold and silver, which are freely traded, at prices that are widely known, so that prospects for sale of any production are virtually assured. For the 2017 LOM plan prepared by Barrick, an average gold price of \$1,000 per ounce was used for four years (2017 to 2020) and \$1,200 per ounce from 2021 onwards. An exchange rate of 1.30 (C\$:US\$) was used. Operations at the Hemlo Mine are cash flow positive within these parameters

CONTRACTS

The Hemlo Mine is a large, modern open pit and underground mining operation, and Barrick is a major international firm with policies and procedures for the letting of contracts. The contracts for smelting and refining are normal contracts for a large producer.

There are numerous contracts underground development contracts to provide services to supplement Barrick's workforce. Major underground contractors include Manroc Developments Inc., who provide Alimak mining services and lateral development. Boart-Longyear will be the diamond drill contractor in 2017. The total contractor workforce at Hemlo varies from approximately 150 to 200 people.

20 ENVIRONMENTAL STUDIES, PERMITTING, AND SOCIAL OR COMMUNITY IMPACT

ENVIRONMENTAL STUDIES

The necessary environmental studies were completed for the operation. Barrick has all of the permits required for operation.

The Phase 6 layback required further studies for the management of waters, changes to the operation of the TMF, air emissions, social impacts, and mine closure.

PROJECT PERMITTING

The active permit list managed by the environmental department at Hemlo is shown in Table 4-4, in Section 4 of this report. The process facilities are permitted for a production rate of 11,500 tpd.

The environmental department at Hemlo also manages the Cedar Creek watershed, which provides water for the mine. Williams Mine commissioned the Actiflo water treatment plant in 2010 to treat water from the TMF for process water use and continued to utilize the storm water recycling infrastructure, which redirects storm water from the open pit operations to the mine site for process water use.

Barrick continues to follow the spill prevention and contingency program to comply with Ontario Regulation 224/07, the cyanide code and Metal Mining Effluent Regulation. The program covers spill prevention, response and contingency planning. Training opportunities for employees and the surface emergency team related to spill response have been ongoing. Hemlo did not have a spill reportable to provincial or federal agencies in 2016.

In support of Barrick's commitment to sustainable development, an Environmental Management System has been developed and implemented to support existing environmental initiatives at the Williams and David Bell sites along with enhancing the long-term feasibility of the operation without compromising the natural environment.

Hemlo has established a procedure for the identification, interpretation and applicability of statutory, regulatory and code-adopted requirements in addition to industry and corporate requirements applicable to the site. All applicable environmental regulations, current federal, provincial and municipal/regional requirements including all permits and certificates of approval obtained for operations are logged in the Intalex database.

The Hemlo environmental group provides continuous monitoring of environmental activities through workplace inspections/audits, which include:

- Regular visits to all work places
- Co-inspection with all levels of supervision
- Tracking and monitoring of all environmental incidents and non-conformance
- Developing and reviewing site standards/procedures
- Implementation of site standards, spill prevention and contingency program and Tailings Operation Maintenance and Surveillance manual

The Hemlo environmental group continues to focus on the site water balance to minimize freshwater usage and water stored in the tailings management facility, while maximizing tailings deposition. In 2016, the Hemlo Mine set an objective and target of 90% recycled water, and achieved 93% recycled water. The reclaim system, Actiflo water plant, and storm-water system reduces reliance on water from Cedar Creek, which during drought conditions, cannot supply all of Williams process water needs.

The Hemlo environmental group also plans to improve its water management strategies. Ongoing evaluation of the overall water balance, tailings deposition management, and ways to reduce the mines water needs will be priorities. In order to store tailings through to 2021, the Williams tailings basin will need to be raised to its maximum elevation of 357.5 MASL according to the design and permit.

Should a decision be made to proceed with the Phase 6 layback, a number of amendments to existing permits and/or approvals will be required from provincial and federal agencies.

SOCIAL OR COMMUNITY REQUIREMENTS

Barrick is a member of the Mining Association of Canada. Within that association, Hemlo Mine participates in the Towards Sustainable Mining initiative.

A stakeholder analysis of the Hemlo operation identifies seven stakeholders; Town of Marathon, Township of Manitouwadge, Township of White River, Pic Mobert First Nation, Biigtigong Nishnaabeg First Nation, Métis Nation of Ontario and Pukaskwa National Park. Hemlo has also initiated a dialogue with Red Sky Independent Nation. An Internal / External Communication Policy is in place and Hemlo representatives meet twice annually to give operational updates to each identified stakeholder.

Socio-economic agreements are in place with local First Nations. Implementation Committees are in place with both First Nations and meet every six weeks. Barrick provides support, financially and in-kind, to various community events and youth sports within all stakeholder communities. A Community Relations Management System is in place and all set up through Intelix management system. Barrick enjoys good relationships with local government. Hemlo management meets at least once per year with provincial and federal elected officials.

NGO engagement is conducted with Lake Superior Bi-National Forum, Mining Watch, and Marathon Residents for Responsible Mining. These NGOs are all interrelated. Hemlo continues to monitor their engagement and involves these organizations as required.

The current socio-economic agreements with the local First Nations expire in 2019, and will require renewal or extension. Hemlo maintains active engagement with the local First Nations. The previous agreements had five-year terms, so it is reasonable to expect that new agreements will be for the same duration.

MINE CLOSURE REQUIREMENTS

AMEC Earth & Environmental was retained to conduct a feasibility level study and supporting engineering and geoscience studies to address the closure requirements for the Williams Mine and David Bell Mine.

Following cessation of Williams operations, the buildings and infrastructure located in the plant site will be demolished and removed. The disturbed plant site area will be re-vegetated with hydro-seed.

The closure of the underground workings will include restriction of access to the site and stability assessment of all surface and underground workings. Shafts and raises will be secured with caps while the A Zone portal will be secured by either sealing off the entrance with barren rock and/or a reinforced cover. It is expected that the portal will be backfilled with non-acid generating (NAG) waste rock over a distance of approximately 20 m to 30 m from the portal entrance. Final closure measures will be certified by a professional engineer.

During operations, tailings will be used to develop beaches required to support the bituminous geomembrane covers at the TMF. During operations or following closure, the tailings beaches will be covered with bituminous geomembrane, maintaining a 50 m setback between the TMF Dam and water cover/pond. The bituminous geomembrane will be covered with a rockfill material. The rockfill layer will prevent wind uplift of the bituminous geomembrane and protect the liner from UV degradation, increasing the service life of the Mine.

Modeling of the TMF water quality suggests that tailings pond water quality will improve after mine operations cease. Discharge from the pond will be replaced by direct precipitation and clean runoff from the TMF catchment, resulting in continuous improvement in the TMF water quality. Current estimates indicate that the TMF discharge will reach concentrations at or below Provincial Water Quality Objectives (PWQO) levels in approximately 17 years. Water treatment will continue until the TMF water is suitable for direct discharge to the receiving environment.

Following decommissioning of the ETP the closure overflow spillway will be constructed in the abutment in between Dam I and Dam A draining towards Lim Lake. Following construction of the closure overflow spillway the Normal Operating Water Level of the TMF will be 355.0 m. The closure overflow spillway was sized to convey the PMP and maintain the minimum 50 m setback between the TMF Dam and water cover/pond.

After mine operations cease, PAG mine waste rock located in the east and west waste rock piles will be re-contoured to 3H:1V slopes. Tree and shrubbery islands will be planted over 25% of the NAG waste rock. Drainage from the waste rock piles will be directed to the open

pit. Forty years after the cessation of operations, all PAG waste rock will be covered with a geosynthetic cover. Tree and shrubbery islands will be constructed over 25% of the geosynthetic cover.

During mining, the crown pillar from the base of the C Zone pit will be removed, resulting in a hydraulic connection between the C Zone pit, the backfilled A Zone pit, and the underground workings. Flooding of the underground workings will occur primarily through runoff and direct precipitation directed towards the C Zone pit, with lesser inputs from groundwater flow directly into the workings. An estimated time until flooding of the workings to the base of the C Zone pit is about six years. Flooding to the maximum allowable water level of 310 MASL involves partial flooding of the C Zone pit and some underground workings, minimizing potential impacts from the underground workings and preventing discharge via the A Zone pit. Flooding to the 310 MASL elevation is estimated to take 43 years after the cessation of mining.

Following cessation of operations, drainage from the waste rock piles will be directed to the C Zone pit. An open channel will be constructed from Pond 300 to the C Zone pit. Pond 102 will provide passive collection of runoff from barren rock storage areas, with drainage 400A to provide passive collection of runoff from barren rock storage areas following cessation of operations. Drainage from Ponds 102 and 400A will be directed to the C Zone Pit. An overflow spillway will be constructed from the C Zone Pit to Cedar Creek. Once the water level inside the C Zone Pit reaches 310 MASL, water from the pit will decant to Cedar Creek, maintaining the water level within the pit and mine workings.

Preliminary mixing models, based on surface water quality data and inflows, combined with available underground water quality data and inflows, indicate that the effluent quality of the pit water to be discharged to Cedar Creek will not exceed the current Certificate of Approval (C of A) limits for the majority of the parameters, and will in fact be at or near PWQO for the metal parameters of interest, with the possible exception of iron, under conditions where the PAG rock begins to produce acid and leach metals.

After mining operations have ceased, it is anticipated that after approximately 40 years, all waters collected and managed at Hemlo will be eligible to be passively discharged, pending regulatory approval. An additional ten years of water quality monitoring is included in the closure cost estimate. Therefore, 50 years post-closure, the only long term commitment remaining at the Hemlo Mine will be routine tailings dam inspections and maintenance.

All potential changes to the operating design of the mine are assessed with closure in mind. Several studies have initiated to facilitate an executable closure plan should the mine life extend beyond 2021. These studies include Socio-Economic, Cultural Heritage, Rock Storage and Geochemistry, Tailings Management, Water Management, Baseline Environmental Assessment, and Permitting Requirements. Through this preliminary work, several opportunities have been identified to optimize any future closure plan amendments regarding an extended mine life. For instance, the possibility of dry-stacking tailings is being investigated, as an alternative to long term water cover.

Table 20-1 summarizes the current mine closure cost estimate.

TABLE 20-1 CLOSURE COSTS
Barrick Gold Corporation - Hemlo Mine

Feasibility Level Estimated Costs	Williams Mine (\$ millions)	David Bell Mine (\$ millions)
Closure Capital	114	3.2
Post Closure (operating)	23	3.1
Total Closure Costs	137	6.3

Notes:

1. Costs include direct, indirect, and contingency for years 2018 to 2056.

21 CAPITAL AND OPERATING COSTS

The Hemlo Mine is an operating open pit and underground gold mine. There are new mine projects and sustaining capital requirements, however, there are no preproduction requirements.

CAPITAL COSTS

Current LOM capital costs for the operation are estimated to be \$301 million, as summarized in Table 21-1. Large capital items for 2017 include tailings dam construction for \$18.8 million, SAG mill shell replacement for \$5.5 million, crown pillar sequencing for \$9.6 million, and a Pre-Feasibility study for underground expansion for \$11.0 million.

TABLE 21-1 LOM CAPITAL COST ESTIMATE
Barrick Gold Corporation - Hemlo Mine

Description	2017 (\$ 000)	LOM Total (\$ 000)
Sustaining Capital		
Open Pit	13,439	59,823
Underground	16,150	52,462
Processing	7,970	16,796
Tailings	18,813	84,236
UG Development & Open Pit Stripping	16,336	75,710
Exploration Drilling	9,588	12,038
Total	82,296	301,066

RPA is of the opinion that the projected capital costs appear to be reasonable.

OPERATING COSTS

Table 21-2 summarizes the reported unit operating costs for 2016, and the projected LOM costs.

TABLE 21-2 OPERATING COST SUMMARY
Barrick Gold Corporation - Hemlo Mine

Item	Units	2016	LOM
Open Pit Mining	\$/t moved in pit	5.46	4.86
Underground Mining	\$/t UG ore	86.56	71.23
Mining Total	\$/t milled	41.10	34.90
Processing	\$/t milled	12.04	12.53
G & A	\$/t milled	7.15	7.78
Total	\$/t milled	60.30	55.21

Open pit costs will decrease on a unit basis as Phase 6 begins, due to slightly higher production rates. Underground unit costs are projected to decrease due to increasing underground production rates over the next five years. Processing and general and administrative (G&A) unit costs are lower in the short-term, then skewed higher by lower volumes once the underground reserves are completed.

WORKFORCE

The personnel as of the end of 2016 is shown in Table 21-3. These figures do not include contractors, which typically number between 150 and 200 people.

TABLE 21-3 SUMMARY OF PERSONNEL BY DEPARTMENT
Barrick Gold Corporation – Hemlo Mine

Department	Hourly	Staff	Total
Open Pit Operations	42	7	49
Open Pit Maintenance	12	3	15
Underground Operations	126	21	147
Underground Maintenance	48	12	57
Mill Operations	27	11	38
Mill & Surface Maintenance	21	22	43
General and Administration	12	82	93
Total	228	154	442

22 ECONOMIC ANALYSIS

Under NI 43-101 rules, producing issuers may exclude the information required for Section 22 Economic Analysis on properties currently in production. RPA notes that Barrick is a producing issuer and the Hemlo Mine is currently in production. RPA has performed an economic analysis of the Hemlo Mine using the estimates presented in this report and confirms that the outcome is a positive cash flow that supports the statement of Mineral Reserves.

23 ADJACENT PROPERTIES

This Technical Report does not include information concerning adjacent properties.

24 OTHER RELEVANT DATA AND INFORMATION

No additional information or explanation is necessary to make this Technical Report understandable and not misleading.

25 INTERPRETATION AND CONCLUSIONS

Based on RPA's site visit, interviews with Hemlo personnel and subsequent review of gathered information, RPA offers the following conclusions:

GEOLOGY AND MINERAL RESOURCES

- The Hemlo deposits are best described as amphibolite-facies, disseminated replacement-stockwork Archean load gold deposits.
- Mineral Resource estimates have been prepared utilizing acceptable estimation methodologies. The classification of Measured, Indicated, and Inferred Mineral Resources conforms to CIM definitions.
- The sampling, sample preparation, and analyses are appropriate for the style of mineralization and Mineral Resource estimation. The current drill hole database is reasonable for supporting the Mineral Resource and Mineral Reserve estimates.
- The EOY2016 open pit and underground resource block models are reasonable for supporting the Mineral Resource and Mineral Reserve estimates.
- The resource modelling procedures are well documented.
- Overall, RPA is of the opinion that Barrick has conducted high quality resource modelling work that exceeds industry standard practice.
- Mineral Resources are reported exclusive of Mineral Reserves and are estimated effective December 31, 2016.
- The open pit Measured plus Indicated Mineral Resources total 56.5 million tonnes grading 0.80 g/t Au, containing 1.45 million oz Au. Inferred Mineral Resources are estimated to be 5.1 million tonnes grading 0.62 g/t Au containing 0.1 million oz Au.
- The underground Measured plus Indicated Mineral Resources total 2.4 million tonnes grading 3.49 g/t Au, containing 0.27 million oz Au. Inferred Mineral Resources are estimated to be 2.6 million tonnes grading 4.5 g/t Au containing 0.4 million oz Au.

MINING AND MINERAL RESERVES

- The Mineral Reserve estimates have been prepared utilizing acceptable estimation methodologies and the classification of Proven and Probable Mineral Reserves conforms to CIM definitions.
- Mine planning for the Hemlo open pit and underground mines follows industry standards.

- RPA considers the selection of mining methods and the design practices to be appropriate for the deposits.
- Open pit Proven and Probable Mineral Reserves, consisting of Phases 5 and 6, total 19.1 million tonnes grading 1.17 g/t Au, containing 0.72 million oz Au.
- Underground Proven and Probable Mineral Reserves total 6.6 million tonnes grading 4.08 g/t Au, containing 0.87 million oz Au.
- The LOM plan is based upon current operating experience. Recovery and cost estimates are based upon actual operating data and engineering to support a Mineral Reserve statement. Economic analysis using these estimates generates a positive cash flow, which supports a statement of Mineral Reserves.
- Mineral Reserves contained in open pit Phase 6 are supported by a Pre-Feasibility Study. Work on a Feasibility Study and integration with plans for the larger operation are in progress.
- Although Phase 6 has the potential to extend the open pit mine life beyond the underground mine life, there is also good potential for further conversion of underground Mineral Resources to Mineral Reserves, and exploration opportunities, which are likely to result in continued blending of open pit and underground feed.
- Equipment purchases scheduled for the current LOM plan are reasonable.
- The workforce is well trained and capable of achieving the necessary production targets established by the engineering department in a safe manner.
- The reconciliation to 2016 production indicates that the open pit resource model is performing well within the normal range of variability (+/-10%), and that the underground resource model is conservative. Overall, Mineral Reserve estimates are within 10% of production results for tonnes and grade.

PROCESSING

- Ore blending is a well-coordinated and standard operation that ensures that the ore going to the processing operation provides optimum results. The milling operations are well run, safe, environmentally sound, and meet industry standards.
- The adjustments made to the process production data and mill head grades based on actual gold production conform to industry standards.
- The recovery estimates appear to be accurate based on recent data, however, large differences between budgeted head grade and actual head grade of ore from the underground mine has the potential to reduce the accuracy.

ENVIRONMENTAL AND COMMUNITY CONSIDERATIONS

- The Hemlo Mine is within environmental compliance. Permits will need to be updated for the open pit expansion, which may take up to two years to obtain.
- The Hemlo Mine has been a major contributor to the local economy, and Barrick will continue to communicate and work with the First Nations and other communities of interest.
- Reclamation estimates reviewed are realistic, in RPA's opinion.
- The David Bell Mine closed in May 2014, and progressive reclamation has been initiated at both that site and the former Golden Giant Mine sites.

RISKS

The mine has been in production for over 30 years and is a mature operation. RPA has not identified any significant risks and uncertainties that could reasonably be expected to affect the reliability or confidence in the exploration information, the Mineral Resource and Mineral Reserve estimates, or associated projected economic outcomes.

26 RECOMMENDATIONS

Based on the site visit, discussions with Hemlo personnel, and subsequent review of available information, RPA offers the following recommendations:

OPEN PIT

- Lithological and alteration mapping should be conducted at every opportunity, in order to refine the geological model.
- Continued mapping of discrete structures is recommended, for the purposes of ground control and the mitigation of water infiltration to the underground.
- Blast movement monitoring using electronic markers appears to be providing improvements in open pit dilution control. This should be supported and continued.

UNDERGROUND

- Resources and reserves are sensitive to cut-off grade. There appears to be a “natural cut-off” in the underground resource, which is a result of the lower grade material found in the vicinity of the higher grade ore lenses. Efforts should continue to be made to reduce the capital and operating costs.
- Significant mineral potential has been recognized in newly-acquired claims west of the current underground workings. This will require a concerted effort in a multi-year exploration program to find and define this potential. RPA recommends continued funding of this exploration effort.
- Continued focus should be given to the Lower B Zone, as this area shows significant potential to extend the mine life, yet will require geotechnical modelling and careful stope sequencing to be successful.
- Further conversion of underground resources and development of exploration potential is required to maintain the balance of open pit and underground feed past 2021.
- The 2016 production versus underground resource model results should be reviewed within the same volume, to assess model performance independent of other estimating factors. Consider adjusting some model input parameters to bring the conservative estimate closer to production results.

PROCESSING

- Ore characterization programs should be continued and supported in order to quantify the milling characteristics and achieve good estimates for plant throughput, gold recovery, and plant operating costs in future LOM plans and financial analyses.

27 REFERENCES

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28 DATE AND SIGNATURE PAGE

This report titled "Technical Report on the Hemlo Mine, Marathon, Ontario, Canada" and dated April 25, 2017, was prepared and signed by the following authors:

(Signed and Sealed) "Jason J. Cox"

Dated at Toronto, ON
April 25, 2017

Jason J. Cox, P.Eng.
Principal Mining Engineer

(Signed and Sealed) "Wayne W. Valliant"

Dated at Toronto, ON
April 25, 2017

Wayne W. Valliant, P.Geo.
Principal Geologist

(Signed and Sealed) "Kathleen Ann Altman"

Dated at Lakewood, CO
April 25, 2017

Kathleen Ann Altman, Ph.D., P.E.
Principal Metallurgist

29 CERTIFICATE OF QUALIFIED PERSON

JASON J. COX

I, Jason J. Cox, P.Eng., as an author of this report entitled "Technical Report on the Hemlo Mine, Marathon, Ontario, Canada", prepared for Barrick Gold Corporation, and dated April 25, 2017, do hereby certify that:

1. I am a Principal Mining Engineer and Executive Vice President, Mine Engineering, with Roscoe Postle Associates Inc. of Suite 501, 55 University Ave Toronto, ON, M5J 2H7.
2. I am a graduate of the Queen's University, Kingston, Ontario, Canada, in 1996 with a Bachelor of Science degree in Mining Engineering.
3. I am registered as a Professional Engineer in the Province of Ontario (Reg. #90487158). I have worked as a Mining Engineer for a total of more than 20 years since my graduation. My relevant experience for the purpose of the Technical Report is:
 - Review and report as a consultant on many mining operations and projects around the world for due diligence and regulatory requirements
 - Feasibility Study project work on several mining projects, including five North American mines
 - Operational experience as Planning Engineer and Senior Mine Engineer at three North American mines
 - Contract Co-ordinator for underground construction at an American mine
4. I have read the definition of "qualified person" set out in National Instrument 43-101 (NI 43-101) and certify that by reason of my education, affiliation with a professional association (as defined in NI 43-101) and past relevant work experience, I fulfill the requirements to be a "qualified person" for the purposes of NI 43-101.
5. I visited the Hemlo Mine on December 14 and 15, 2016.
6. I am responsible for Sections 15, 16, 18, 19, 22, and parts of Sections 1 to 6, 21, 24, 25, 26, and 27 of the Technical Report.
7. I am independent of the Issuer applying the test set out in Section 1.5 of NI 43-101.
8. I have had no prior involvement with the property that is the subject of the Technical Report.
9. I have read NI 43-101, and the Technical Report has been prepared in compliance with NI 43-101 and Form 43-101F1.

10. At the effective date of the Technical Report, to the best of my knowledge, information, and belief, the Technical Report contains all scientific and technical information that is required to be disclosed to make the Technical Report not misleading.

Dated this 25th day of April, 2017

(Signed and Sealed) "Jason J. Cox"

Jason J. Cox, P.Eng.

WAYNE W. VALLIANT

I, Wayne W. Valliant, P.Geo., as an author of this report entitled "Technical Report on the Hemlo Mine, Marathon, Ontario, Canada", prepared for Barrick Gold Corporation, and dated April 25, 2017, do hereby certify that:

1. I am Principal Geologist with Roscoe Postle Associates Inc. of Suite 501, 55 University Ave Toronto, ON, M5J 2H7.
2. I am a graduate of Carleton University, Ottawa, Ontario, Canada in 1973 with a Bachelor of Science degree in Geology.
3. I am registered as a Geologist in the Province of Ontario (Reg. #1175). I have worked as a geologist for a total of 41 years since my graduation. My relevant experience for the purpose of the Technical Report is:
 - Review and report as a consultant on more than fifty mining operations and projects around the world for due diligence and resource/reserve estimation
 - General Manager of Technical Services for corporation with operations and mine development projects in Canada and Latin America
 - Superintendent of Technical Services at three mines in Canada and Mexico
 - Chief Geologist at three Canadian mines, including two gold mines
4. I have read the definition of "qualified person" set out in National Instrument 43-101 (NI 43-101) and certify that by reason of my education, affiliation with a professional association (as defined in NI 43-101) and past relevant work experience, I fulfill the requirements to be a "qualified person" for the purposes of NI 43-101.
5. I visited the Hemlo Mine on December 14 and 15, 2016.
6. I am responsible for Sections 7 to 12, 14, and 23 and parts of Sections 1 to 6, 24, 25, 26, and 27 of the Technical Report.
7. I am independent of the Issuer applying the test set out in Section 1.5 of NI 43-101.
8. I have had no prior involvement with the property that is the subject of the Technical Report.
9. I have read NI 43-101, and the Technical Report has been prepared in compliance with NI 43-101 and Form 43-101F1.
10. At the effective date of the Technical Report, to the best of my knowledge, information, and belief, the Technical Report contains all scientific and technical information that is required to be disclosed to make the Technical Report not misleading.

Dated this 25th day of April, 2017

(Signed and Sealed) "Wayne W. Valliant"

Wayne W. Valliant, P.Geo.

KATHLEEN ANN ALTMAN

I Kathleen Ann Altman, P.E., as an author of this report entitled "Technical Report on the Hemlo Mine, Marathon, Ontario, Canada", prepared for Barrick Gold Corporation, and dated April 25, 2017, do hereby certify that:

1. I am Principal Metallurgist and Director, Mineral Processing and Metallurgy with RPA (USA) Ltd. of Suite 505, 143 Union Boulevard, Lakewood, Co., USA 80228.
2. I am a graduate of the Colorado School of Mines in 1980 with a B.S. in Metallurgical Engineering. I am a graduate of the University of Nevada, Reno Mackay School of Mines with an M.S. in Metallurgical Engineering in 1994 and a Ph.D. in Metallurgical Engineering in 1999.
3. I am registered as a Professional Engineer in the State of Colorado (Reg. #37556) and a Qualified Professional Member of the Mining and Metallurgical Society of America (Member #01321QP). I have worked as a metallurgical engineer for a total of 34 years since my graduation. My relevant experience for the purpose of the Technical Report is:
 - Review and report as a metallurgical consultant on numerous mining operations and projects around the world for due diligence and regulatory requirements.
 - I have worked for operating companies, including the Climax Molybdenum Company, Barrick Goldstrike, and FMC Gold in a series of positions of increasing responsibility.
 - I have worked as a consulting engineer on mining projects for approximately 15 years in roles such a process engineer, process manager, project engineer, area manager, study manager, and project manager. Projects have included scoping, prefeasibility and feasibility studies, basic engineering, detailed engineering and start-up and commissioning of new projects.
 - I was the Newmont Professor for Extractive Mineral Process Engineering in the Mining Engineering Department of the Mackay School of Earth Sciences and Engineering at the University of Nevada, Reno from 2005 to 2009.
4. I have read the definition of "qualified person" set out in National Instrument 43-101 (NI 43-101) and certify that by reason of my education, affiliation with a professional association (as defined in NI 43-101) and past relevant work experience, I fulfill the requirements to be a "qualified person" for the purposes of NI 43-101.
5. I visited the Hemlo Mine on December 14 and 15, 2016.
6. I am responsible for Sections 13, 17, and 20 and parts of Sections 1 to 6, 21, 25, 26, and 27 of the Technical Report.
7. I am independent of the Issuer applying the test set out in Section 1.5 of NI 43-101.
8. I have prepared a previous audit and internal technical report on the property that is the subject of the Technical Report.
9. I have read NI 43-101, and the Technical Report has been prepared in compliance with NI 43-101 and Form 43-101F1.

10. At the effective date of the Technical Report, to the best of my knowledge, information, and belief, the Technical Report contains all scientific and technical information that is required to be disclosed to make the Technical Report not misleading.

Dated this 25th day of April, 2017

(Signed and Sealed) “Kathleen Ann Altman”

Kathleen Ann Altman, Ph.D., P.E.